Polylogarithms and their Arithmetic Bridges

Volume 4: Gamma Grids, CM Periods and Herglotz Arithmetic

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About this volume

Gamma grids and exact certificates lead into complex multiplication, class products and genus ratios. Herglotz functions then connect the period and lattice methods to real-quadratic arithmetic and arithmetic obstructions.

The four volumes form one collective manuscript. They share definitions, notation, source evidence and a bibliography. Chapter, theorem and equation numbers are retained from the unified manuscript so references in research reports remain stable. Cross-volume references link to the companion PDF; keep the four PDF files together when downloading them.

Volume 1	Polylogarithm Identities and Arithmetic Reductions
Volume 2	Signed Kernels, Zero Geometry and Experimental Discovery
Volume 3	Hurwitz Jets, Stieltjes Calculus and Gamma Correlations
Volume 4	Gamma Grids, CM Periods and Herglotz Arithmetic

A rigorous analytic proof, a finite exact certificate and an independently repeated numerical diagnostic carry different evidence. Computation drives conjecture formation and exposes missing structure; successful numerical recognition does not itself prove an identity or arithmetic independence. The source inventory and editorial ledger record the accepted corrections and the scope of material still undergoing analytic reconciliation.

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Chapter 6

Gamma grids and exact certificates

An experiment proposes a relation; a certificate explains why it holds. For gamma values at rational points, taking logarithms turns reflection and multiplication into rational rows. Their row combinations yield an exact, terminating reduction procedure relative to a specified law matrix. This computational completeness must be distinguished from the open arithmetic completeness conjecture.

6.1 The lattice model

Work on the denominator-N grid, in coordinates $x_k = \log\Gamma(k/N)$, $k = 1, \dots, N-1$. The two classical law families become exact $\mathbb{Q}$-linear rows with symbolic right-hand sides over the atoms $\log\pi$, $\log n$, $\log\sin(\pi k/N)$:

- reflection, for $k < N/2$:

$$x_k + x_{N-k} = \log\pi - \log\sin\Big(\frac{\pi k}{N}\Big);$$

- Gauss multiplication, for each divisor $d \geq 2$ of N and $k = 1, \dots, N/d$:

$$\sum_{j=0}^{d-1} x_{k+jN/d} \;-\; [dk < N]\, x_{dk} = \frac{d-1}{2}\big(\log 2 + \log\pi\big) + \Big(\frac{1}{2} - \frac{dk}{N}\Big)\log d,$$

  the instance at $z = k/N$ of the Gauss multiplication theorem [11],

$$\prod_{j=0}^{d-1} \Gamma\Big(z + \frac{j}{d}\Big) = (2\pi)^{(d-1)/2}\, d^{\,1/2-dz}\, \Gamma(dz),$$

  with the convention $x_N := \log\Gamma(1) = 0$ in the boundary case $dk = N$.

The raw matrix has $\lfloor (N-1)/2 \rfloor + \sigma(N) - N$ rows and $(N-1)$ columns, before redundant rows are removed. One exact row reduction of the augmented block $[M \mid I]$ over $\mathbb{Q}$ (columns ordered numerator-descending, so pivots land on large numerators) yields simultaneously:

1. a reduction $\Gamma(k/N) = \exp(\text{elementary}) \cdot \prod \Gamma(\text{basis})^{\mathbb{Q}}$ for *every* non-basis k at once, the elementary side computed exactly as (combination matrix) $\cdot$ (right-hand-side vector) — no `FullSimplify` anywhere;

2. the grid's *basis* — its free formal gamma coordinates — and with it a termination guarantee: the lattice rank says the number of free coordinates in this presented system.

Proposition 6.1 (Completeness relative to the law matrix)**.** *For a fixed denominator grid and a fixed finite family of reflection and multiplication rows, exact row reduction computes every linear consequence of those rows. The free columns give a basis of the formal quotient by that row space, and the augmented matrix records a certificate for each reduction.*

Proof. Elementary row operations preserve the row space and are invertible. The reduced echelon matrix therefore spans exactly the same consequences. Each pivot equation is a recorded rational combination of the original rows. The nonpivot columns are free in the presented quotient. □

Unrestricted completeness of the standard relations for gamma monomials at rational arguments is Rohrlich's conjecture. Koblitz–Ogus prove a sufficient algebraicity criterion, not its converse. Thus irreducibility of the numerical gamma basis is conditional, and a finite-grid computation also needs care if it is compared with relations using larger grids. See the discussion of standard relations in Rivoal and Roques, *Hadamard products of algebraic functions* (Section 3) [34].

Result 6.2 (benchmarks and coverage). Implementations: `tools/gamma-reducer.wl` (Wolfram) and `tools/gamma_reducer.py` (pure Python: exact $\mathbb{Q}$ arithmetic via `fractions.Fraction`, sparse atom-coefficient right-hand sides, mpmath verification). Both produce *identical* output (same rules, same bases).

workload (generation only)	Wolfram	Python
denominator-120 grid (103 rules, 16-element basis)	0.42 s	0.056 s
all 151 legacy-database grids, denominators ≤ 308 (15 588 rules)	75.3 s	7.44 s
mpmath verification of all 15 588 rules	—	8.32 s

Coverage of the 550 recovered `GammaRules.wdx` arguments: 550/550. The original prover needed multi-hour sessions for 550 rules with no terminus; either implementation beats it by orders of magnitude *with* a completeness guarantee. Sample output (the sine values collapse to radicals on their own):

$$\Gamma\Big(\frac{5}{12}\Big) = \frac{2^{2/3}\sqrt{\pi}\,\Gamma\big(\frac{1}{12}\big)}{(1+\sqrt{3})\,\Gamma\big(\frac{1}{6}\big)}.$$

Remark 6.3. The row-combination matrix produced by the augmented reduction is, at the same time, a *multiplier certificate* in the sense of Section 6.5: the reducer's output is born proved, not merely verified.

6.2 Trig-root Γ identities: mechanism and census

The prompt for this investigation was the identity

$$\frac{\Gamma\big(\frac{2}{7}\big)\Gamma\big(\frac{11}{42}\big)}{\Gamma\big(\frac{1}{21}\big)} = \frac{8\sin\big(\frac{\pi}{7}\big)\sqrt{\pi\,\sin\big(\frac{\pi}{21}\big)\sin\big(\frac{4\pi}{21}\big)\sin\big(\frac{5\pi}{21}\big)}}{2^{1/42}\,7^{1/3}\,3^{9/28}},$$

of the kind that has surprised mathematicians before the mechanism was recognized [historical experiment: verified RootReduce-exactly against the legacy table].

Result 6.4 (the mechanism, in lattice terms). When the lattice solve expresses a target $\log\Gamma(k/N)$, the elimination may pass through a combination in which the target appears with multiplicity $D > 1$ — i.e. the relation natively proves $\Gamma(k/N)^D = (\text{elementary}) \cdot (\Gamma\text{-monomial})$. Dividing by D puts coefficient $1/D$ on the elementary $\log\sin$ atoms, and exponentiating yields a D-th root of a sine product; positivity of all factors fixes the branch. The root is a property of the *derivation path*, not of the value: the same $\Gamma(11/42)$ carries a square root in the harvested spelling above and cube roots in the lattice's own reduction — both exactly correct.

The surprise dissolves further at grids whose sines are radical-expressible: for $N = 30, 60$ the renderer collapses $\sqrt{\sin\cdots}$ into golden-ratio radicals and nothing looks trig-rooted at all. The phenomenon is only *visible* when the sine arguments have large totient (21, 39, 78, …), where nested square-root expressions need not exist; general radical expressibility is a different question.

Result 6.5 (census, denominator grids $N \le 200$, primitive arguments). 856 lattice reductions carry fractional $\log\sin$ coefficients. By root degree D (the lcm of the coefficient denominators):

D	2	3	4	6	8	16
rules	616	46	106	26	42	20

Demanding *integral* Γ-exponents (so only the trigonometric part is rooted, as in the prompt identity) leaves exactly 10 rules, all $D = 4$, at $\Gamma(71/78)$, $\Gamma(85/156)$, $\Gamma(65/114)$, $\Gamma(167/198)$ and conjugate-numerator siblings. The $D = 16$ cases live on the $N = 168$ grid with 26 sines under the radical.

Result 6.6 (showpieces). [historical experiment: verified to 30+ digits in both implementations] The fourth-root champion ($N = 78$; $\varphi(78) = 24$, so square-root-only expressions are unavailable):

$$\Gamma\Big(\frac{71}{78}\Big) = 2\cdot 2^{55/312}\,\pi\,\sqrt{\cos\tfrac{\pi}{78}}\;\cos^{1/8}\tfrac{\pi}{39}\;\cos^{3/4}\tfrac{\pi}{26}\;\cos^{5/8}\tfrac{2\pi}{39}\;\cos^{11/8}\tfrac{4\pi}{39}\;\big(\cos\tfrac{\pi}{13}\sec\tfrac{7\pi}{78}\big)^{3/8}\,\Gamma\Big(\frac{1}{39}\Big)$$
$$\Big/\Big[\,13^{1/48}\,\cos^{1/4}\tfrac{5\pi}{78}\;\cos^{3/2}\tfrac{3\pi}{26}\;\cos^{7/8}\tfrac{5\pi}{39}\;\cos^{5/8}\tfrac{11\pi}{78}\;\Gamma\Big(\frac{2}{39}\Big)\Gamma\Big(\frac{5}{78}\Big)\Big],$$

and the lattice's cube-root reduction of the prompt's own $\Gamma(11/42)$:

$$\Gamma\Big(\frac{11}{42}\Big) = \frac{7^{1/6}\sqrt{\pi}\;\Gamma\big(\frac{1}{42}\big)\Gamma\big(\frac{1}{21}\big)\big(\sec\frac{\pi}{42}\sec\frac{\pi}{21}\big)^{2/3}\sec^2\frac{\pi}{14}\,\big(\cos\frac{2\pi}{21}\sec\frac{5\pi}{42}\big)^{1/3}}{8\cdot 2^{5/21}\,3^{3/7}\,\Gamma\big(\frac{2}{21}\big)\Gamma\big(\frac{3}{14}\big)}.$$

6.3 The trigamma analogue

The same construction applies one weight up. On the grid $y_k = \psi^{(1)}(k/N)$ the lattice rows are the reflection law $\psi^{(1)}(k/N) + \psi^{(1)}(1 - k/N) = \pi^2\csc^2(\pi k/N)$ and the multiplication theorem $\psi^{(1)}(dk/N) = d^{-2}\sum_{j=0}^{d-1}\psi^{(1)}(k/N + j/d)$, with the pin $\psi^{(1)}(1) = \pi^2/6$ wherever an argument hits 1; the elementary atoms are π^2 and the $\pi^2\csc^2(\pi k/N)$ values.

Result 6.7. `tools/psi_reducer.py` derives the complete trigamma reduction tables for grids $N = 3, \ldots, 100$: 3428 rules, all mpmath-verified, in 7.7 s total, with basis dimensions equal to $\varphi(N)/2$ throughout — the same dimension count as the cyclotomic-polylogarithm lattice, which the exact DFT bridge between the two worlds explains (the DFT provides a change of algebraic coordinates between the value systems).

This subsumes, by a factor of ∼50, a legacy 70-entry hand/PSLQ-era workbook of $\psi^{(1)}(k/n)$ closed forms (denominators 2–16), which is preserved separately as a harvested store.

6.4 The harvest: legacy notebooks, verified and taxonomized

A kernel-free pipeline (notebook box parser, custom `TraditionalForm`/`InputForm` interpreter, mpmath 60-digit verification) mined the owner's OneDrive notebook corpus into verified identity stores:

- `harvested-gammaprover-rules.wl` — the recovered 550-rule saturation-prover database, every rule verified in-kernel to $|\,\mathrm{N}[\Gamma(\mathrm{arg}) - \text{closed form}, 40]\,| < 10^{-30}$, zero failures;
- `harvested-gamma-values.wl` — 508 verified Γ reductions from four further notebook families (including a 451-rule table recovered from a single 700 KB output cell, and 637 sine-product reductions from a working-table pair identified during the sweep);
- `harvested-trigamma-values.wl` — all 70 identities of the trigamma workbook, mpmath-verified at 60 digits and re-verified in-kernel.

Result 6.8 (novelty taxonomy). Measured against the 670-rule `FunctionTransformations` Γ table [12] (denominators ≤ 120) that anchors the corpus: 314 distinct harvested arguments lie beyond the table — 280 ceiling-breakers from the prover database (denominators 123–308) plus 34 arguments that only the notebook harvest reaches: one further ceiling-breaker, $\Gamma(67/160)$, and 33 upper-half reflection twins of table arguments. (Eight more harvest ceiling-breakers duplicate prover-database arguments; an earlier recorded decomposition "$280 + 8 + 34$" double-counted them.)

6.5 Multiplier certificates

Before this program, every identity in the project's stores carried one badge: *numerically verified* ($\gtrsim 25$ digits). The certificate program adds a qualitatively stronger badge: *derived* — a recorded, machine-checkable certificate exhibiting the value as an exact linear combination of verified functional-equation instances, replayed with no search and no numerics in the algebraic replay path of the historical checker. Its numerical branch guards remain outside the exact certificate guarantee.

6.5.1 The model

A certificate stores law *names* plus concrete substitution strings (parsed through a restricted substitution grammar; strings alone do not guarantee safe or valid instantiation) and two exact multiplier vectors over a canonical real/imaginary splitting of the instances. Validity means the claim lhs − rhs equals the recorded linear combination *identically*: every inert Li/ψ atom coefficient cancels `RootReduce`-exactly, and the elementary residual is killed by a dedicated sound zero test (certified principal-branch log normalization, certified multiplicative log-basis reduction, formal coefficient comparison). Numerics appear in exactly two proposal/guard roles: proposing root-of-unity decompositions that exact algebra then certifies, and a numerical per-instance branch guard. That guard can reject bad instances, but is not an exact domain proof; branch validity is a separate obligation. Linear systems subsume orbit words, fixed-point arguments, and multi-value systems in one format.

A `derived` certificate proves its claim *relative to* a declared axiom layer: (a) the classical laws of the project's functional-equation store [1] — Euler reflection, inversion, Landen, duplication/triplication/multiplication, the five-term family, the Kummer-class relations, Schwarz conjugation, the cyclotomic polygamma sum, and the polygamma reflection/multiplication laws, each numerically verified at recorded test points; (b) definitional pins ($\mathrm{Li}_n(1) = \zeta(n)$, $\psi^{(m)}(1) = (-1)^{m+1} m!\,\zeta(m+1)$,

and the two Catalan pins below); (c) Euler's $\zeta(2k) \in \mathbb{Q}\pi^{2k}$ and trigonometric values at rational multiples of π; (d) `RootReduce` as the exact algebraic-number oracle.

6.5.2 Findings the linear algebra produced

Finding 6.9 (Schwarz conjugation is a necessary axiom). The anharmonic orbit of i is the six-point family $\{i, -i, 1 \pm i, (1 \pm i)/2\}$, and inversion and duplication at i produce the same sum relation. The full system of reflection/inversion/Landen/duplication instances over the orbit is rank-deficient by *exactly one* dimension, and the missing direction is precisely $\mathrm{Li}_2(\bar{z}) = \overline{\mathrm{Li}_2(z)}$. No combination of the single-variable transformation laws separates $\mathrm{Li}_2(i)$ from $\mathrm{Li}_2(-i)$ — the transformation laws are Möbius/algebraic in the argument and blind to conjugation. Obvious in hindsight; the linear algebra *measured* it.

Finding 6.10 (Catalan is definitional; the rest is not). $\beta(2)$ admits no derivation from the stored laws: it enters every certificate that needs it through one of two recorded definitional pins, both termwise series identities,

$$\mathrm{Im}\,\mathrm{Li}_2(i) = G \qquad \text{or} \qquad \psi^{(1)}\!\left(\tfrac{1}{4}\right) - \psi^{(1)}\!\left(\tfrac{3}{4}\right) = 16\,G.$$

By contrast $\mathrm{Li}_2(-1) = -\pi^2/12$ (duplication at $z = 1$), the four golden-ratio dilogarithm values, $\mathrm{Li}_3(\frac{1}{2})$, the golden trilogarithm $\mathrm{Li}_3(1/\varphi^2)$, and $\mathrm{Li}_2(2)$ are all honestly derived; the flagship compaction is $\mathrm{Li}_2(\frac{1}{2})$ as a *one-instance* certificate (Euler reflection at its own fixed point), and each golden-ratio dilogarithm carries exactly the classical four-equation linear system, the coincidence $1 - 1/\varphi = 1/\varphi^2 = (1/\varphi)^2$ doing all the work. Within this law vocabulary, the tested imaginary orbit uses Catalan as its single additional named constant. The whole imaginary line also carries $\mathrm{Ti}_2(y)$; this finite orbit calculation is not a universal reduction theorem.

Finding 6.11 (prime denominators need the full multiplication row). For the cyclotomic values $\mathrm{Li}_n(e^{2\pi i p/q})$ the polygamma vocabulary pins the ψ-grid via proper-divisor multiplication instances when q is composite; for *prime* q the only ψ-sum relation is the full row $d = q$ — the multiplication theorem at $z = 1/q$,

$$\psi^{(m)}(1) = q^{-m-1} \sum_{k=1}^{q} \psi^{(m)}\!\left(\frac{k}{q}\right),$$

whose $k = q$ term is $\psi^{(m)}(1)$ itself — and the DFT coefficients are the source of the algebraic irrational coefficients — e.g. the $\sqrt{5}\,\zeta(3)$ terms in the $q = 5$ trilogarithm closed forms. Omitting $d = q$ cost precisely the prime-q certificates and nothing else.

Finding 6.12 (a trusted layer must be applied, not assumed). In the historical checker, a trigonometric normalization step was omitted: the $q = 8$ certificates initially failed because $\csc^2(\pi/8)$ survived as an opaque atom in the formal zero test while being numerically zero all along. The fix is policy, not code volume: trig at rational multiples of π is part of the declared trusted base and is applied explicitly during law instantiation and residual checking. A trusted constant layer is only trusted if the checker actually *invokes* it; silent reliance on auto-evaluation makes the trust boundary version-dependent.

6.5.3 Coverage ledger

At the repository snapshot, the certificate ledger replayed by the independent checker under the regression guard is 212 **derived** + 4 **definitional, zero failures**:

store	derived / attempted	notes
curated values	$45 + 4$ definitional	every concrete curated value
legacy polylog table, order-2 real	82/83	one honest holdout (below)
legacy Γ table, denominators ≤ 40	85/100	complete-lattice law pool

The Γ certificates use the complete denominator-q relation lattice of Section 6.1 as the law pool (20/20 on the pilot slice, versus 3/20 for the earlier heuristic pool). The polylog order-2 sweep was driven by three cumulative generator upgrades — atom-closure five-term pairs with structured one-parameter pair families $(x, 1-x)$, $(x, x/(x-1))$, $(x, -x)$; square-root argument seeding for duplication ladders; and a height-weighted cluster retry that pools orbit vocabulary across related rules without ever using a related *claim* (no circularity) — the largest certificate being 80 instances.

Search observation 6.13 (the honest holdout). $\mathrm{Li}_2(-7/10)$, the deepest reduction ladder in its slice (10 residual dilogarithms), remains rank-deficient even at deep search caps: the orbit pairs demand an internal 17-flavoured cancellation that no implemented pair family reaches. The natural next lever is second-order stray matching or ladder relations beyond the stored law set. On the Γ side, the frontier is 15 no-certs of $q = 40$ grade (sine-product exact algebra blow-ups) and the $q > 40$ tail, awaiting cyclotomic-unit arithmetic or certificate transport through the exact DFT bridge to the lighter ψ-side atoms.

Chapter 7

Polygamma row sums at complex multiplication points

Rational gamma grids account for periods that reappear in modular forms. At imaginary quadratic arguments the polygamma series becomes a row of a lattice Eisenstein sum. Symmetry isolates elementary components; modular periods and class-field coordinates account for the non-elementary ones. Individual values and full row sums require different evidence.

7.1 Lattice normalization and row sums

For even $w \geq 4$ and $\Im\tau > 0$, define the absolutely convergent

$$G_w(\tau) = \sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}} \frac{1}{(m+n\tau)^w} = 2\zeta(w)E_w(\tau).$$

Pair positive and negative n, and use the defining polygamma series:

$$G_w(\tau) = 2\zeta(w) + \frac{2}{(w-1)!}\sum_{n\geq 1}\left[\psi^{(w-1)}(n\tau) + \psi^{(w-1)}(1-n\tau)\right]. \tag{7.1}$$

The hexagonal point in this chapter is $\rho = e^{i\pi/3}$. The modular invariant $\mathfrak{g}_2 = E_4/\eta^8$ below is unrelated to the second Stieltjes constant. Algebraic fits and historical numerical residuals are labeled as such; class polynomials are formed from normalized modular functions, rather than untreated modular-form values.

7.2 The elementary layer: reflection composed with conjugation

Before any CM input, one family of components is fixed by symmetry alone. The Schwarz reflection $\psi^{(m)}(\bar{z}) = \overline{\psi^{(m)}(z)}$ turns the reflection law into a statement about a single real component, leaving the other component untouched.

Result 7.1 (vertical-line trigamma law). On the line $\operatorname{Re} z = \frac{1}{2}$ the reflection law alone evaluates the real part of the trigamma function at every height: for all real t,

$$\operatorname{Re}\psi^{(1)}\Big(\tfrac{1}{2} + it\Big) = \frac{\pi^2}{2\cosh^2(\pi t)}.$$

In particular $\operatorname{Re}\psi^{(1)}\big(\frac{1+i}{2}\big) = \pi^2/(2\cosh^2(\pi/2))$ [historical experiment: residual 0 at 50 dps]. Off the symmetric line, combining reflection, conjugation and the recurrence gives e.g.

$$\operatorname{Re}\psi^{(1)}(i) = -\frac{1}{2} - \frac{\pi^2}{2\sinh^2\pi} \qquad \text{[historical experiment: residual 0 at 50 dps].}$$

The proof of the first identity is one line: on $\operatorname{Re} z = \frac{1}{2}$ one has $1 - z = \bar{z}$, so the reflection law $\psi^{(1)}(z) + \psi^{(1)}(1-z) = \pi^2/\sin^2(\pi z)$ reads $2\operatorname{Re}\psi^{(1)}(z) = \pi^2/\sin^2(\pi z)$ there.

The imaginary parts on this line stay opaque — they are odd-row lattice data in the sense of Section 7.12.

7.2.1 The FoxTrot identity demystified

MathWorld's *Digamma Function* page [17] calls the following identity "surprising"; it arises from the FoxTrot series $\sum(-1)^{n-1}n^2/(n^3+1)$, whose partial fractions hit the roots of $n^2 - n + 1$:

$$-\psi\big(\tfrac{\omega}{2}\big) - \psi\big(-\tfrac{\omega^2}{2}\big) + \psi\big(\tfrac{1+\omega}{2}\big) + \psi\big(\tfrac{1-\omega^2}{2}\big) = 2\pi\ \operatorname{sech}\Big(\tfrac{\sqrt{3}\,\pi}{2}\Big), \qquad \omega = e^{i\pi/3}. \tag{7.2}$$

Law 7.2 (line law for digamma differences)**.** *For every shift $c \in (0,1)$, on the line $\operatorname{Re} z = (1-c)/2$,*

$$\operatorname{Re}\big[\psi(z+c) - \psi(z)\big] = \pi\ \operatorname{Re}\cot(\pi\bar{z}).$$

[historical experiment: verified for $c = \frac{1}{2}, \frac{1}{3}, \frac{2}{3}$ at generic heights t]

The mechanism is the same one-liner: on that line the shift satisfies $z + c = 1 - \bar{z}$, so reflection composed with conjugation eliminates the digamma values, leaving an elementary sinh / cos closed form. The FoxTrot identity (7.2) is the instance $c = \frac{1}{2}$, $z = \frac{1}{4} + i\frac{\sqrt{3}}{4}$: its four arguments pair into $2\operatorname{Re}[\psi(z+\frac{1}{2}) - \psi(z)]$, and on $\operatorname{Re} z = \frac{1}{4}$ one has $\pi\operatorname{Re}\cot(\pi\bar{z}) = \pi\operatorname{sech}(2\pi t)$. The identity is thus the point $t = \sqrt{3}/4$ on a line of identities, itself one member of a two-parameter family — no complex multiplication required.

Differentiating in t ladders the law up the polygamma tower [historical experiment: weights $2, 3$ verified at 10^{-50}]:

$$\begin{aligned}
\operatorname{Im}\big[\psi^{(1)}(\tfrac{3}{4}+it) - \psi^{(1)}(\tfrac{1}{4}+it)\big] &= 2\pi^2\operatorname{sech}(2\pi t)\tanh(2\pi t),\\
\operatorname{Re}\big[\psi^{(2)}(\tfrac{3}{4}+it) - \psi^{(2)}(\tfrac{1}{4}+it)\big] &= 8\pi^3\operatorname{sech}^3(2\pi t) - 4\pi^3\operatorname{sech}(2\pi t).
\end{aligned}$$

More generally, for $0 < c < 1$, let $z = (1-c)/2 + it$ and $R_c(t) = \pi\sin(\pi c)/(\cosh(2\pi t) + \cos(\pi c))$. Reflection and conjugation give $\operatorname{Re}(\psi(z+c) - \psi(z)) = R_c(t)$. Differentiating m times yields

$$\operatorname{Re}\Big(\mathrm{i}^m[\psi^{(m)}(z+c) - \psi^{(m)}(z)]\Big) = R_c^{(m)}(t).$$

Thus the real part is determined for even m and the imaginary part for odd m. With weight $w = m+1$, these are the real component at odd weight and imaginary component at even weight. Their complementary components are not determined by this reflection identity; that leaves open reductions by other methods. This shifted-difference statement is distinct from individual polygamma values on $\operatorname{Re} z = 1/2$.

7.3 Maximal symmetry: vanishing identities and Γ-periods

At the two CM points with extra symmetry — $\tau = i$ (square lattice, $j = 1728$) and $\tau = \rho = e^{i\pi/3}$ (hexagonal lattice, $j = 0$) — classical theory pins every G_{2k} exactly. The square lattice admits no weight-6 form ($i \cdot (\mathbb{Z} + \mathbb{Z}i) = \mathbb{Z} + \mathbb{Z}i$ forces $G_6(i) = i^{-6}G_6(i) = -G_6(i)$), and the hexagonal lattice no weight-4 form; $G_4(i)$ and $G_6(\rho)$ are the lemniscatic and equianharmonic period powers — monomials in $\Gamma(1/4)$, resp. $\Gamma(1/3)$, by Chowla–Selberg [13].

7.3.1 Vanishing identities

Result 7.3 (the CM gems).

$$\sum_{n\geq 1}\Big[\psi^{(5)}(ni) + \psi^{(5)}(1-ni)\Big] = -120\,\zeta(6) \qquad \text{[historical experiment: residual } 1.7\times 10^{-49}\text{]},$$

$$\sum_{n\geq 1}\Big[\psi^{(3)}(n\rho) + \psi^{(3)}(1-n\rho)\Big] = -6\,\zeta(4), \qquad \rho = e^{i\pi/3} \qquad \text{[historical experiment: residual } 6.8\times 10^{-50}\text{]}.$$

Each individual bracket is a transcendental-looking complex number (elementary in coth / csch of $\pi n\sqrt{3}/2$ and the like after the bilateral cotangent expansion, but nobody would guess the totals); the sums collapse to rational multiples of $\zeta(6)$, $\zeta(4)$ *because lattice rotation multiplies the corresponding value by a nontrivial root of unity.* Via (7.1) these are exactly the statements $G_6(i) = 0$ and $G_4(\rho) = 0$.

7.3.2 Chowla–Selberg made polygamma-visible

Result 7.4 (period identities).

$$2\zeta(4) + \frac{1}{3}\sum_{n\geq 1}\Big[\psi^{(3)}(ni) + \psi^{(3)}(1-ni)\Big] = \frac{\Gamma(1/4)^8}{960\,\pi^2} \qquad \text{[historical experiment: residual } 2.1\times 10^{-50}\text{]},$$

$$2\zeta(6) + \frac{1}{60}\sum_{n\geq 1}\Big[\psi^{(5)}(n\rho) + \psi^{(5)}(1-n\rho)\Big] = \frac{\Gamma(1/3)^{18}}{8960\,\pi^6} \qquad \text{[historical experiment: residual } 2.0\times 10^{-60}\text{]}.$$

Polygamma values at purely imaginary (resp. hexagonal) points, summed over the integer multiples of the point, know the lemniscate constant and its equianharmonic sibling. The atoms $\Gamma(1/4)^2/\pi$ and $\Gamma(1/3)$ are precisely the CM-period currency of the repository's hypergeometric reducer (`cmPeriodReduce` in `src/Hypergeometric`), so these identities tie the polygamma world directly to that corpus.

7.3.3 The complete low-weight table

The symmetry argument and the classical Hurwitz recursion [15]

$$(2n+1)(2n-1)(n-3)\,G_{2n} = 3\sum_{m=2}^{n-2}(2m-1)(2n-2m-1)\,G_{2m}G_{2n-2m}$$

extend both subsections through weight 12 with no new constants [historical experiment: all entries verified, residuals 10^{-60}–10^{-263}]:

weight $2k$	square lattice $\tau = i$	hexagonal lattice $\tau = \rho$
4	$\Gamma(1/4)^8/(960\pi^2)$	$\mathbf{0}$
6	$\mathbf{0}$	$\Gamma(1/3)^{18}/(8960\pi^6)$
8	$\frac{3}{7}G_4(i)^2 = 3\,\Gamma(1/4)^{16}/(7\cdot 960^2\pi^4)$	$\mathbf{0}$
10	$\mathbf{0}$	$\mathbf{0}$
12	$\frac{1}{143}\big(18\,G_4(i)^3 + 25\,G_6(i)^2\big)$ type	$\frac{25}{143}G_6(\rho)^2 \propto \Gamma(1/3)^{36}$

Every bold zero is a vanishing identity $\sum_{n\geq 1}[\psi^{(w-1)}(n\tau) + \psi^{(w-1)}(1-n\tau)] = -(w-1)!\,\zeta(w)$, and every non-zero entry is a $\psi^{(w-1)}$-sum evaluation. The pattern is exhaustive: $G_{2k}(i) = 0$ iff $2k \not\equiv 0 \pmod 4$, $G_{2k}(\rho) = 0$ iff $2k \not\equiv 0 \pmod 6$, and all non-zero values are $\mathbb{Q}$-monomials in the two period powers — so the *entire* Eisenstein tower over both lattices is polygamma-explicit.

7.4 Beyond maximal symmetry: $2i$ and $i\sqrt{2}$

The same row-sum at the next two CM points, with closed forms found by PSLQ [historical experiment: 120-dps canaries]:

Theorem 7.5.

$$G_4(2i) = \frac{11}{15360}\,\frac{\Gamma(1/4)^8}{\pi^2} \quad \textit{[historical experiment: canary } 0 \textit{ at } 120 \textit{ dps]},$$

$$G_4(i\sqrt{2}) = \frac{\Gamma(1/8)^4\,\Gamma(3/8)^4}{4608\,\pi^2} \quad \textit{[historical experiment: canary } 3.9\times 10^{-121}\textit{]}.$$

The weight-6 row at the same points [historical experiment: 160-dps canaries]:

$$G_6(2i) = \frac{\Gamma(1/4)^{12}}{2^{14}\cdot 5\cdot \pi^3}, \qquad G_6(i\sqrt{2}) = \frac{\Gamma(1/8)^6\,\Gamma(3/8)^6}{2^{12}\cdot 3^3\cdot 5\cdot \pi^3}.$$

The first value confirms the isogeny picture: $2i$ sits over the same field $\mathbb{Q}(i)$, so only the rational factor changes ($G_4(i)/G_4(2i) = 16/11$). The second uses the period coordinate associated with discriminant -8, field $\mathbb{Q}(\sqrt{-2})$ — the polygamma sum $\sum_n[\psi^{(3)}(ni\sqrt{2}) + \psi^{(3)}(1-ni\sqrt{2})]$ evaluates in $\Gamma(1/8)\Gamma(3/8)$. The fundamental unit $1+\sqrt{2}$ was offered in the PSLQ basis and declined (exponent 0). Together with Section 7.3 this makes three distinct Chowla–Selberg periods polygamma-visible.

7.5 Class number 2: discriminants -20 and -15

At $\mathbb{Q}(\sqrt{-5})$ (class number 2) a monomial-only logarithmic basket can miss the value: the algebraic factor in G_4/Ω^4 lives in the Hilbert class field and is not a monomial. The structure that works is the Hecke/Damerell pair [14] over the two reduced forms x^2+5y^2 ($\tau_1 = i\sqrt{5}$) and $2x^2+2xy+3y^2$ ($\tau_2 = (-1+i\sqrt{5})/2$).

Theorem 7.6 (disc -20)**.**

$$\frac{G_4(\tau_1)}{G_4(\tau_2)} = \frac{29+12\sqrt{5}}{44} = \frac{(3+2\sqrt{5})^2}{44} \quad \textit{[historical experiment: canary } 0 \textit{ at } 160 \textit{ dps]},$$

$$G_4(\tau_1)\,G_4(\tau_2) = \frac{11\,\varphi^4\,\Gamma(3/20)^8\,\Gamma(7/20)^8}{2^{14}\cdot 3^4\cdot 5^4\cdot \pi^4} \quad \textit{[historical experiment: canary } 5.3\times 10^{-160}\textit{]},$$

where φ is the golden ratio. The ratio is algebraic of degree 2 with norm $N(29+12\sqrt{5}) = 121 = 11^2$; the class product is a period monomial carrying the prime 11 and a golden unit.

Theorem 7.7 (disc -15)**.** *With $\tau_1 = (1+i\sqrt{15})/2$ for $x^2+xy+4y^2$ and the conjugate pair $\tau_{2,3} = (\pm 1 + i\sqrt{15})/4$ representing $2x^2+xy+2y^2$, write $\widehat{G}_4 = \frac{1}{2}\big[G_4(\tau_2)+G_4(\tau_3)\big] = \operatorname{Re} G_4(\tau_2)$ for the class-pair average. Then [historical experiment: 160-dps canaries]:*

$$\frac{G_4(\tau_1)}{\widehat{G}_4} = \frac{2\,(4+\sqrt{5})^2}{77}, \qquad G_4(\tau_1)\,\widehat{G}_4 = \frac{77\left(\Gamma(\frac{1}{15})\Gamma(\frac{2}{15})\Gamma(\frac{4}{15})\Gamma(\frac{8}{15})\right)^4}{2^{13}\cdot 3^6\cdot 5^5\cdot \pi^4}.$$

Unlike disc -20, the non-principal value here is genuinely complex (the form $(2,1,2)$ has $|\tau_2|=1$, where conjugation acts nontrivially on G_4); the real-part projection matches the representative convention used in these formulas; its period factor is fixed in the proof below.

The Γ-block in Result 7.7 is exactly the set of residues $\{1,2,4,8\}$ with $\chi_{-15}=+1$; the golden unit is declined by PSLQ (exponent 0); and the prime content $77 = 7\cdot 11$ of the product matches the ratio's denominator ($N(4+\sqrt{5}) = 11$). The heuristic that emerges, used at every later discriminant: *run the ratio first; its norm names the primes the class-product basis needs.*

7.6 Class number 3: discriminant -23

$\mathbb{Q}(\sqrt{-23})$ has $h=3$, with Hilbert class field generated over $\mathbb{Q}(\sqrt{-23})$ by a root of x^3-x-1. The three reduced forms give $\tau_1 = (1+i\sqrt{23})/2$ and $\tau_{2,3} = (\pm 1 + i\sqrt{23})/4$; $G_4(\tau_1)$ is real and $G_4(\tau_{2,3})$ form a conjugate pair.

Theorem 7.8 (disc -23 class product)**.** *[historical experiment: canary 1.4×10^{-160}]*

$$G_4(\tau_1)\,G_4(\tau_2)\,G_4(\tau_3) = \frac{11\cdot 17\,P^4}{2^{24}\cdot 3^6\cdot 23^5\cdot \pi^{16}}, \qquad P = \prod_{\substack{k \bmod 23 \\ (\frac{k}{23})=+1}} \Gamma\Big(\frac{k}{23}\Big) \quad \textit{(11 factors).}$$

Theorem 7.9 (the $\mathfrak{g}_2$ class polynomial from row-sums)**.** *Let $P(X) = X^3+155X^2+650X+23375$. For each of the three singular moduli j_Q, define ξ_Q as the unique root of P satisfying $\xi_Q^3 = j_Q$. Thus ξ_Q equals $E_4(\tau_Q)/\eta(\tau_Q)^8$ times a cube root of unity, with its phase fixed by this algebraic condition. The three ξ_Q have the exact class polynomial*

$$P(X) = X^3+155X^2+650X+23375.$$

Its constant term $23375 = 5^3\cdot 11\cdot 17$ identifies the same prime support as the proved class product. The cube-image identity and uniqueness of the phase choice are proved below in Proposition 7.21.

Search observation 7.10 (weight-4 sums are not Galois-clean). The dimensionless symmetric functions e_1^3/e_3 and e_2^3/e_3^2 of the raw G_4 values were not recognized as rational in the source search (height 10^{30}, 300 digits). Unlike the value *product* (a norm) and the weight-0 $\mathfrak{g}_2$ invariants, sums of weight-4 CM values across ideal classes are not Galois-stable — the period correction under $\mathrm{Gal}(H/K)$ is nontrivial at weight 4. Class polynomials must be formed from $\mathfrak{g}_2 = E_4/\eta^8$ (or j), not from the G_4 values themselves.

7.7 Class number 4: discriminant -39 and genus structure

The four reduced forms of discriminant -39, namely $(1,1,10)$, $(2,\pm 1,5)$, $(3,3,4)$, give $\tau \in \{(1+i\sqrt{39})/2,\ (\pm 1+i\sqrt{39})/4,\ (3+i\sqrt{39})/6\}$.

Theorem 7.11 (disc -39 class product). *[historical experiment: 160-dps canary]*

$$\prod_{i=1}^{4} G_4(\tau_i) = \frac{17 \cdot 23 \cdot 29\, P^4}{2^{28} \cdot 3^9 \cdot 5^4 \cdot 13^8 \cdot \pi^{16}}, \qquad P = \prod_{\substack{k \bmod 39 \\ \chi_{-39}(k)=+1}} \Gamma\Big(\frac{k}{39}\Big) \quad \textit{(12 factors).}$$

At this discriminant no cube root of a singular modulus can belong to the Hilbert class field: Proposition 7.21 proves its degree is twelve, while that class field has degree eight over $\mathbb{Q}$. This supplies an exact obstruction to every cube-root phase choice. The weight-zero j-invariant still gives the Hilbert class polynomial directly.

Theorem 7.12 (Hilbert class polynomial of disc -39).

$$H_{-39}(x) = x^4 + 331531596\,x^3 - 429878960946\,x^2 + 109873509788637459\,x + 20919104368024767633$$

[historical experiment: $e_1, \ldots, e_3$ PSLQ-exact integers; e_4 integral to 10^{-137}].

Worth dwelling on: a quartic with 20-digit integer coefficients recovered from sums of polygamma values at quadratic-irrational complex points. The polygamma row-sum is a perfectly serviceable special-function front door to the classical CM tower.

Theorem 7.13 (genus structure inside the quartet). *The class group is $\mathbb{Z}/4$, with the ambiguous form $(3,3,4)$ as the order-2 class. The genus-wise ratio — principal genus $\{1, A^2\}$ over $\{A, A^3\}$ — is exactly quadratic, in the real part of the genus field $\mathbb{Q}(\sqrt{-3}, \sqrt{13})$; its precise period cancellation and positive-root extraction are proved by Theorem 7.23 [historical experiment: 160-dps canary; $\sqrt{3}$, $\sqrt{39}$ declined by PSLQ]:*

$$\frac{G_4(\tau_1)\,G_4(\tau_4)}{G_4(\tau_2)\,G_4(\tau_3)} = \frac{103299 + 4440\sqrt{13}}{181424}, \qquad N = \Big(\frac{9}{16}\Big)^2, \qquad 181424 = 16 \cdot 17 \cdot 23 \cdot 29.$$

The denominator reuses the class-product primes 17, 23, 29.

The raw cross-class ratio $G_4(\tau_1)/G_4(\tau_4)$ was separately reported as a quartic candidate. No degree-two polynomial was found up to height 10^{12} in that search. Neither its quartic candidate nor that unsuccessful search establishes its minimal polynomial or degree.

7.8 Class number 5: discriminant -47

The five reduced forms $(1,1,12)$, $(2,\pm1,6)$, $(3,\pm1,4)$ give the quintet $\tau \in \{(1+i\sqrt{47})/2,\ (\pm1+i\sqrt{47})/4,\ (\pm1+i\sqrt{47})/6\}$.

Theorem 7.14 (disc -47 class product). *[historical experiment: 200-dps canary]*

$$\prod_{i=1}^{5} G_4(\tau_i) = \frac{11^2 \cdot 23 \cdot 29\, P^4}{2^{52} \cdot 3^6 \cdot 47^9 \cdot \pi^{36}}, \qquad P = \prod_{\substack{k \bmod 47 \\ (\frac{k}{47})=+1}} \Gamma\Big(\frac{k}{47}\Big) \quad \textit{(23 factors).}$$

Theorem 7.15 (Hilbert class polynomial of disc -47). *The row-sum j-values deliver the quintic Hilbert class polynomial [historical experiment: all symmetric functions integral to 10^{-129} or better]:*

$$\begin{aligned} H_{-47}(x) = x^5 &+ 2257834125\,x^4 - 9987963828125\,x^3 + 5115161850595703125\,x^2 \\ &- 14982472850828613281250\,x + 16042929600623870849609375. \end{aligned}$$

The CM tower is now polygamma-explicit through class number 5 with one uniform recipe: *row-sum → class-product log-PSLQ (norm-forecast primes) → j-symmetric-function integrality.*

7.9 Exact CM row products from class polynomials

The manuscript's complex-multiplication chapter records striking products of weight-four polygamma row sums at discriminants $-15, -20, -23, -39, -47$. Its displayed numerical residuals establish useful evidence, but do not explain the prime factors or provide proofs of the gamma products. The following argument supplies that missing step. It also evaluates the corresponding products at every even weight by a finite computation with a Hilbert class polynomial.

The analytic ingredients are classical: the Kronecker limit formula, the Lerch–Chowla–Selberg formula, and the algebra of modular forms. The contribution here is an explicit, uniformly normalized consequence for the manuscript's row sums, accompanied by certified class-polynomial data and new displayed weight-six evaluations. We do not claim that the underlying CM period theory is new.

7.9.1 Conventions and the main theorem

Let $D = -d < -4$ be a fundamental discriminant, let $\chi_D(r) = (D/r)$ be its primitive quadratic Dirichlet character, and let $h = h(D)$. Take the standard reduced primitive positive definite forms

$$\mathcal{Q}_D = \{(a,b,c) : b^2 - 4ac = D,\ |b| \le a \le c,\ b \ge 0 \text{ if } |b| = a \text{ or } a = c\}.$$

There is one representative for each proper ideal class. Write

$$\tau_Q = \frac{-b + i\sqrt{d}}{2a}, \qquad A_D = \prod_{Q \in \mathcal{Q}_D} a, \qquad P_D = \prod_{\substack{1 \le r < d \\ \chi_D(r) = 1}} \Gamma(r/d), \tag{7.3}$$

$$H_D(X) = \prod_{Q \in \mathcal{Q}_D} (X - j(\tau_Q)), \qquad C_d = \Phi_d(1), \qquad \varphi_d = \varphi(d).$$

Thus $H_D \in \mathbb{Z}[X]$ is monic; $C_d = p$ if d is a positive power of a prime p, and $C_d = 1$ otherwise. All fractional powers of positive real numbers below denote their positive values.

For even $w \ge 4$, use exactly the lattice normalization of the manuscript:

$$G_w(\tau) = \sum_{(m,n) \ne (0,0)} (m + n\tau)^{-w} = 2\zeta(w) E_w(\tau). \tag{7.4}$$

The modular discriminant is $\Delta(\tau) = \eta(\tau)^{24}$; in particular $j = E_4^3/\Delta$ and $j - 1728 = E_6^2/\Delta$.

Lemma 7.16 (The polynomial attached to an even weight)**.** *For every even integer $w \ge 4$, there is a unique monic polynomial $F_w \in \mathbb{Q}[X]$ of degree w such that*

$$\frac{E_w(\tau)^{12}}{\Delta(\tau)^w} = F_w(j(\tau)). \tag{7.5}$$

It is effectively computable using exact rational Fourier coefficients.

Proof. The left side is a weight-zero modular function, holomorphic on the upper half-plane since Δ has no zeros there, with a pole of order w at the cusp and no other poles. The field of modular functions is $\mathbb{C}(j)$, and a modular function with no finite poles is a polynomial in j. Since $E_w = 1 + O(q)$, $\Delta = q + O(q^2)$, and $j = q^{-1} + 744 + O(q)$, this polynomial is monic of degree w. Successively removing its Laurent coefficients at $q^{-w}, \dots, q^0$ computes all coefficients. The input coefficients are rational, hence so are the polynomial coefficients. Equivalently, one can work in the exact graded algebra $\mathbb{Q}[E_4, E_6]$. □

Set

$$\mathcal{R}_{D,w} = \mathrm{Res}_X(H_D(X), F_w(X)).$$

Our resultant convention is $\mathrm{Res}(H_D, F_w) = \prod_Q F_w(j(\tau_Q))$, since H_D is monic. No unidentified sign or leading coefficient is implicit.

7.9.2 An exact generator for the modular weight polynomial

Normalize the Eisenstein series to have constant coefficient one:

$$E_w(\tau) = 1 - \frac{2w}{B_w} \sum_{m \geq 1} \sigma_{w-1}(m) q^m, \qquad q = e^{2\pi i \tau},$$

and set

$$\Delta = q \prod_{m \geq 1} (1 - q^m)^{24} = \frac{E_4^3 - E_6^2}{1728}, \qquad j = \frac{E_4^3}{\Delta}.$$

For each even $w \geq 4$, let $a \in \{0, 1, 2\}$, $b \in \{0, 1\}$, and $c \geq 0$ be the unique integers satisfying

$$w = 4a + 6b + 12c. \tag{7.6}$$

The excluded weight two is precisely the exceptional small weight in this residue prescription.

The level-one graded-ring structure [88] gives a unique monic polynomial $P_w \in \mathbb{Q}[X]$ of degree c such that

$$E_w = E_4^a E_6^b \Delta^c P_w(j). \tag{7.7}$$

This can be computed using only rational arithmetic. Write $P_w(X) = \sum_{r=0}^{c} p_r X^r$ and expand the basis

$$E_4^{a+3r} E_6^b \Delta^{c-r}, \qquad 0 \leq r \leq c.$$

The basis element indexed by r has first term q^{c-r} with coefficient one. Successive coefficient comparison at $q^0, q^1, \ldots, q^c$ therefore determines $p_c, p_{c-1}, \ldots, p_0$ by unit triangular elimination. Since E_w has constant term one, $p_c = 1$.

Consequently the polynomial in the CM norm formula is explicitly

$$\boxed{F_w(X) = X^{4a}(X - 1728)^{6b} P_w(X)^{12}.} \tag{7.8}$$

Indeed, raising (7.7) to the twelfth power and using $E_4^3 = j\Delta$, $E_6^2 = (j - 1728)\Delta$, and (7.6) gives $E_w^{12}/\Delta^w = F_w(j)$. The degree of F_w is exactly w and its leading coefficient is one.

Finite certificates. The companion generator computes Δ independently from its Euler product and verifies its normalization against $E_4^3 - E_6^2$. For every even w from four through twenty-four, it verifies exactly the coefficients of $q^0, \ldots, q^w$ in

$$E_w^{12} = \sum_{r=0}^{w} [X^r] F_w(X)\, E_4^{3r} \Delta^{w-r}. \tag{7.9}$$

Both sides are holomorphic modular forms of weight $12w$ on $\mathrm{SL}_2(\mathbb{Z})$. The level-one Sturm bound is w, so these finitely many rational equalities certify the identity. In characteristic zero this bound follows directly from the valence formula [88, Theorem 2.11]: a nonzero weight-$12w$ holomorphic modular form has order at infinity at most w.

The JSON artifact stores the exact rational coefficients, the compact factorization, the degree and monicity checks, and the finite coefficient certificates. It contains the full range, including

$$F_{14}(X) = X^8(X-1728)^6, \tag{7.10}$$

$$F_{16}(X) = X^4\left(X - \frac{3456000}{3617}\right)^{12}, \tag{7.11}$$

$$F_{18}(X) = (X-1728)^6\left(X - \frac{9504000}{43867}\right)^{12}, \tag{7.12}$$

$$F_{20}(X) = X^8\left(X - \frac{209520000}{174611}\right)^{12}, \tag{7.13}$$

$$F_{22}(X) = X^4(X-1728)^6\left(X - \frac{35424000}{77683}\right)^{12}, \tag{7.14}$$

$$F_{24}(X) = \left(X^2 - \frac{340364160000}{236364091}X + \frac{30710845440000}{236364091}\right)^{12}. \tag{7.15}$$

For example,

$$E_{14} = E_4^2E_6, \qquad E_{16} = \frac{1617E_4^4 + 2000E_4E_6^2}{3617} = E_4\Delta\left(j - \frac{3456000}{3617}\right),$$

which directly explain (7.10)–(7.11). The generator implements a standard modular-form construction, supplying explicit reusable coefficients and proof certificates for the CM recipe.

7.9.3 CM nonvanishing with a rational principal-lattice bound

The possible zero denominators in the product and genus formulas can be excluded uniformly. Non-elliptic CM nonvanishing is classical, through Kohnen's conjugation argument [90]; see the introduction of Chiriac–Jorza [91] for its place in zero theory. The following proof supplies an explicit principal-point bound, certified using rational arithmetic. It applies to nonmaximal orders as well.

Theorem 7.17 (Nonvanishing at every non-elliptic CM point)**.** *Let $D < -4$ be any quadratic-order discriminant. For every CM point τ of discriminant D and every even integer $w \geq 4$,*

$$G_w(\tau) \neq 0.$$

For the principal reduced point

$$\tau_D = \begin{cases} i\sqrt{|D|}/2, & D \equiv 0 \pmod 4, \\ (-1+i\sqrt{|D|})/2, & D \equiv 1 \pmod 4, \end{cases}$$

one has the weight- and discriminant-uniform bound $|G_w(\tau_D)| > 7/20$.

Proof. The principal lattice has, for $n \neq 0$, squared lengths bounded below by one of the two integer quadratic forms

$$q_7(m,n) = m^2 - mn + 2n^2, \qquad q_8(m,n) = m^2 + 2n^2.$$

Indeed the odd principal discriminant has $|D| \geq 7$, and the even one has $|D| \geq 8$. Both forms are at least two when $n \neq 0$. Hence for every even $w \geq 4$ the off-axis absolute sum is bounded by

$$T_d = \sum_{n\neq 0}\sum_{m\in\mathbb{Z}} q_d(m,n)^{-2}, \qquad d = 7, 8.$$

We prove $T_7 < 33/20$ and $T_8 < 11/8$ by finite rational sums and explicit infinite-tail bounds.

For any $A > 0$ and real shift s, layer-cake integration gives

$$\sum_{m\in\mathbb{Z}} \frac{1}{((m-s)^2+A^2)^2} \le A^{-4} + \frac{\pi}{2A^3}.$$

To see this, each superlevel set of the even decreasing function is an interval. Its number of translated lattice points is at most its length plus one. Integrating that inequality bounds the lattice sum by the integral of the function plus its maximum. The integral is $\pi/(2A^3)$.

Use $N = 10$, $M = 20$ in the finite rectangle. The imaginary coordinate in both lattices is greater than one. Thus the part with $|n| > N$, summed over every m, is less than

$$\frac{2}{3N^3} + \frac{2}{N^2},$$

by $\pi < 4$, $\sum_{n>N} n^{-4} \le 1/(3N^3)$ and $\sum_{n>N} n^{-3} \le 1/(2N^2)$. For $1 \le |n| \le N$ and $|m| > M$, the squared length is at least $(|m| - N/2)^2$. Integral comparison bounds this part by $4N/[3(M - N/2)^3]$. The finite certificate therefore verifies

$$2\sum_{n=1}^{10}\sum_{m=-20}^{20} q_7(m,n)^{-2} + \frac{31}{1500} + \frac{8}{2025} < \frac{33}{20},$$

and the same expression with q_8 is less than $11/8$. Every term and comparison in these two finite inequalities is rational; the full fractions are recorded in `verification/CM-nonvanishing.json`.

The $n = 0$ axis contributes exactly $2\zeta(w) > 2$. The triangle inequality now gives $|G_w(\tau_D)| > 2 - 33/20 = 7/20$ in the odd case, and the stronger $5/8$ in the even case.

Suppose a CM value at the same discriminant vanished. By (7.5), its singular modulus would be a zero of the rational polynomial F_w. The class polynomial H_D is the minimal polynomial of that singular modulus, including for a nonmaximal order [86, 16]. Thus H_D divides F_w, and the principal singular modulus is also a zero. Since Δ never vanishes on the upper half-plane, this would make $G_w(\tau_D) = 0$, contradicting the bound. Modular transformations multiply a positive-weight value by a nonzero factor, so the conclusion covers every representative. □

Consequently all class and genus resultants in the following formulas are nonzero at $D < -4$. Their positive roots and quotients are defined at every even weight. The excluded discriminants $-3, -4$ retain their familiar elliptic vanishing patterns.

7.9.4 Class products in every even weight

Theorem 7.18 (CM product formula at every even weight)**.** *With the preceding definitions, for every even $w \ge 4$,*

$$\prod_{Q\in\mathcal{Q}_D} |G_w(\tau_Q)| = \left(\frac{|B_w|}{w!}\right)^h \left(\frac{A_D}{d^h}\right)^{w/2} C_d^{w/4} (2\pi)^{w(2h-\varphi_d)/4} P_D^w \, |\mathcal{R}_{D,w}|^{1/12}. \tag{7.16}$$

Here the Bernoulli numbers satisfy $B_2 = 1/6$. Every factor and the resultant are nonzero by the preceding CM nonvanishing theorem. The identity itself also applies to a zero product when extended to an elliptic discriminant in the order formula below.

The absolute values are essential in a uniform statement. Selecting one standard representative for each class need not make the product of positive-weight modular forms real. Section 7.9.7 gives the phase information needed for the manuscript's examples.

7.9.5 Derivation of the period factor

Put

$$R_D = \prod_{r=1}^{d-1} \Gamma(r/d)^{\chi_D(r)}.$$

We first establish the precise normalization

$$\prod_{Q \in \mathcal{Q}_D} |\Delta(\tau_Q)| = \frac{A_D^6 R_D^6}{(2\pi d)^{6h}}. \tag{7.17}$$

This is the fundamental-discriminant case of the classical Lerch–Chowla–Selberg formula. For a modern primary treatment with all normalization factors, see Barquero-Sanchez et al. [84] and Cohen[85]. Here is a derivation that makes the powers of 2, π, and d checkable.

For $\operatorname{Re} s > 1$, define the full Epstein sum

$$Z(\tau, s) = \sum_{(m,n) \neq (0,0)} \frac{(\operatorname{Im} \tau)^s}{|m + n\tau|^{2s}}.$$

The class decomposition of the Dedekind zeta function gives

$$\sum_{Q \in \mathcal{Q}_D} Z(\tau_Q, s) = 2 \left(\frac{\sqrt{d}}{2} \right)^s \zeta(s) L(s, \chi_D). \tag{7.18}$$

Indeed $Q(m, -n) = a|m + n\tau_Q|^2$ (the sign change in n does not affect the full lattice sum), while $a \operatorname{Im} \tau_Q = \sqrt{d}/2$; the factor 2 is the number of units, since $D < -4$.

The Kronecker limit formula at zero reads

$$Z(\tau, 0) = -1, \qquad Z'(\tau, 0) = -2 \log\bigl(2\pi \sqrt{\operatorname{Im} \tau} |\eta(\tau)|^2\bigr).$$

Also $L(0, \chi_D) = h$, and the finite Hurwitz-zeta expression for L and Lerch's identity give

$$L'(0, \chi_D) = -h \log d + \log R_D.$$

Differentiate (7.18) at zero, using $\zeta(0) = -1/2$ and $\zeta'(0) = -\frac{1}{2} \log(2\pi)$. After collecting terms, one obtains

$$\prod_{Q} \sqrt{\operatorname{Im} \tau_Q} |\eta(\tau_Q)|^2 = (4\pi\sqrt{d})^{-h/2} R_D^{1/2}. \tag{7.19}$$

Finally, $\prod_Q \operatorname{Im} \tau_Q = (\sqrt{d}/2)^h / A_D$. Dividing (7.19) by the square root of this product and raising to the twelfth power proves (7.17).

Because χ_D is odd, the positive and negative character residues are interchanged by $r \mapsto d - r$. Euler reflection therefore gives

$$R_D = \frac{P_D^2}{\pi^{\varphi_d/2}} \prod_{\chi_D(r)=1} \sin(\pi r/d) = \frac{P_D^2 \sqrt{C_d}}{(2\pi)^{\varphi_d/2}}. \tag{7.20}$$

To verify the last equality, square the sine product: it becomes the product over all reduced residues, because paired sines are equal. The identity $|1 - e^{2\pi i r/d}| = 2 \sin(\pi r/d)$ and the cyclotomic polynomial evaluated at 1 give its square as $C_d / 2^{\varphi_d}$. Every sine is positive, fixing the square-root sign.

Proof of Theorem 7.18. Multiply (7.5) over the class representatives and take absolute values:

$$\prod_Q |E_w(\tau_Q)| = \left(\prod_Q |\Delta(\tau_Q)|\right)^{w/12} |\mathcal{R}_{D,w}|^{1/12}.$$

Substitute (7.17), then $2\zeta(w) = |B_w|(2\pi)^w/w!$ and (7.20). Directly collecting exponents gives (7.16). No logarithm is taken of E_w, so the same argument applies when a factor vanishes. □

7.9.6 A finite arithmetic recipe and useful low weights

The general formula is particularly simple in the first few weights:

w	$F_w(X)$	$\lvert\mathcal{R}_{D,w}\rvert^{1/12}$
4	X^4	$\lvert H_D(0)\rvert^{1/3}$
6	$(X-1728)^6$	$\lvert H_D(1728)\rvert^{1/2}$
8	X^8	$\lvert H_D(0)\rvert^{2/3}$
10	$X^4(X-1728)^6$	$\lvert H_D(0)\rvert^{1/3}\lvert H_D(1728)\rvert^{1/2}$
12	$(X-432000/691)^{12}$	$\lvert H_D(432000/691)\rvert$

For the last row, use $E_{12} = (441E_4^3 + 250E_6^2)/691 = \Delta(j - 432000/691)$. This table is an exact algebraic calculation. In particular, the prime factors of a class product are no longer guessed from a ratio: at weights four and six they are determined by the factorizations of $H_D(0)$ and $H_D(1728)$, together with the explicitly displayed elementary factors.

Specializing the theorem to weight four gives

$$\prod_Q |G_4(\tau_Q)| = \frac{|H_D(0)|^{1/3} A_D^2 C_d}{180^h d^{2h} 2^{\varphi_d} \pi^{\varphi_d - 2h}} P_D^4. \tag{7.21}$$

At weight six it gives

$$\prod_Q |G_6(\tau_Q)| = \frac{A_D^3 C_d^{3/2} |H_D(1728)|^{1/2}}{30240^h d^{3h}} (2\pi)^{3h - 3\varphi_d/2} P_D^6. \tag{7.22}$$

7.9.7 The phase and the exceptional class-pair average

For a form strictly inside the reduced domain with $b \neq 0$, the form with $-b$ also occurs. Their G_w values are complex conjugates, so their product is positive unless a factor is zero. Boundary forms with $b = 0$ or $b = a < c$ give real G_4 and G_6, and these values are positive. For G_4 this follows from its positive value high on each boundary ray, continuity, and the fact that E_4 has no zero there except the hexagonal endpoint; for G_6 the same argument uses the unique elliptic zero at i. The excluded discriminants $-3, -4$ are exactly the exceptional endpoints. These are the usual zero-divisor consequences of the valence formula.

The remaining boundary type is $a = c$ and $0 < b < a$. Then $|\tau_Q| = 1$. Modularity and conjugation imply $\overline{E_4(\tau_Q)} = \tau_Q^4 E_4(\tau_Q)$. Following the unit arc from i and using the absence of an interior zero fixes the sign and yields

$$G_4(\tau_Q) = -\tau_Q^{-2} |G_4(\tau_Q)| = \left(\frac{d - b^2}{4a^2} - \frac{ib\sqrt{d}}{2a^2}\right) |G_4(\tau_Q)|. \tag{7.23}$$

Thus an unqualified equality between a raw class product and its positive gamma norm is not valid for arbitrary discriminants. Recent detailed work on the analogous eta-phase issue is Cohen[85]; the elementary modular calculation above suffices for the present cases.

For $D=-15$ the nonprincipal form is $(2,1,2)$. Consequently

$$G_4((-1+i\sqrt{15})/4)=\frac{7-i\sqrt{15}}{8}|G_4((-1+i\sqrt{15})/4)|.$$

The manuscript uses its real part, which is $7/8$ of the absolute value. This explains its factor 7 exactly; averaging a conjugate pair must not be confused with taking one class representative in a norm.

7.9.8 Proofs of the manuscript's five class products

The needed class-polynomial constants and leading-coefficient products are as follows. The supplied interval certificate proves these are the exact Hilbert class polynomials, independently of numerical recognition.

D	h	A_D	$H_D(0)$	$H_D(1728)$
-15	2	2	$-3^6 5^3 11^3$	$3^6 7^4 11^2$
-20	2	2	$-2^{12} 5^3 11^3$	$-2^{16} 11^2 19^2$
-23	3	4	$5^9 11^3 17^3$	$7^6 11^4 19^2 23$
-39	4	12	$3^{15} 17^3 23^3 29^3$	$3^{12} 7^8 19^4 23^2$
-47	5	36	$5^{15} 11^6 23^3 29^3$	$11^6 19^4 23^4 31^2 43^2 47$

The signs and factorizations in this table are exact integer arithmetic; they are also recorded in the machine receipt.

Corollary 7.19 (The five recorded class products are proved)**.** *The product formulas in Theorems 7.7, 7.6, 7.8, 7.11 and 7.14 hold exactly, with the real-part projection specified at discriminant* -15.

Proof. For $D=-23,-39,-47$ there are no unit-arc boundary representatives. Interior pairs are conjugates and all remaining weight-four boundary values are positive, so the absolute norm is the raw product. Substitute the factorizations in the preceding table into (7.21); these give exactly the three displayed class products. For $D=-15$, the real-part factor $7/8$ in (7.23) gives the recorded factor 7.

For $D=-20$, the norm first gives

$$\prod_Q G_4(\tau_Q)=\frac{11P_{-20}^4}{2^{14}3^4 5^5\pi^4}.$$

The multiplication and reflection formulas give

$$\frac{\Gamma(1/20)\Gamma(9/20)}{\Gamma(3/20)\Gamma(7/20)}=4\,5^{1/4}\sin(3\pi/20)\sin(7\pi/20)=5^{1/4}\varphi,$$

where $\varphi=(1+\sqrt{5})/2$. Indeed apply the fivefold multiplication formula at $1/20$ and reflect the factors at $13/20$ and $17/20$. Thus $P_{-20}=5^{1/4}\varphi\Gamma(3/20)^2\Gamma(7/20)^2$; substitution proves the remaining displayed formula. No period independence or numerical recognition is used in this deduction. □

7.9.9 Three explicit weight-six extensions

The same exact class data now produce formulas that were not displayed in the baseline manuscript:

$$\prod_{Q\in\mathcal{Q}_{-23}} G_6(\tau_Q) = \frac{11^2\cdot 19\,P_{-23}^6}{2^{33}3^9 5^3 23^7\pi^{24}}, \tag{7.24}$$

$$\prod_{Q\in\mathcal{Q}_{-39}} G_6(\tau_Q) = \frac{19^2\cdot 23\,P_{-39}^6}{2^{38}3^{15}5^4 13^{12}\pi^{24}}, \tag{7.25}$$

$$\prod_{Q\in\mathcal{Q}_{-47}} G_6(\tau_Q) = \frac{11^3 19^2 23^2\cdot 31\cdot 43\,P_{-47}^6}{2^{73}3^9 5^5 7^5 47^{13}\pi^{54}}. \tag{7.26}$$

They follow by inserting the factorizations of $H_D(1728)$ in (7.22); positivity was already established. For example, at $D=-23$ the square root is $7^3 11^2 19\sqrt{23}$, and $C_{23}^{3/2}=23^{3/2}$. Their product contributes 23^2, which changes 23^9 in the denominator to 23^7. This calculation is a useful check on the normalization.

Every factor here is a polygamma row sum. For example,

$$G_6(\tau) = 2\zeta(6) + \frac{1}{60}\sum_{n\ge 1}\left[\psi^{(5)}(n\tau)+\psi^{(5)}(1-n\tau)\right].$$

Consequently (7.24)–(7.26) are exact identities for finite products of exponentially convergent polygamma-row combinations. Weight twelve gives a further family whose entire arithmetic factor is the rational number $|H_D(432000/691)|$.

7.9.10 How the class-polynomial data are certified

The program `cm_norms.py` enumerates reduced forms by exact integer arithmetic. It then evaluates all singular moduli in rectangular complex intervals using 120 decimal digits of directed interval arithmetic and 70 terms of the Lambert series

$$E_4 = 1+240\sum_{n\ge1}\frac{n^3q^n}{1-q^n}, \qquad E_6 = 1-504\sum_{n\ge1}\frac{n^5q^n}{1-q^n}, \qquad j=\frac{1728E_4^3}{E_4^3-E_6^2}.$$

The interval engine encloses π, $\sqrt{d}$, and the complex exponential. Its output endpoints are finite binary rationals; the program converts them to exact Python fractions before every final containment comparison.

Reduced points satisfy $\operatorname{Im}\tau\ge\sqrt{3}/2$, hence $|q|\le e^{-\pi\sqrt{3}}<1/200$. This last strict inequality needs no floating-point assumption: use $\pi>314/100$, $\sqrt{3}>173/100$, their product greater than $27/5$, and the rational partial sum $\sum_{k=0}^{15}(27/5)^k/k!>200$. For $r=1/200$ and $p=3,5$, the omitted tail satisfies

$$\left|\sum_{n>N}\frac{n^pq^n}{1-q^n}\right| \le \frac{(N+1)^p r^{N+1}\mathcal{A}_p(r)}{(1-r)^{p+2}}, \tag{7.27}$$

where

$$\mathcal{A}_3(r)=1+4r+r^2, \qquad \mathcal{A}_5(r)=1+26r+66r^2+26r^3+r^4.$$

Indeed $n=N+1+j\le(N+1)(j+1)$, and $\sum_{j\ge0}(j+1)^p r^j=\mathcal{A}_p(r)/(1-r)^{p+1}$. The code adds the resulting error bound in both real and imaginary coordinates before forming j or multiplying the class polynomial.

Classical CM theory proves in advance that each coefficient of H_D is an integer. The interval product encloses every coefficient in an interval strictly contained in its asserted integer plus $(-1/2, 1/2)$; its imaginary interval contains zero. Thus the unique possible integer is proved, rather than merely recognized to many digits. The maximum real interval widths were

$D = -15$	$D = -20$	$D = -23$	$D = -39$	$D = -47$
$1.36 \cdot 10^{-109}$	$3.95 \cdot 10^{-108}$	$1.83 \cdot 10^{-103}$	$2.80 \cdot 10^{-95}$	$1.51 \cdot 10^{-88}$

The displayed bounds have been rounded upward. Exact interval endpoints, all polynomial coefficients, and the integer factorizations are saved in `cm_norms_receipt.json`.

An additional 25 numerical comparisons test the gamma norm formula at weights $4, 6, 8, 10, 12$ for each of the five discriminants. They have relative residuals below $1.1 \cdot 10^{-119}$ at 120 digits. These comparisons are regression checks: they are not interval proofs of the gamma identities, whose proof is Theorem 7.18. Similarly, the certified-polynomial computation is a numerical proof conditional on the standard CM integrality theorem and on the interval library's directed-rounding implementation; it is not a Lean formalization. This evidence boundary is explicit and reproducible.

7.9.11 Extension to arbitrary quadratic orders

There is also a direct version for $D = D_0 f^2$, where D_0 is fundamental, including the unit exceptions $D_0 = -3, -4$. Let $h = h(D)$ and $d = |D|$, and let $\mathrm{CS}(D_0)$ be Cohen's normalized gamma quotient

$$\mathrm{CS}(D_0) = \left(\prod_{r=1}^{|D_0|-1} \Gamma(r/|D_0|)^{\chi_{D_0}(r)} \right)^{w(D_0)/(2h(D_0))}.$$

Define the explicit conductor factor

$$c(D_0, f) = \prod_{p \mid f} p^{e(p)}, \qquad e(p) = \frac{(1 - p^{-v_p(f)})(1 - \chi_{D_0}(p))}{(1 - 1/p)(p - \chi_{D_0}(p))}.$$

The order form of Lerch–Chowla–Selberg[84, 85] gives the following immediate extension of the proof:

$$\prod_{Q \in \mathcal{Q}_D} |G_w(\tau_Q)| = (2\zeta(w))^h A_D^{w/2} \left(\frac{c(D_0, f)\mathrm{CS}(D_0)}{2\pi d} \right)^{hw/2} |\operatorname{Res}(H_D, F_w)|^{1/12}. \tag{7.28}$$

Thus nonmaximal orders require an explicit conductor correction, not a new unidentified period basket. Setting $f = 1$ and $D < -4$ recovers the main theorem.

Corollary 7.20 (The four class-number-one row evaluations)**.** *The four formulas at $2i$ and $i\sqrt{2}$ in Theorem 7.5 hold exactly.*

Proof. The additional coefficient certificates give $H_{-8}(X) = X - 8000$ and $H_{-16}(X) = X - 287496$, each with one reduced class. At $D = -8$ insert $H_{-8}(0)$ and $H_{-8}(1728)$ into the fundamental formula; $C_8 = 2$ and $P_{-8} = \Gamma(1/8)\Gamma(3/8)$. This gives the two recorded coefficients. At $D = -16$ use (7.28) with $D_0 = -4$, $f = 2$, $c(-4, 2) = \sqrt{2}$ and

$$\mathrm{CS}(-4) = \left(\frac{\Gamma(1/4)}{\Gamma(3/4)} \right)^2 = \frac{\Gamma(1/4)^4}{2\pi^2}.$$

The weight-four and weight-six resultant roots are $287496^{1/3} = 66$ and $(287496 - 1728)^{1/2} = 378\sqrt{2}$. Substitution gives the stated factors $11/15360$ and $1/(2^{14}5)$. These principal imaginary-axis row sums are positive by the principal-point proof, so absolute values may be removed. The two new class polynomials are certified by directed intervals and CM integrality at 120 digits and 70 Lambert terms, with exact recorded endpoints. □

Proposition 7.21 (Two exact Weber degree statements)**.** *At discriminant -23, the phase-corrected cubic coordinate defined in Theorem 7.9 is unique and belongs to the same cubic field as its singular modulus. At discriminant -39, every cube root of a singular modulus has degree twelve over $\mathbb{Q}$, and none belongs to the Hilbert class field of degree eight over $\mathbb{Q}$.*

Proof. Exact polynomial arithmetic gives

$$\mathrm{Res}_Y(Y^3 + 155Y^2 + 650Y + 23375,\, Y^3 - X) = -H_{-23}(X).$$

The three cube images are therefore the three distinct singular moduli. For each one exactly one root of the cubic has that image. Since $[\mathbb{Q}(j_Q) : \mathbb{Q}] = 3$ and $j_Q = \xi_Q^3$, the cubic equation forces $\mathbb{Q}(\xi_Q) = \mathbb{Q}(j_Q)$. The quotient of ξ_Q by E_4/η^8 is necessarily a cube root of unity, giving the precise phase correction.

For $D = -39$, the integer polynomial $H_{-39}(Y^3)$ reduces modulo five to

$$f(Y) = Y^{12} + Y^9 + 4Y^6 + 4Y^3 + 3.$$

The independent finite-field certificate verifies

$$Y^{5^{12}} \equiv Y \pmod{f}, \qquad \gcd(f, Y^{5^6} - Y) = \gcd(f, Y^{5^4} - Y) = 1.$$

Rabin's criterion proves f irreducible over $\mathbf{F}_5$; its monic integer lift is thus irreducible over $\mathbb{Q}$. Every possible phase has cube j_Q, so it is a root of this degree-twelve polynomial. Such a field cannot be a subfield of the degree-eight Hilbert class field. The resultant, the modular witness and its independent standard-library Frobenius/gcd replay are retained in the CM verification receipts. □

7.9.12 Research directions and scope

The theorem suggests several concrete continuations.

1. Extend the exact polynomial F_w generator supplied here and combine it with certified class-polynomial software such as Sutherland's CRT methods [86]. This extends the product database to large weights and class numbers without integer-relation searches.

2. Determine signed, rather than absolute, products for all weights by tracking the phases of boundary representatives and the zero signs of E_w along the unit arc. The present phase proof covers the displayed weight-four and weight-six examples; a general norm alone does not settle this problem.

3. Use factorizations of $H_D(0)$ and $H_D(1728)$ to compare the prime support with arithmetic formulas for norms of singular-modulus differences. This gives an explanation for the prime patterns observed in the manuscript, beyond using those patterns to build a PSLQ basis.

4. Extend the certified computations to class-by-class algebraic factors and genus-wise products. A full class norm does not by itself prove class ratios. Section 7.10 supplies the additional genus-character argument for the recorded ratios at $D = -15, -20, -39$; extending it beyond these examples remains a natural direction.

5. Build a uniform certificate format containing the discriminant, reduced forms, exact H_D, exact F_w, resultant, conductor factor, phase convention, gamma residue set, and analytic tail bound. Each claimed identity then has a finite, reviewable provenance chain.

None of these norm computations proves linear or algebraic independence of individual gamma values or individual polygamma rows. Nor does it imply that arbitrary sums of raw positive-weight CM values are class invariants. The result is a precise product theorem with completely specified representative dependence.

7.10 Genus characters prove the recorded CM ratios

A class product and a class ratio contain different information. The preceding norm formula controls the former. A nontrivial genus character supplies the missing signed product and proves all three recorded weight-four ratios at discriminants $-15, -20, -39$. The same calculation gives a theorem for every even weight and three explicit weight-six ratios.

The analytic mechanism is classical: twist the Kronecker limit formula by a genus character and use Kronecker's factorization of its Hecke L-function. The character factorization and unit normalization are stated explicitly in Masri[89, eqs. (1.3)–(1.5)]. We give the normalization argument because confusing the ordinary and narrow real-quadratic class numbers, or overlooking the units at discriminants -3 and -4, changes the answer.

7.10.1 A genus ratio theorem in arbitrary even weight

Suppose that the fundamental discriminant $D < -4$ has a factorization

$$D = eD_-, \qquad e > 1, \qquad D_- < 0,$$

where e, D_- are coprime fundamental discriminants. Let $\varepsilon : \mathrm{Cl}(D) \longrightarrow \{1, -1\}$ be the corresponding nontrivial genus character. Concretely, if a form Q represents an integer n coprime to D, then $\varepsilon(Q) = \chi_e(n)$. This value is independent of the represented integer chosen. Retain the reduced representatives τ_Q and the normalizations of Section 7.9, and put

$$A_\varepsilon = \prod_{Q \in \mathcal{Q}_D} a_Q^{\varepsilon(Q)}, \qquad H_D^{\pm}(X) = \prod_{\varepsilon(Q)=\pm 1} (X - j(\tau_Q)), \tag{7.29}$$

$$\lambda_- = L(0, \chi_{D_-}) = \frac{2h(D_-)}{\omega(D_-)}, \qquad \omega(D_-) = |\mathcal{O}_{D_-}^\times|.$$

Here $h(D_-)$ is the imaginary-quadratic class number. Let $h(e)$ be the *ordinary* real-quadratic class number, and let $u_e > 1$ be the least unit greater than one in the real embedding of $\mathbb{Q}(\sqrt{e})$. Its norm is allowed to be -1. In particular,

$$\lambda_{-3} = \frac{1}{3}, \qquad \lambda_{-4} = \frac{1}{2};$$

replacing either of these values by its imaginary class number would give an incorrect result.

Both genus factors are monic of degree $h(D)/2$, with $H_D = H_D^+ H_D^-$. The two sign sets are unions of genera; in the three examples below they are the two genera themselves. Their coefficients lie in the ring of integers of $\mathbb{Q}(\sqrt{e})$. Indeed, the kernel of ε fixes the quadratic genus extension $\mathbb{Q}(\sqrt{D}, \sqrt{e})$ of $\mathbb{Q}(\sqrt{D})$; the two products are invariant under this kernel. Complex conjugation inverts ideal classes, and inversion preserves ε, so both products are real. Their coefficients therefore belong to the real quadratic subfield. The coefficients are algebraic integers because the singular moduli are algebraic integers.

Theorem 7.22 (Genus ratios at every even weight)**.** *For every even $w \geq 4$, all factors are nonzero by Theorem 7.17, and*

$$\prod_{Q \in \mathcal{Q}_D} |G_w(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^{w/2}\, u_e^{-wh(e)\lambda_-} \left| \frac{\mathrm{Res}(H_D^+, F_w)}{\mathrm{Res}(H_D^-, F_w)} \right|^{1/12}. \tag{7.30}$$

The polynomials F_w and the resultant convention are those of Lemma 7.16. All real roots on the right are positive roots. Every numerator and denominator is nonzero.

Proof. Write $d = |D|$ and

$$Z(\tau, s) = \sum_{(m,n) \neq (0,0)} \frac{(\operatorname{Im}\tau)^s}{|m + n\tau|^{2s}}.$$

The quadratic-form and ideal-class zeta correspondence, with two units because $D < -4$, gives

$$\sum_Q \varepsilon(Q) Z(\tau_Q, s) = 2 \left(\frac{\sqrt{d}}{2} \right)^s L(s, \chi_e) L(s, \chi_{D_-}). \tag{7.31}$$

Here we used Kronecker's factorization $L_{\mathbb{Q}(\sqrt{D})}(s, \varepsilon) = L(s, \chi_e) L(s, \chi_{D_-})$. For completeness, at an unramified split prime the two quadratic characters have the same sign and give the two equal prime-ideal factors; at an inert prime their signs are opposite and their product is $(1 - p^{-2s})^{-1}$, the factor for the principal ideal (p). At a ramified prime, coprimality of e, D_- leaves precisely the factor of the unramified Dirichlet character. Thus the Euler products agree in $\operatorname{Re} s > 1$, and continuation gives the identity.

Since χ_e is nontrivial and even, $L(0, \chi_e) = 0$. Differentiate (7.31) at zero and use

$$Z'(\tau, 0) = -2 \log\bigl(2\pi \sqrt{y}\, |\eta(\tau)|^2\bigr), \qquad y = \operatorname{Im}\tau.$$

The constant disappears because $\sum_Q \varepsilon(Q) = 0$. We obtain

$$\sum_Q \varepsilon(Q) \log\bigl(\sqrt{y_Q} |\eta(\tau_Q)|^2\bigr) = -\lambda_- L'(0, \chi_e). \tag{7.32}$$

Since $y_Q = \sqrt{d}/(2a_Q)$, its signed logarithmic sum is $\sum_Q \varepsilon(Q) \log y_Q = -\log A_\varepsilon$. Consequently

$$\prod_Q |\Delta(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^6 \exp\bigl(-12\lambda_- L'(0, \chi_e)\bigr). \tag{7.33}$$

To fix the real-quadratic normalization, the completed function

$$\Lambda_e(s) = (e/\pi)^{s/2} \Gamma(s/2) L(s, \chi_e)$$

satisfies $\Lambda_e(s) = \Lambda_e(1 - s)$. Its values at zero and one give $2L'(0, \chi_e) = \sqrt{e}\, L(1, \chi_e)$. The real-quadratic class number formula, in the ordinary-class-number and least-unit convention specified above, is

$$L(1, \chi_e) = \frac{2h(e) \log u_e}{\sqrt{e}}.$$

Thus $L'(0, \chi_e) = h(e) \log u_e$, and (7.33) becomes

$$\boxed{\prod_Q |\Delta(\tau_Q)|^{\varepsilon(Q)} = A_\varepsilon^6 u_e^{-12h(e)\lambda_-}.} \tag{7.34}$$

Finally, apply $|E_w(\tau_Q)|^{12} = |\Delta(\tau_Q)|^w |F_w(j(\tau_Q))|$ to the signed product. The factors $2\zeta(w)$ cancel because the plus and minus character fibers have the same cardinality. The remaining products of $F_w(j)$ are exactly the stated resultants. Taking the positive twelfth root proves the formula. □

In particular, write $R_w(D) = \prod_Q |G_w(\tau_Q)|^{\varepsilon(Q)}$. The two smallest weights give the especially convenient identities

$$R_4(D)^3 = A_\varepsilon^6 u_e^{-12h(e)\lambda_-} \left| \frac{H_D^+(0)}{H_D^-(0)} \right|, \tag{7.35}$$

$$R_6(D)^2 = A_\varepsilon^6 u_e^{-12h(e)\lambda_-} \left| \frac{H_D^+(1728)}{H_D^-(1728)} \right|. \tag{7.36}$$

Genus theory identifies the field of the factors $H_D^\pm$; these equations additionally account for the periods of the raw modular forms. Claiming a quadratic raw ratio requires verifying the root extraction. We now perform that step exactly.

7.10.2 The three recorded ratios, now theorems

The required character and unit data are as follows. A plus or minus sign attached to a form denotes its genus-character value.

D	(e, D_-)	plus forms	A_ε	u_e	λ_-
-15	$(5,-3)$	$(1,1,4)$	$1/2$	$(1+\sqrt{5})/2$	$1/3$
-20	$(5,-4)$	$(1,0,5)$	$1/2$	$(1+\sqrt{5})/2$	$1/2$
-39	$(13,-3)$	$(1,1,10),(3,3,4)$	$3/4$	$(3+\sqrt{13})/2$	$1/3$

All the other reduced forms have minus sign. These signs can be checked by represented integers: at $D=-15$, the nonprincipal form represents 2 and $\chi_5(2) = -1$; at $D=-20$, the nonprincipal form represents 7 and $\chi_5(7) = -1$; at $D=-39$, the forms with $a=2$ have $\chi_{13}(2) = -1$, whereas $(3,3,4)$ represents 10 and $\chi_{13}(10) = 1$. The class numbers $h(5) = h(13) = 1$ follow from the real-quadratic Minkowski bound $\sqrt{e}/2 < 2$. The displayed units are fundamental: checking $x^2 - ey^2 = \pm 4$ below their sizes leaves no smaller positive unit.

The genus factors are explicitly

$$H_{-15}^\pm(X) = X + \frac{191025 \pm 85995\sqrt{5}}{2}, \tag{7.37}$$

$$H_{-20}^\pm(X) = X - 632000 \mp 282880\sqrt{5}, \tag{7.38}$$

$$H_{-39}^\pm(X) = X^2 + (165765798 \pm 45975573\sqrt{13})X + \frac{63399280527 \pm 17399806263\sqrt{13}}{2}. \tag{7.39}$$

Multiplication gives the certified H_D polynomials of Section 7.9. The correspondence between these quadratic-field factors and the displayed genera is also certified: for each CM root, evaluate the other factor in a directed interval enclosure and exclude zero. The attached receipt records the exact rational interval endpoints, so the assignment is not a numerical guess about which radical sign to choose.

Theorem 7.23 (Exact weight-four genus ratios)**.** *With the character data in the table,*

$$R_4(-15) = \frac{(4+\sqrt{5})^2}{44}, \tag{7.40}$$

$$R_4(-20) = \frac{29+12\sqrt{5}}{44}, \tag{7.41}$$

$$R_4(-39) = \frac{103299+4440\sqrt{13}}{181424}. \tag{7.42}$$

The first equation is an absolute-value ratio; the precise real-part version in the manuscript is

$$\frac{G_4((1+i\sqrt{15})/2)}{\operatorname{Re} G_4((1+i\sqrt{15})/4)} = \frac{2(4+\sqrt{5})^2}{77}. \tag{7.43}$$

For $D=-20$ and $D=-39$, the positive products represented by $R_4(D)$ are exactly the products and ratios in the manuscript, without an additional phase factor.

Proof. Substitute (7.37)–(7.39) and the table into (7.35). In each of the two quadratic fields, direct multiplication verifies that the cube of the claimed positive number is the right-hand side. For example, for $D=-20$ this is the explicit identity

$$\left(\frac{29+12\sqrt{5}}{44}\right)^3 = \frac{1}{64}\left(\frac{1+\sqrt{5}}{2}\right)^{-6}\left|\frac{-632000-282880\sqrt{5}}{-632000+282880\sqrt{5}}\right|.$$

The identical finite verification with exponents -4 and -4, and leading-coefficient ratios $1/2$ and $3/4$, proves the other two cases. The attached exact-arithmetic script verifies all three equalities in $\mathbb{Q}(\sqrt{5})$ or $\mathbb{Q}(\sqrt{13})$ before any decimal evaluation. Positivity makes the cube root unique.

For $D=-15$, the phase calculation of Section 7.9.7 gives $\operatorname{Re} G_4((1+i\sqrt{15})/4) = \frac{7}{8}|G_4((1+i\sqrt{15})/4)|$. Multiplying (7.40) by $8/7$ proves (7.43); translation by one and reflection in the imaginary axis account for the manuscript's positive-real-part representatives. The phase discussion in the preceding section also shows that the numerator and denominator genus products at -20 and -39 are positive: boundary representatives contribute positive real G_4, and the two interior representatives at -39 form a conjugate pair. □

This supplies the missing proofs for the ratios originally displayed in the manuscript's Results `cm:res:d15`, `cm:res:d20`, and `cm:res:genus`. Combined with the proved total products, it also determines the individual absolute values at class number two, and the absolute products over each genus at class number four: if T_w is the full positive product, the two genus products are $\sqrt{T_w R_w}$ and $\sqrt{T_w/R_w}$.

7.10.3 Three explicit weight-six identities

The same genus polynomials have further consequences beyond the recorded weight-four data.

Theorem 7.24 (Weight-six genus ratios)**.** *With the same genera and representatives,*

$$R_6(-15) = \frac{29+12\sqrt{5}}{88}, \tag{7.44}$$

$$R_6(-20) = \frac{601+252\sqrt{5}}{1672}, \tag{7.45}$$

$$R_6(-39) = \frac{1323+324\sqrt{13}}{1472}. \tag{7.46}$$

The last two again hold for the corresponding raw genus products. At discriminant -15 *an explicit real-part identity is*

$$\frac{G_6((1+i\sqrt{15})/2)}{\operatorname{Re} G_6((1+i\sqrt{15})/4)} = \frac{2(29+12\sqrt{5})}{121}. \tag{7.47}$$

Proof. Use (7.36). Exact arithmetic in the relevant quadratic field shows that its right-hand sides are, respectively,

$$\frac{1561+696\sqrt{5}}{7744} = \left(\frac{29+12\sqrt{5}}{88}\right)^2,$$
$$\frac{678721+302904\sqrt{5}}{2795584} = \left(\frac{601+252\sqrt{5}}{1672}\right)^2,$$
$$\frac{3115017+857304\sqrt{13}}{2166784} = \left(\frac{1323+324\sqrt{13}}{1472}\right)^2.$$

The roots on the right are positive. The valence formula gives the only zero of E_6 in the standard fundamental domain at i; none of the CM representatives under consideration is this point. Along the boundary rays $\operatorname{Re}\tau = 0$ and $\operatorname{Re}\tau = -1/2$ above the relevant representatives, E_6 is real, has no zero, and tends to one at infinity, so its values are positive. Conjugate interior points again contribute a positive product. This proves the raw-product assertion at $-20, -39$.

On the unit arc $|\tau| = 1$ with $\pi/2 < \arg\tau \le 2\pi/3$, modularity and conjugation imply that $\tau^3 G_6(\tau)$ is real. It is positive at the hexagonal endpoint, where $\tau^3 = 1$ and $G_6 > 0$, and cannot change sign before i. Thus $G_6(\tau) = \tau^{-3}|G_6(\tau)|$ on this open arc. At $\tau = (-1+i\sqrt{15})/4$,

$$\tau^{-3} = \frac{11+3i\sqrt{15}}{16}, \qquad \operatorname{Re} G_6(\tau) = \frac{11}{16}|G_6(\tau)|.$$

Reflection preserves the real part. Multiplying (7.44) by $16/11$ proves (7.47). □

The quadratic norms provide useful exact checks:

$$N(R_4(-15)) = N(R_4(-20)) = \frac{1}{16}, \qquad N(R_4(-39)) = \frac{81}{256},$$

and

$$N(R_6(-15)) = N(R_6(-20)) = \frac{1}{64}, \qquad N(R_6(-39)) = \frac{729}{4096}.$$

These follow by conjugating $\sqrt{e} \mapsto -\sqrt{e}$ in the displayed exact formulas. The identities are explicit consequences of the classical genus-period mechanism; no assertion of historical priority for individual numerical evaluations is intended.

7.10.4 A fixed field contains all even-weight genus ratios

The low-weight identities imply a uniform field-degree theorem. For a fixed genus character, increasing the weight introduces no additional algebraic field extension beyond the two seed ratios.

Theorem 7.25 (Uniform algebraic degree of genus ratios)**.** *Under the assumptions of Theorem 7.22, put* $K = \mathbb{Q}(\sqrt{e})$ *and*

$$\mathcal{K}_{D,\varepsilon} = K(R_4(D), R_6(D)).$$

Then

$$[\mathcal{K}_{D,\varepsilon} : K] \le 6, \qquad [\mathcal{K}_{D,\varepsilon} : \mathbb{Q}] \le 12,$$

and every ratio $R_w(D)$ with even $w \ge 4$ belongs to this same field. The degree bound is independent of w and of the imaginary class number $h(D)$.

Proof. The genus period ratio

$$B_{D,\varepsilon} = A_\varepsilon^6 u_e^{-12h(e)\lambda_-}$$

belongs to K. Indeed,

$$12h(e)\lambda_- = \frac{24h(e)h(D_-)}{\omega(D_-)} \in \mathbb{Z},$$

because $\omega(D_-)$ is 2, 4, or 6. Equations (7.35) and (7.36) therefore give $R_4(D)^3 \in K$ and $R_6(D)^2 \in K$. The seed ratios are nonzero: E_4 vanishes only at the elliptic orbit of discriminant -3, and E_6 only at that of discriminant -4, whereas $D < -4$ is fundamental. Adjoining the two positive seed ratios has degree at most $3 \cdot 2 = 6$ over K.

To prove that this one field suffices, use the reduced modular polynomial of (7.7):

$$E_w = E_4^a E_6^b \Delta^m P_w(j), \qquad a \in \{0,1,2\}, \quad b \in \{0,1\}, \quad w = 4a + 6b + 12m, \tag{7.48}$$

where $m \ge 0$ and $P_w \in \mathbb{Q}[X]$ is monic of degree m. The algebraic reason is short: the weight condition forces every monomial of $E_w \in \mathbb{Q}[E_4, E_6]$ to have exponents $a+3r, b+2s$ with $r+s = m$. Substituting $E_4^3 = \Delta j$ and $E_6^2 = \Delta(j - 1728)$ gives the displayed form; the constant term $E_w = 1 + O(q)$ makes P_w monic.

The signed absolute product gives the formula without fractional root extraction

$$R_w(D) = R_4(D)^a R_6(D)^b B_{D,\varepsilon}^m \left| \frac{\mathrm{Res}(H_D^+, P_w)}{\mathrm{Res}(H_D^-, P_w)} \right|. \tag{7.49}$$

The resultant ratio is in K, because both genus factors have coefficients there. Its absolute value is also in K, which is a real field. The CM nonvanishing theorem makes every resultant at these points nonzero, so the quotient is defined at every even weight. This proves the field assertion. □

Corollary 7.26 (Quadratic closure for the recorded discriminants)**.** *For $D = -15, -20, -39$, every genus ratio $R_w(D)$ with even $w \ge 4$ lies in $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{5})$, or $\mathbb{Q}(\sqrt{13})$, respectively. Each is computable in that quadratic field using the reduced polynomial P_w, whose degree is at most $\lfloor w/12 \rfloor$.*

Proof. Theorems 7.23 and 7.24 place both seed ratios in the indicated quadratic field. Formula (7.49) completes the proof. □

For example, $E_{14} = E_4^2 E_6$ gives $R_{14}(D) = R_4(D)^2 R_6(D)$. At weight 16, $a = m = 1, b = 0$ and $P_{16}(X) = X - 3456000/3617$, so

$$R_{16}(D) = R_4(D) B_{D,\varepsilon} \left| \frac{H_D^+(3456000/3617)}{H_D^-(3456000/3617)} \right|.$$

The equal degrees of the two genus factors cancel the signs in the linear-polynomial resultants. This formula, together with the three explicit pairs $H_D^\pm$, generates further exact quadratic identities using rational arithmetic alone.

7.10.5 Certificates and further directions

The companion `cm_genus.py` imports the directed-interval Lambert-series evaluator from `cm_norms.py`. It first certifies the integral Hilbert polynomials, then verifies the displayed genus factorizations by exact symbolic multiplication. For every reduced form it records a represented integer witnessing the genus character, and proves that the opposite genus factor excludes zero at its CM j-value. It next checks the cubic and square identities exactly. The receipt `cm_genus_receipt.json` separates these proof steps from six ordinary 120-digit Fourier-series regressions. A second checker, `verify_genus_arithmetic.py`, independently verifies all six radical equalities and their positive-root choices using only exact standard-library rational arithmetic in the quadratic fields. The numerical regressions have relative discrepancies below 3.1×10^{-120} and are independent consistency checks, not the justification for replacing the manuscript's numerical evidence by theorems.

The genus theorem answers the three recorded ratio questions while leaving several precise extensions open for this project.

1. Extend the genus-ratio certificate to nonmaximal quadratic orders, including the local conductor factors in the twisted Kronecker factorization. The untwisted conductor correction proved earlier cannot simply be copied into a signed genus product.

2. Beyond the three discriminants of Corollary 7.26, classify the discriminants and weights for which the raw ratio $R_w(D)$ itself belongs to $\mathbb{Q}(\sqrt{e})$, rather than merely having a controlled algebraic power in that field. Exact root extraction and representative phases must be part of the statement.

3. For class groups with several independent genus characters, combine their signed formulas by finite Fourier inversion to recover the product over every genus. This determines genus products but does not distinguish classes inside one genus.

4. Investigate twists by nonreal class characters. Their Hecke L-functions no longer reduce to a pair of quadratic Dirichlet L-functions; the resulting period data measure exactly the information absent from genus theory.

7.11 Norms, exact quadratic degree and oscillating genus ratios

The fixed-field formula gives sharper arithmetic information at multiples of twelve. Together with the lattice bounds used for CM nonvanishing, it also proves that the recorded quadratic fields are attained at every even weight. These are further consequences developed from the incoming genus and modular-polynomial results.

Theorem 7.27 (Quadratic norm at every weight divisible by twelve)**.** *Under the hypotheses of the genus theorem, put $K = \mathbb{Q}(\sqrt{e})$ and let σ be its nontrivial automorphism. For every integer $m \geq 1$,*

$$R_{12m}(D) \in K_{>0}, \qquad \sigma(R_{12m}(D)) = \frac{A_\varepsilon^{12m}}{R_{12m}(D)} > 0, \qquad N_{K/\mathbb{Q}}(R_{12m}(D)) = A_\varepsilon^{12m}.$$

Thus $A_\varepsilon^{-6m} R_{12m}(D)$ is a totally positive norm-one element of K. No integrality or unit assertion is implicit.

Proof. In (7.49), weight $12m$ has $a = b = 0$. Write $B = A_\varepsilon^6 u_e^{-t}$, where $t = 12h(e)\lambda_- = 24h(e)h(D_-)/\omega(D_-)$. This is an even integer: the unit counts $\omega(D_-) = 2, 4, 6$ give respectively factors $12, 6, 4$. Since $N(u_e) = \pm 1$,

$$\sigma(B) = A_\varepsilon^6 u_e^t, \qquad N(B) = A_\varepsilon^{12}.$$

The two genus factors are exchanged by σ. Since P_{12m} has rational coefficients, their resultant quotient $x \in K^\times$ satisfies $\sigma(x) = 1/x$. Both real embeddings of x have the same sign, so $\sigma(|x|) = 1/|x|$. This last conclusion uses the reciprocal identity; absolute value does not generally commute with arbitrary field embeddings. Now $R_{12m} = B^m|x|$, and multiplication proves all statements. The CM nonvanishing theorem makes every quotient nonzero. □

Theorem 7.28 (Exact quadratic degree at all even weights)**.** *For $D = -15, -20, -39$ and every even $w \geq 4$, respectively,*

$$R_w(D) \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}, \quad R_w(D) \in \mathbb{Q}(\sqrt{5}) \setminus \mathbb{Q}, \quad R_w(D) \in \mathbb{Q}(\sqrt{13}) \setminus \mathbb{Q}.$$

Each ratio is totally positive in its quadratic field, and

$$N(R_w(-15)) = N(R_w(-20)) = 2^{-w}, \qquad N(R_w(-39)) = (3/4)^w.$$

Proof. Quadratic closure follows from the preceding seed theorem. The explicit seed ratios are totally positive and have norms A_ε^4 and A_ε^6. Formula (7.49) and the reciprocal resultant argument just used therefore give total positivity and norm A_ε^w at every even weight. If a ratio were rational, its positivity and norm would force $R_w = A_\varepsilon^{w/2}$.

We exclude that value by uniform rational lattice bounds. At $D = -15$, the absolute fourth-power lattice sum for $(2, 1, 2)$ is less than 8; at $D = -20$, that for $(2, 2, 3)$ is less than 5. Every nonzero vector in a reduced lattice has length at least one, so these are upper bounds for the absolute sums at all even $w \geq 4$. Combine them with the principal lower bound $7/20$ to get

$$R_w(-15) > 7/160 > 2^{-5}, \qquad R_w(-20) > 7/100 > 2^{-4}.$$

They exclude rationality for $w \geq 10$ and $w \geq 8$, respectively. At $D = -39$, the weight-twelve tails after removing the two axis unit vectors are less than $1/2, 1/2, 3/4$ at forms $(1, 1, 10), (2, 1, 5), (3, 3, 4)$. All remaining vectors have length greater than one. The same bounds hold at every $w \geq 12$, and give

$$R_w(-39) > \frac{(3/2)(5/4)}{(5/2)^2} = 3/10 > (3/4)^6 \geq (3/4)^{w/2}.$$

The conjugate pair of forms with $a = 2$ has equal absolute values, which explains the squared denominator.

These five lattice caps are certified by the finite rectangle $|n| \leq 10$, $|m| \leq 20$ and the tail estimates from the nonvanishing proof. For a twelfth-power sum the layer-cake integral is $63\pi/(256A^{11})$; using $\pi < 4$ gives the n-tail $2/(11N^{11}) + 63/(320N^{10})$. The m-tail is $4N/[11(M - N/2)^{11}]$, with the additional axis tail $2/(11M^{11})$. The fourth-power cap at $(2, 1, 2)$ uses the rational imaginary lower bound $9/10$; all other caps use one. The exact fractions and comparisons are retained in `verification/CM-genus-extensions.json`.

The only smaller weights left are $4, 6, 8$, and where needed 10. The seed formulas at $4, 6$ have nonzero radical coefficients, and $R_8 = R_4^2$, $R_{10} = R_4 R_6$ have positive radical coefficients as well. Thus none is rational. This proves exact degree two at every stated weight. □

7.11.1 New explicit quadratic evaluations

The reduced polynomials give nine further formulas at weights $12, 16, 18$. For instance,

$$R_{12}(-15) = \frac{34829 + 15576\sqrt{5}}{1216}, \tag{7.50}$$

$$R_{16}(-15) = \frac{307761 + 137632\sqrt{5}}{511744}, \tag{7.51}$$

$$R_{18}(-15) = \frac{64494289 + 28842228\sqrt{5}}{193434112}. \tag{7.52}$$

For the other two discriminants the complete coefficients are recorded as rational pairs in `verification/CM-genus-extensions.json`. These are exact evaluations by the resultant recipe, rather than decimal fits. In particular the first has norm $1/4096$ even though its positive embedding is approximately 57.28; the other embedding is very small.

7.11.2 A quadratic sequence with a full interval of limit points

The large first value is part of a structural phenomenon. Let $\tau = (-1 + i\sqrt{15})/4 = e^{i\theta}$, and write $r_m = R_{12m}(-15)$.

Theorem 7.29 (The cluster set at discriminant -15)**.** *The finite limit points of r_m, as $m \to \infty$, are precisely $[1/2, \infty)$. Also*

$$\liminf_{m\to\infty} r_m = 1/2, \qquad \limsup_{m\to\infty} r_m = \infty, \qquad \sigma(r_m) = 2^{-12m}/r_m \longrightarrow 0.$$

Thus exact quadratic degree at every weight coexists with dense limiting behavior in one embedding and exponential decay in the other.

Proof. The principal lattice has only the unit vectors ± 1, and all other vectors have length greater than one. Its row sum tends to two. In the second lattice the unit vectors are exactly $\pm 1, \pm\tau$; the next length is $\sqrt{3/2}$. Absolute convergence at weight four therefore gives the additive estimate

$$G_{12m}(\tau) = 2 + 2\tau^{-12m} + O((3/2)^{-6m}), \qquad |G_{12m}(\tau)| = 4|\cos(6m\theta)| + O((3/2)^{-6m}).$$

The error is additive; it is not divided by a small cosine.

The angle θ/π is irrational. Otherwise τ would be a root of unity, making $\tau + \tau^{-1} = -1/2$ an algebraic integer, which is impossible for that rational number. Thus $6m\theta$ is dense modulo π. Along a subsequence with $|\cos(6m\theta)| \to c > 0$, $r_m \to 1/(2c)$, giving every finite limit in $[1/2, \infty)$. Along a subsequence with the cosine tending to zero, the denominator tends to zero while the numerator tends to two; CM nonvanishing makes each term defined, so $r_m \to \infty$. The upper bound $|G_{12m}(\tau)| \le 4 + o(1)$ excludes finite limit points below $1/2$. Finally the norm theorem gives the conjugate formula, and the positive lower bound on r_m proves its convergence to zero. □

7.12 The opaque components

The components *not* fixed by reflection-conjugation or by the even-row mechanism — $\operatorname{Im}\psi^{(1)}(i)$, $\operatorname{Re}\psi(i)$ beyond γ, $\operatorname{Im}\psi^{(1)}\left(\frac{1+i}{2}\right)$ — are *odd-row* lattice data: for example

$$\operatorname{Im}\psi^{(1)}(i) = -2\sum_{k\ge 1} \frac{k}{(k^2+1)^2}$$

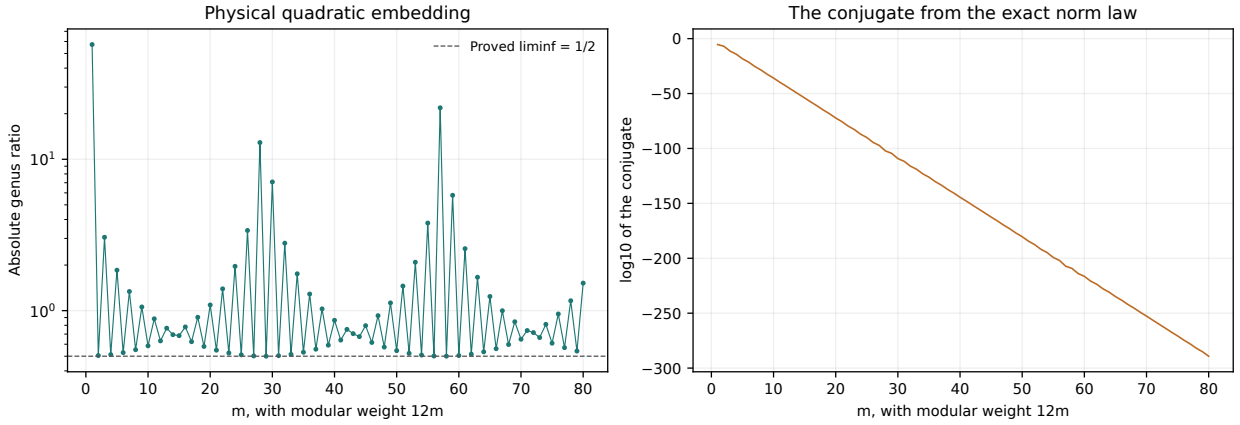


Figure 7.1: Numerical illustrations of the proved genus-ratio behavior at $D = -15$. The physical embedding has large oscillations; its quadratic conjugate is computed from the exact norm law. Finite lattice sums supply the plotted values and explicit analytic tail budgets. The cluster theorem, not these finitely many points, proves the full interval of limit points.

is an odd-power analogue of the coth family, which the bilateral (even) cotangent expansion cannot reach. These are Eichler-integral/quasi-modular territory — the E_2-flavoured layer; compare $\sum_{k\geq 1} k/(e^{2\pi k}-1) = \frac{1}{24} - \frac{1}{8\pi}$, which is $E_2(i) = 3/\pi$ in Lambert-series clothing.

Search observation 7.30. A PSLQ basket for these constants against products drawn from

$$\{1,\ \pi,\ \coth\pi,\ \pi^2/\sinh^2\pi,\ \pi\coth\pi,\ \dots\}$$

found nothing at small coefficient heights — consistent with genuinely new constants. Recorded as an honest negative; the Lambert-series face ($\sum \sigma_5(n)q^n$ at $q = e^{-2\pi}$ and $q = -e^{-\pi\sqrt{3}}$) is the natural next probe surface.

7.13 Summation order, differentiation and certified tails

Absolute convergence made the preceding weights straightforward. Weight two needs a specified order of summation. Once that boundary is handled, Ramanujan's differential algebra turns the existing CM period seeds into all derivative evaluations, with explicit exponential tail estimates.

7.14 Regularized modular rows and all derivative evaluations

The modular draft obtains even-weight Eisenstein series by pairing polygamma values along lattice rows. Its weight-two question requires care: the paired rows converge rapidly, although the ungrouped lattice sum is not absolutely convergent. We first prove an explicit bound for every differentiated row. We then determine the change of summation order directly, before using any modular transformation law. Finally, Ramanujan's differential system evaluates all the resulting derivative rows at the square and hexagonal CM points.

Throughout this section, let

$$\tau \in \mathbb{H} := \{z \in \mathbb{C} : \operatorname{Im} z > 0\}, \qquad q = e^{2\pi i\tau}, \qquad r = |q| = e^{-2\pi \operatorname{Im}\tau} < 1, \qquad \rho = \tau_{\mathrm{hex}} = \frac{1+i\sqrt{3}}{2}.$$

For even $w \geq 2$ and an integer $j \geq 0$, put

$$\mathcal{T}_{w,j}(\tau) = \sum_{n\geq 1} n^j \left[\psi^{(w-1+j)}(n\tau) + (-1)^j \psi^{(w-1+j)}(1-n\tau)\right]. \tag{7.53}$$

Convergence will follow from the next theorem. The sign on the reflected term is part of the definition; differentiating that term changes its sign at every step.

7.14.1 Rational row formulas and explicit tail bounds

The negative-index polylogarithms appearing below are elementary rational functions. For $d \geq 0$, define

$$R_d(t) = \mathrm{Li}_{-d}(t), \qquad R_0(t) = \frac{t}{1-t}, \qquad R_{d+1}(t) = t\frac{d}{dt}R_d(t).$$

These relations are identities of rational functions, since on $|t| < 1$ they follow by differentiating $R_d(t) = \sum_{\ell\geq 1} \ell^d t^\ell$. In particular,

$$R_1(t) = \frac{t}{(1-t)^2}, \qquad R_2(t) = \frac{t(1+t)}{(1-t)^3}, \qquad R_3(t) = \frac{t(1+4t+t^2)}{(1-t)^4}.$$

Theorem 7.31 (Rational rows and an explicit remainder)**.** *For w, j, τ as above,*

$$\psi^{(w-1+j)}(n\tau) + (-1)^j \psi^{(w-1+j)}(1-n\tau) = (2\pi\mathrm{i})^{w+j}\,\mathrm{Li}_{1-w-j}(q^n), \tag{7.54}$$

$$\mathcal{T}_{w,j}(\tau) = (2\pi\mathrm{i})^{w+j} \sum_{n\geq 1} n^j\,\mathrm{Li}_{1-w-j}(q^n). \tag{7.55}$$

The series converges normally on every compact subset of $\mathbb{H}$. If $\mathcal{T}_{w,j}^{[N]}$ is its sum through $n = N$, where $N \geq 0$, then

$$\left|\mathcal{T}_{w,j}(\tau) - \mathcal{T}_{w,j}^{[N]}(\tau)\right| \leq (2\pi)^{w+j}(N+1)^j \frac{\mathrm{Li}_{-j}(r)}{r}\,\mathrm{Li}_{1-w-j}\left(r^{N+1}\right). \tag{7.56}$$

Thus the bound contains only rational functions of r and r^{N+1}, together with an explicit power of π.

Proof. For $\mathrm{Im}\, z > 0$, splitting the absolutely convergent bilateral power sum into its nonnegative and negative indices gives

$$\sum_{m\in\mathbb{Z}} (m+z)^{-w} = \frac{\psi^{(w-1)}(z) + \psi^{(w-1)}(1-z)}{(w-1)!}. \tag{7.57}$$

Here evenness of w is essential. The classical cotangent expansion and its partial-fraction interpretation [49] read

$$\pi\cot(\pi z) = -\pi\mathrm{i} - 2\pi\mathrm{i}\sum_{\ell\geq 1} e^{2\pi\mathrm{i}\ell z}, \qquad \pi\cot(\pi z) = \lim_{M\to\infty} \sum_{m=-M}^{M} \frac{1}{m+z}.$$

Differentiate $w-1$ times. On compact subsets of the upper half-plane the exponential series and its derivatives converge uniformly. After the first derivative, the bilateral series and all its subsequent derivatives also converge normally. Since $(-1)^{w-1} = -1$, the two minus signs cancel and yield

$$\psi^{(w-1)}(z) + \psi^{(w-1)}(1-z) = (2\pi\mathrm{i})^w \sum_{\ell\geq 1} \ell^{w-1} e^{2\pi\mathrm{i}\ell z}.$$

Differentiating another j times with respect to z proves (7.54). Multiplication by n^j gives the summand in (7.55).

To prove the remainder bound, expand the rational polylogarithm in its nonnegative majorizing series. The absolute value of the omitted rows is at most

$$(2\pi)^{w+j}\sum_{\ell\geq 1}\ell^{w+j-1}\sum_{n>N}n^j r^{n\ell}. \tag{7.58}$$

Write $n = N+1+a$. For $a\geq 0$ and $\ell\geq 1$,

$$n^j \leq (N+1)^j(a+1)^j, \qquad r^{a\ell}\leq r^a.$$

Consequently

$$\sum_{n>N}n^j r^{n\ell} \leq (N+1)^j r^{(N+1)\ell}\sum_{a\geq 0}(a+1)^j r^a = (N+1)^j r^{(N+1)\ell}\frac{\mathrm{Li}_{-j}(r)}{r}.$$

Substitution in (7.58) proves (7.56). In particular, the majorant is finite. On a compact subset of $\mathbb{H}$, one has $r\leq r_0<1$, and each of its power series with nonnegative coefficients is bounded by its value at r_0. This proves normal convergence, for every fixed w, j, and justifies all subsequent termwise differentiations. □

For $j=0$, the tail formula simplifies to

$$\left|\mathcal{T}_{w,0}-\mathcal{T}_{w,0}^{[N]}\right| \leq \frac{(2\pi)^w}{1-r}\,\mathrm{Li}_{1-w}\left(r^{N+1}\right).$$

The case $w=2$, which will be particularly useful, is

$$\left|\mathcal{T}_{2,0}-\mathcal{T}_{2,0}^{[N]}\right| \leq \frac{4\pi^2 r^{N+1}}{(1-r)(1-r^{N+1})^2}. \tag{7.59}$$

Theorem 7.31 also proves

$$\frac{d^j}{d\tau^j}\mathcal{T}_{w,0}(\tau) = \mathcal{T}_{w,j}(\tau). \tag{7.60}$$

Computationally, (7.54) is preferable to evaluating its two polygamma terms separately. Their algebraic asymptotic contributions cancel, leaving the exponentially small paired row. The rational formula performs that cancellation exactly before numerical evaluation.

7.14.2 The weight-two ordering correction from telescoping

For $w\geq 4$, the ordinary absolutely convergent lattice series is

$$G_w(\tau) = \sum_{(m,n)\in\mathbb{Z}^2\setminus\{(0,0)\}}(m+n\tau)^{-w} = 2\zeta(w)+\frac{2}{(w-1)!}\mathcal{T}_{w,0}(\tau).$$

For $w=2$, specify the order explicitly:

$$G_2(\tau) = \sum_{n\in\mathbb{Z}}\left(\sum_{m\in\mathbb{Z}}{}'(m+n\tau)^{-2}\right) = \frac{\pi^2}{3}+2\mathcal{T}_{2,0}(\tau), \tag{7.61}$$

$$C_2(\tau) = \sum_{m\in\mathbb{Z}}\left(\sum_{n\in\mathbb{Z}}{}'(m+n\tau)^{-2}\right). \tag{7.62}$$

The prime omits only $(m,n)=(0,0)$. In the first expression, each inner row and the outer series of completed rows converge absolutely. The reverse expression exists as well: changing the names and signs of the one-dimensional indices gives

$$C_2(\tau)=\tau^{-2}G_2(-1/\tau). \tag{7.63}$$

This is a coordinate identity for the prescribed iterated sums, not an assumption of modularity. The series of absolute values over all lattice points diverges, so an interchange of the two orders still requires a calculation.

Theorem 7.32 (Summation-order anomaly)**.** *For every $\tau\in\mathbb{H}$,*

$$C_2(\tau)=G_2(\tau)-\frac{2\pi\mathrm{i}}{\tau}. \tag{7.64}$$

Consequently

$$G_2(-1/\tau)=\tau^2G_2(\tau)-2\pi\mathrm{i}\tau,\qquad G_2(\tau+1)=G_2(\tau). \tag{7.65}$$

Proof. We prove the difference of the two orders by replacing their common part with an absolutely convergent series. Put

$$h(z)=\frac{1}{z}-\frac{1}{z+1},\qquad f(z)=\frac{1}{z^2}-h(z)=\frac{1}{z^2(z+1)},$$

and define

$$F(\tau)=\sum_{\substack{n\in\mathbb{Z}\setminus\{0\}\\ m\in\mathbb{Z}}} f(m+n\tau).$$

No pole occurs in this sum. The real-linear map $(m,n)\mapsto m+n\tau$ is invertible, so for some $c_\tau>0$, $|m+n\tau|\ge c_\tau\sqrt{m^2+n^2}$. Outside a finite set, $|f(z)|\le 2|z|^{-3}$; hence F converges absolutely. It can therefore be summed in either order.

For each fixed $n\neq 0$, the series $\sum_m h(m+n\tau)$ is absolutely convergent and its symmetric partial sum telescopes:

$$\sum_{m=-M}^{M} h(m+n\tau)=\frac{1}{-M+n\tau}-\frac{1}{M+1+n\tau}\longrightarrow 0.$$

Separating the row $n=0$ proves

$$G_2(\tau)=\frac{\pi^2}{3}+F(\tau). \tag{7.66}$$

For the reverse order, define for integer m

$$A(m)=\lim_{K\to\infty}\sum_{0<|n|\le K}\frac{1}{m+n\tau}.$$

This symmetric limit exists: pairing n and $-n$ gives an $O(n^{-2})$ summand. The cotangent partial-fraction formula gives

$$A(m)=\frac{\pi}{\tau}\cot\Big(\frac{\pi m}{\tau}\Big)-\frac{1}{m}\quad(m\neq 0),\qquad A(0)=0. \tag{7.67}$$

The value at zero is also the removable limit of the expression on the right. For each fixed m, the n-series of $h(m+n\tau)$ is absolutely convergent. Taking its symmetric partial sums shows

$$\sum_{n\neq 0}h(m+n\tau)=A(m)-A(m+1).$$

It follows that

$$C_2(\tau) - \frac{\pi^2}{3} - F(\tau) = \lim_{M\to\infty} \sum_{m=-M}^{M} \sum_{n\neq 0} h(m+n\tau)$$
$$= \lim_{M\to\infty} \big(A(-M) - A(M+1)\big).$$

The extraction of F here is legitimate by its absolute convergence; no rearrangement of the original double power sum has been made.

Since $\operatorname{Im}(1/\tau) < 0$,

$$\cot(\pi m/\tau) \longrightarrow \mathrm{i} \quad (m \to +\infty), \qquad \cot(\pi m/\tau) \longrightarrow -\mathrm{i} \quad (m \to -\infty).$$

Equation (7.67) therefore gives $A(-M) - A(M+1) \to -2\pi\mathrm{i}/\tau$. Combining this with (7.66) proves (7.64). Equation (7.63) now gives the first transformation in (7.65). Periodicity follows directly from the q-series in (7.55). □

The anomaly is classical; Romik and Scherer discuss it together with more general shape-dependent summation procedures [57]. The proof above fixes both its sign and the normalization needed for the draft's polygamma sums.

Corollary 7.33 (The missing weight-two CM rows)**.** *One has*

$$G_2(\mathrm{i}) = \pi, \qquad G_2(\rho) = \frac{2\pi}{\sqrt{3}}, \tag{7.68}$$

$$\mathcal{T}_{2,0}(\mathrm{i}) = \frac{\pi}{2} - \frac{\pi^2}{6}, \qquad \mathcal{T}_{2,0}(\rho) = \frac{\pi}{\sqrt{3}} - \frac{\pi^2}{6}. \tag{7.69}$$

Proof. At $\tau = \mathrm{i}$, the first transformation in (7.65) gives $G_2(\mathrm{i}) = -G_2(\mathrm{i}) + 2\pi$. For ρ, use $-1/\rho = \rho - 1$ and periodicity. The same transformation becomes

$$(\rho^2 - 1)G_2(\rho) = 2\pi\mathrm{i}\rho.$$

Since $\rho^2 - 1 = \mathrm{i}\sqrt{3}\rho$, this proves the second value in (7.68). Substitution into (7.61) proves (7.69). □

7.14.3 CM seeds and the complete differential tower

Normalize the Eisenstein series by

$$E_2 = \frac{3G_2}{\pi^2}, \qquad E_4 = \frac{G_4}{2\zeta(4)}, \qquad E_6 = \frac{G_6}{2\zeta(6)}.$$

The seed values, in precisely this normalization, are

$$(E_2(\mathrm{i}), E_4(\mathrm{i}), E_6(\mathrm{i})) = \left(\frac{3}{\pi}, \frac{3\Gamma(1/4)^8}{64\pi^6}, 0\right), \tag{7.70}$$

$$(E_2(\rho), E_4(\rho), E_6(\rho)) = \left(\frac{2\sqrt{3}}{\pi}, 0, \frac{27\Gamma(1/3)^{18}}{512\pi^{12}}\right). \tag{7.71}$$

We have already proved the weight-two entries. The zero entries follow by rotating the absolutely convergent lattices. Multiplication by i fixes the square lattice and sends its weight-six sum to

its negative; multiplication by ρ fixes the hexagonal lattice and multiplies its weight-four sum by $\rho^{-4} \neq 1$.

For completeness, the classical period integrals fix the nonzero entries without numerical identification. The standard inverse Weierstrass integral [49], with $g_2 = 60G_4$ and $g_3 = 140G_6$, gives positive real periods

$$p_{\mathrm{sq}} = 2\int_1^\infty \frac{dx}{\sqrt{4x^3-4x}} = 2\int_0^1 \frac{du}{\sqrt{1-u^4}} = \frac{1}{2}B\Big(\frac{1}{4},\frac{1}{2}\Big) = \frac{\Gamma(1/4)^2}{2\sqrt{2\pi}},$$
$$p_{\mathrm{hex}} = 2\int_1^\infty \frac{dx}{\sqrt{4x^3-4}} = \frac{1}{3}B\Big(\frac{1}{6},\frac{1}{2}\Big) = \frac{\Gamma(1/3)^3}{2^{4/3}\pi}.$$

The corresponding lattices are $p_{\mathrm{sq}}(\mathbb{Z}+\mathrm{i}\mathbb{Z})$ for $(g_2,g_3)=(4,0)$, and $p_{\mathrm{hex}}(\mathbb{Z}+\rho\mathbb{Z})$ for $(g_2,g_3)=(0,4)$. Their square and equianharmonic shapes follow from the rotations of the respective cubics; the real period integral fixes the scale and orientation. The first substitution above is $x = u^{-2}$; in the second beta integral use $t = x^{-3}$. Euler's reflection and duplication formulas give the displayed gamma evaluations. Scaling the lattice invariants consequently yields

$$G_4(\mathrm{i}) = \frac{p_{\mathrm{sq}}^4}{15} = \frac{\Gamma(1/4)^8}{960\pi^2}, \qquad G_6(\rho) = \frac{p_{\mathrm{hex}}^6}{35} = \frac{\Gamma(1/3)^{18}}{8960\pi^6}.$$

Using $2\zeta(4) = \pi^4/45$ and $2\zeta(6) = 2\pi^6/945$ proves the remaining entries of (7.70)–(7.71).

Ramanujan's classical differential identities [58] are

$$\theta E_2 = \frac{E_2^2 - E_4}{12}, \qquad \theta E_4 = \frac{E_2E_4 - E_6}{3}, \qquad \theta E_6 = \frac{E_2E_6 - E_4^2}{2}, \qquad \theta = \frac{1}{2\pi\mathrm{i}}\frac{d}{d\tau}. \tag{7.72}$$

Encode these identities by the rational derivation

$$\mathcal{V} = \frac{X^2-Y}{12}\frac{\partial}{\partial X} + \frac{XY-Z}{3}\frac{\partial}{\partial Y} + \frac{XZ-Y^2}{2}\frac{\partial}{\partial Z}, \qquad P_0 = X, \quad P_{j+1} = \mathcal{V}P_j. \tag{7.73}$$

For example,

$$P_1 = \frac{X^2-Y}{12},$$
$$P_2 = \frac{X^3-3XY+2Z}{72},$$
$$P_3 = \frac{X^4-6X^2Y+8XZ-3Y^2}{288}.$$

Theorem 7.34 (All differentiated weight-two rows)**.** *Set $\mathcal{S}_j(\tau) = \mathcal{T}_{2,j}(\tau)$. For every $\tau \in \mathbb{H}$,*

$$\mathcal{S}_0(\tau) = \frac{\pi^2}{6}\big(E_2(\tau) - 1\big), \tag{7.74}$$
$$\mathcal{S}_j(\tau) = \frac{\pi^2}{6}(2\pi\mathrm{i})^j P_j(E_2(\tau), E_4(\tau), E_6(\tau)), \qquad j \geq 1. \tag{7.75}$$

In particular, substitution of (7.70)–(7.71) evaluates every $\mathcal{S}_j(\mathrm{i})$ and $\mathcal{S}_j(\rho)$ by a finite polynomial calculation over $\mathbb{Q}$ and the indicated gamma periods.

Proof. Equation (7.61) and the definition of E_2 give (7.74). By the chain rule and (7.72), every polynomial $P \in \mathbb{Q}[X,Y,Z]$ satisfies

$$\theta\big(P(E_2,E_4,E_6)\big) = (\mathcal{V}P)(E_2,E_4,E_6).$$

Induction from $P_0 = X$ therefore gives $\theta^j E_2 = P_j(E_2, E_4, E_6)$. Differentiate (7.74) $j \geq 1$ times and use (7.60). The constant term disappears, and $d/d\tau = 2\pi\mathrm{i}\theta$ contributes the factor $(2\pi\mathrm{i})^j$. This proves (7.75); the CM specializations use the established seed values. □

For clarity, the first two differentiated evaluations are

$$\sum_{n\geq 1} n\left[\psi''(n\mathrm{i}) - \psi''(1-n\mathrm{i})\right] = \frac{\mathrm{i}\pi}{4} - \frac{\mathrm{i}\Gamma(1/4)^8}{768\pi^3}, \tag{7.76}$$

$$\sum_{n\geq 1} n\left[\psi''(n\rho) - \psi''(1-n\rho)\right] = \frac{\mathrm{i}\pi}{3}, \tag{7.77}$$

$$\sum_{n\geq 1} n^2\left[\psi'''(n\mathrm{i}) + \psi'''(1-n\mathrm{i})\right] = -\frac{\pi}{4} + \frac{\Gamma(1/4)^8}{256\pi^3}, \tag{7.78}$$

$$\sum_{n\geq 1} n^2\left[\psi'''(n\rho) + \psi'''(1-n\rho)\right] = -\frac{2\sqrt{3}\pi}{9} - \frac{\Gamma(1/3)^{18}}{1024\pi^8}. \tag{7.79}$$

These follow by substituting the seed values into P_1, P_2, so their derivation contains no integer-relation search.

Corollary 7.35 (Polynomial degree bounds for the CM row formulas)**.** *Put $A = \Gamma(1/4)^8/\pi^4$ and $B = \Gamma(1/3)^{18}/\pi^9$. For each $j \geq 1$, there are polynomials $U_j \in \mathbb{Q}[t]$ and $V_j \in \mathbb{Q}(\sqrt{3})[t]$ such that*

$$\frac{\mathcal{S}_j(\mathrm{i})}{\mathrm{i}^j\pi} = U_j(A), \qquad \frac{\mathcal{S}_j(\rho)}{\mathrm{i}^j\pi} = V_j(B),$$

with

$$\deg U_j \leq \left\lfloor \frac{j+1}{2} \right\rfloor, \qquad \deg V_j \leq \left\lfloor \frac{j+1}{3} \right\rfloor.$$

Proof. Give X, Y, Z weights $2, 4, 6$. Each term of $\mathcal{V}$ raises weight by two, so P_j is homogeneous of weight $2j+2$. At the square point, the surviving monomials $X^a Y^b$ have $a + 2b = j + 1$. On inserting the seeds into (7.75) and dividing by $\mathrm{i}^j\pi$, each becomes a rational multiple of A^b. At the hexagonal point, the surviving monomials $X^a Z^c$ satisfy $a + 3c = j + 1$, and the same substitution produces an element of $\mathbb{Q}(\sqrt{3})B^c$. The stated degree bounds follow. □

7.14.4 What this resolves and how it is verified

Corollary 7.33 resolves the weight-two row-sum question posed in the modular draft. Theorem 7.34 extends it to all polynomially weighted derivative rows, with the explicit error estimate of Theorem 7.31. These are evaluations of specified infinite aggregates. They do not evaluate an isolated component such as $\operatorname{Im}\psi'(\mathrm{i})$, nor do they prove that such a component is an Eichler integral. Any proposed identification of that isolated constant needs its own exact transform or functional equation.

The accompanying `code/modular_rows.py` constructs the rational functions R_d and the polynomials P_j by exact arithmetic. It checks individual rows against a separate complex-polygamma implementation, compares the CM formulas for $j = 0, \ldots, 4$ with their rapidly convergent row sums, and checks representative tails. The output `data/modular_rows.json` records the exact coefficients, analytic truncation bounds, and numerical residuals. Its decimal calculations use ordinary high-precision arithmetic; they are numerical checks, not outward-rounded interval certificates. A machine enclosure can be obtained by evaluating the rational formulas and (7.56) with directed interval arithmetic. The identities themselves are proved above and do not depend on the numerical checks.

Chapter 11

Herglotz arithmetic and quadratic regulators

The Herglotz function joins the digamma series to finite dilogarithm sums. At rational arguments this supplies exact evaluations; at quadratic units it connects continued-fraction cycles with real-quadratic Kronecker limits. The same experimental strategy thus leads from a finite cyclotomic sum to regulator candidates in class fields.

11.1 The function and the exact dilogarithm sum

For $x > 0$ define

$$F(x) = \sum_{n\geq 1} \frac{\psi(nx) - \log(nx)}{n}, \qquad J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t}\, dt.$$

The series is locally uniformly convergent on the positive line and extends analytically to the cut plane $\mathbb{C} \setminus (-\infty, 0]$. Radchenko–Zagier [35] relate the two functions by

$$J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t}\, dt, \qquad J(x) = F(2x) - 2F(x) + F(x/2) + \frac{\pi^2}{12x}, \tag{11.1}$$

which relates the companion J to the Herglotz function F. For the rational-argument results the engine is Radchenko–Zagier's Theorem 3, an *exact, finite* evaluation obtained from the second integral representation of F via the distribution relations:

$$F\Big(\tfrac{P}{Q}\Big) - F(1) = \sum_{\alpha^P=1} \Big(\sum_{\substack{\beta^Q=1 \\ \beta\neq 1}} f(\alpha,\beta) - \sum_{\substack{\beta^P=1 \\ \beta\neq 1}} f(\alpha,\beta) \Big), \quad f(\alpha,\beta) = \mathrm{Li}_2\Big(\tfrac{\beta}{\beta-1}\Big) - \mathrm{Li}_2\Big(\tfrac{\alpha-\beta}{1-\beta}\Big). \tag{11.2}$$

The finite sum is an exact analytic identity, with compatible principal branches at its arguments. Floating-point evaluation still has rounding and cancellation errors. It is a useful independent comparator for the defining integral, not a zero-error numerical oracle.

11.2 Quadratic-unit values: Tables 1 & 2, and two errata

For real quadratic units the relevant object is the companion $\mathcal{J}(x) = J(x) - \frac{\log^2 2}{2} + \frac{\pi^2}{24}(x - \frac{1}{x})$, and the arithmetic atoms are $S_{p,q} = \log \varepsilon_p \, \log \varepsilon_q$ where ε_d is the fundamental unit of $\mathbb{Q}(\sqrt{d})$ for a fundamental

discriminant $d > 1$ and $\varepsilon_1 := 2$. Tables 1 and 2 of the source paper express, for the 45 known one-class-per-genus cases, the quantities $4\big(\mathcal{J}(n+\sqrt{n^2-1}) - n\frac{\pi^2}{24}\big)$ and $2\big(\mathcal{J}(n+\sqrt{n^2+1}) - n\frac{\pi^2}{24}\big)$ as integer combinations of the $S_{p,q}$.

We re-implemented $\mathcal{J}$ and $S_{p,q}$ in PARI/GP (`intnum` integrals, `bnfinit` regulators) and evaluated both sides of all 45 rows at 200-digit precision. All 16 rows of Table 2 and 27 of the 29 rows of Table 1 agree to better than 10^{-80}. The two failures are $O(1)$:

$$n = 43\text{: LHS} - \text{RHS} = -13.0160\ldots, \qquad n = 61\text{: LHS} - \text{RHS} = -2.9011\ldots.$$

Each resolves to a single exact correction, confirmed against the high-DPI scan to be a typo in the printed paper rather than an OCR artifact.

Proposition 11.1 (Two corrections to Table 1)**.**

- $n = 43$ *($n^2 - 1 = 2^3 \cdot 3 \cdot 7 \cdot 11$): the printed right-hand side overshoots by exactly* $S_{33,56}$ *$\big((LHS - RHS)/S_{33,56} = -1.000\ldots$ to 110 digits$\big)$; the spurious* $+S_{33,56}$ *term should be deleted.*
- $n = 61$ *($n^2 - 1 = 2^3 \cdot 3 \cdot 5 \cdot 31$): the printed* $S_{60,1032}$ *should read* $S_{60,248}$*. The printed subscript* $1032 = 2^3{\cdot}3{\cdot}43$ *carries the prime* 43*, foreign to* $n^2{-}1 = 3720$*, whereas* $60{\cdot}248 = 14880 = 4{\cdot}3720$ *is the correct genus splitting.*

A brute-force search over all single edits (delete a term, change a coefficient in $[-4, 4]$*, or replace a subscript by any admissible fundamental-discriminant pair) returns these as the* unique *fixes. With both applied, all* 45 *rows verify to better than* 10^{-80}*.*

The quadratic-unit tables concern specified arithmetic families. They do not imply that every even-trace unit has a pure-logarithm evaluation: outside genus characters the Kronecker-limit formula introduces additional class-group coordinates. Those coordinates are developed below.

11.3 Rational arguments: a family of theorems

The paper writes out only two rational-argument cases: $F(n)$ (pure logarithms, its eq. 23) and the lone fractional example $F(2/5)$ (its eq. 24). We now give the general laws.

11.3.1 Two reductions on the line $\mathrm{Re} = \frac{1}{2}$

Everything rests on one observation: for a unit $\beta = e^{i\theta}$,

$$\frac{\beta}{\beta - 1} \text{ has real part } \tfrac{1}{2}, \qquad \arg\frac{\beta}{\beta-1} = \frac{\theta}{2} - \frac{\pi}{2}, \qquad \Big|\frac{\beta}{\beta-1}\Big| = \frac{1}{2\sin(\theta/2)}. \tag{11.3}$$

By the reflection $\mathrm{Li}_2(z) + \mathrm{Li}_2(1-z) = \frac{\pi^2}{6} - \log z \log(1-z)$ with $1 - z = \bar{z}$ on $\mathrm{Re}\, z = \frac{1}{2}$,

$$2\,\mathrm{Re}\,\mathrm{Li}_2\Big(\tfrac{1}{2} + iy\Big) = \frac{\pi^2}{6} - \log^2|z| - (\arg z)^2. \tag{11.4}$$

The second reduction handles the $\alpha = -1$ part of (11.2): $\frac{-1-\beta}{1-\beta} = -i\cot\frac{\theta}{2}$ is purely imaginary, and for real y

$$\mathrm{Re}\,\mathrm{Li}_2(iy) = \tfrac{1}{4}\,\mathrm{Li}_2(-y^2). \tag{11.5}$$

11.3.2 $F(1/q)$: pure logarithms

Theorem 11.2. *For all $q \geq 2$, with $m = \lfloor (q-1)/2 \rfloor$,*

$$F\Big(\tfrac{1}{q}\Big) - F(1) = -\frac{(q-1)(3q-2)}{24q}\,\pi^2 - \sum_{k=1}^{m} \log^2\Big(2\sin\tfrac{\pi k}{q}\Big) \;-\; [\,q \text{ even}\,]\,\tfrac{1}{2}\log^2 2.$$

Proof. With $P = 1$ only $\alpha = 1$ occurs and the inner β^P-sum is empty, so (11.2) reads $F(1/q) - F(1) = \sum_{\beta^q=1,\beta\neq 1}(\mathrm{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6})$. Pairing β with $\bar\beta$ and applying (11.4) gives the π^2 coefficient $-\big[\frac{q-1}{12} + \sum_{k=1}^{m}(\frac{k}{q} - \frac{1}{2})^2\big] = -\frac{(q-1)(3q-2)}{24q}$ and the log-sine sum. For even q the self-conjugate root $\beta = -1$ contributes $\mathrm{Li}_2(\frac{1}{2}) = \frac{\pi^2}{12} - \frac{1}{2}\log^2 2$. □

11.3.3 $F(2/q)$: the general theorem

Theorem 11.3. *For every $q \geq 3$,*

$$F\Big(\tfrac{2}{q}\Big) - F(1) = -\frac{(q-1)(q-2)}{12q}\pi^2 + \log^2 2 - 2\sum_{k=1}^{\lfloor (q-1)/2\rfloor} \log^2\Big(2\sin\tfrac{\pi k}{q}\Big) - \tfrac{1}{2}\sum_{k=1}^{\lfloor (q-1)/2\rfloor} \mathrm{Li}_2\Big(-\cot^2\tfrac{\pi k}{q}\Big),$$

where for even q the two sums run to $k = \frac{q}{2} - 1$ and a further $-\log^2 2$ is added.

Proof (odd q). Here $\alpha \in \{1, -1\}$ and the inner β^P-sum is the single term $\beta = -1$. For the first part, $f(1,\beta) + f(-1,\beta) = 2\,\mathrm{Li}_2(\frac{\beta}{\beta-1}) - \frac{\pi^2}{6} - \mathrm{Li}_2(\frac{-1-\beta}{1-\beta})$. Summing over $\beta \neq 1$ and using (11.4) for the first pieces and (11.5) (with $\frac{-1-\beta}{1-\beta} = -i\cot\frac{\pi k}{q}$, pairing $k \leftrightarrow q-k$) for the last gives $2S_1 - (q-1)\frac{\pi^2}{6} - \frac{1}{2}\sum_k \mathrm{Li}_2(-\cot^2\frac{\pi k}{q})$, where $S_1 = \sum_{k=1}^{(q-1)/2}[\frac{\pi^2}{6} - \log^2(2\sin\frac{\pi k}{q}) - \pi^2(\frac{k}{q} - \frac{1}{2})^2]$. The boundary gives $-\sum_\alpha f(\alpha,-1) = +\log^2 2$ from $\mathrm{Li}_2(\frac{1}{2})$. Collecting, the $\frac{\pi^2}{6}$ terms cancel and the π^2 coefficient is $-2\sum_{k=1}^{(q-1)/2}(\frac{k}{q} - \frac{1}{2})^2$, which equals $-\frac{(q-1)(q-2)}{12q}$ by an elementary identity (verified symbolically). The even-q case differs only in that $\beta = -1$ is now a genuine q-th root, contributing the extra $-\log^2 2$. □

The algebraic dilogarithm arguments make the field structure explicit. At $q = 5$ the symmetric sum can be reduced further as described below. Unsuccessful searches at larger conductors do not prove that no other closed form exists; the finite-sum theorem remains valid at every q.

11.3.4 The $F(n/(n+1))$ family and the $F(p/2)$ reciprocity

Theorem 11.4. *For all $n \geq 2$, with $S(q) = \sum_{k=1}^{q-1}\log^2(2\sin\frac{\pi k}{q})$,*

$$F\Big(\tfrac{n}{n+1}\Big) - F(1) = -\frac{3n^2+3n+2}{24n(n+1)}\pi^2 + \mathrm{Li}_2\Big(\tfrac{n}{n+1}\Big) + \tfrac{1}{2}\log^2\tfrac{n+1}{n} - \tfrac{1}{2}S(n+1) + \tfrac{1}{2}S(n),$$

the single dilogarithm $\mathrm{Li}_2(n/(n+1))$ appearing with coefficient exactly $+1$ (the source paper's "bilinear-log" Example 2, made explicit). The π^2 coefficient equals $-\frac{1}{8} - \frac{1}{12n(n+1)}$.

Proof. The two functional equations of F are

$$F(x) - F(x+1) - F(x/(x+1)) = -F(1) + \mathrm{Li}_2(1/(x+1)),$$
$$F(x) + F(1/x) - 2F(1) = \tfrac{1}{2}\log^2 x - \frac{\pi^2}{6x}(x-1)^2.$$

The second equation and Theorem 11.2 give

$$F(n) - F(1) = -\frac{(n-1)(n-2)}{24n}\pi^2 + \frac{1}{2}\log^2 n + \frac{1}{2}S(n).$$

Subtract the cases n and $n+1$ in the first equation, and apply Euler reflection to $\mathrm{Li}_2(1/(n+1))$. Collecting the logarithms and rational π^2 coefficients yields the displayed formula. □

Theorem 11.5 (Reciprocity)**.** *For all odd $p \geq 3$,*

$$F\Big(\tfrac{p}{2}\Big) + F\Big(\tfrac{2}{p}\Big) - 2F(1) = -\frac{(p-2)^2}{12p}\pi^2 + \tfrac{1}{2}\log^2 \tfrac{p}{2}.$$

The $\mathrm{Li}_2(-\cot^2 \frac{\pi k}{p})$ terms of $F(p/2)$ are exactly minus those of $F(2/p)$ and cancel in the sum; combined with Theorem 11.3 this yields $F(p/2)$ in closed form.

Proof. Substitute $x = p/2$ into the second functional equation above. □

11.3.5 The derivative law

Theorem 11.6 (Radchenko–Zagier Proposition 4, $P = 2$)**.** *For all $q \geq 3$, with $T(q) := \frac{2}{q}F'(2/q) - (1 + \log 2)$,*

$$T(q) = \frac{q^2-4}{24q}\pi^2 + W(q), \qquad W(q) = \sum_{k=1}^{q-1} C(2k/q)\ \mathrm{Cl}_2\Big(\tfrac{2\pi k}{q}\Big),$$

where $C(t) = \cot(\pi t)$ for $t \notin \mathbb{Z}$ and $C(t) = 0$ for $t \in \mathbb{Z}$, with an extra $-\log 2$ for even q. This assigned value at integers is part of the formula; a literal cotangent there would be undefined. The sum provides a Clausen representation, without an independence assertion.

11.3.6 Rational-argument samples and the companion integral

The source's proposed criterion $\mathrm{rad}(q) \mid 30$ is false as a necessary condition for a reduction to rational dilogarithms: Theorem 11.2 gives a pure-logarithm answer for $F(1/7)$. Numerator, denominator and the available functional equations must all be considered. The formal five-term classification is proved in the following section. The following experimentally recovered samples are useful:

$$\begin{aligned}
F(3/4) - F(1) &= -\frac{19}{144}\pi^2 - \frac{3}{4}\log^2 2 - 2\log 2\log 3 + \frac{7}{4}\log^2 3 + 2\,\mathrm{Li}_2(1/3),\\
F(4/5) - F(1) &= \frac{3}{80}\pi^2 + \frac{11}{4}\log^2 2 - \frac{1}{2}\log^2 5 - \mathrm{Li}_2(1/5) - \log^2(2\sin(\pi/5)) - \log^2(2\sin(2\pi/5)),\\
J(2/3) &= -\frac{\pi^2}{144} + \frac{3}{4}\log^2 2,\\
J(2/5) &= \frac{11\pi^2}{240} + \frac{3}{4}\log^2 2 - 2\log^2 \phi.
\end{aligned}$$

The last two values follow by combining the F evaluations through (11.1). Their existence does not prove that these are the only q for which $J(2/q)$ simplifies.

11.4 From rational examples to an exact formal criterion

The denominator-only condition was false, but the newer cyclotomic argument supplies a precise replacement. Its equivalences concern the specified five-term calculus. Successful cases give actual analytic evaluations; an unsuccessful formal test is not a proof of numerical period independence.

11.5 Cyclotomic symbols and formal reduction

We use additive notation for a multiplicative group after rationalization:

$$\mathcal{A}(K) = K^\times \otimes_{\mathbb{Z}} \mathbb{Q}, \qquad \langle uv\rangle = \langle u\rangle + \langle v\rangle.$$

Roots of unity map to zero. Exterior powers below are over $\mathbb{Q}$ unless otherwise specified.

Definition 11.7 (Pre-Bloch group and formal reduction)**.** *Let $\mathcal{P}(K)$ be the $\mathbb{Q}$-vector space on symbols $[z]$, $z \in K$, with $[0] = [1] = 0$, modulo the five-term relations. Its differential is*

$$\partial[z] = \langle z\rangle \wedge \langle 1-z\rangle \quad (z \neq 0, 1), \qquad \partial[0] = \partial[1] = 0, \qquad \mathcal{B}(K) = \ker \partial.$$

For an invariant dilogarithm element $\xi \in \mathcal{P}(K)$, a formal logarithmic reduction *means $\xi = 0$ in $\mathcal{P}(K)$. A* formal rational-dilogarithm reduction *means that ξ belongs to the image of $\mathcal{P}(\mathbb{Q})$. Upon evaluation, five-term relations carry products of algebraic logarithms and rational multiples of π^2; these are the terms suppressed by this terminology.*

For a finite Galois number field $K/\mathbb{Q}$, two standard rationalized facts will be used:

$$\mathcal{B}(K)^{\mathrm{Gal}(K/\mathbb{Q})} = \mathcal{B}(\mathbb{Q}) = 0, \qquad \partial\mathcal{P}(\mathbb{Q}) = \Lambda^2\mathcal{A}(\mathbb{Q}). \tag{11.6}$$

The first is Bloch-group descent and the torsion of the Bloch group of $\mathbb{Q}$; the second is equivalent to the torsion of $K_2(\mathbb{Q})$. These are established algebraic K-theory inputs, not new results of this article; see [46, 45]. The same inputs underpin the symbol argument in [35, §7.2].

Let $\zeta_q = e^{2\pi i/q}$ and, for $(a, q) = 1$, set

$$\beta_q(a) = \sum_{k=1}^{q-1} \langle 1-\zeta_q^{ak}\rangle \wedge \langle 1-\zeta_q^{k}\rangle \in \Lambda^2\mathcal{A}(\mathbb{Q}(\zeta_q)). \tag{11.7}$$

We put $\beta_1 = \beta_2 = 0$. The term $k = 0$ is excluded: there is no multiplicative-group element $\langle 0\rangle$. For coprime positive p, q, the finite cyclotomic dilogarithm sum for $F(p/q) - F(1)$ defines a Galois-invariant element $\xi_{p/q}$ with

$$\partial\xi_{p/q} = \beta_q(p) - \beta_p(q) - \langle p\rangle \wedge \langle q\rangle. \tag{11.8}$$

One can take $K = \mathbb{Q}(\zeta_{pq})$. We retain this classical normalization throughout; the rational wedge in (11.8) matters for the distinction between logarithms and rational dilogarithms.

Lemma 11.8 (Elementary symmetries)**.** *For every conductor and every invertible residue a,*

$$\beta_q(-a) = \beta_q(a), \qquad \beta_q(a^{-1}) = -\beta_q(a).$$

Each $\beta_q(a)$ is Galois invariant.

Proof. The identity $1 - \zeta_q^{-ak} = -\zeta_q^{-ak}(1-\zeta_q^{ak})$ proves the first claim after roots of unity are killed. In the sum for $\beta_q(a^{-1})$, replace k by ak and reverse the exterior factors. Finally, every Galois automorphism multiplies k by a unit modulo q, permuting the summation set. □

11.6 The all-conductor relation theorem

The following is the main algebraic theorem.

Theorem 11.9 (Complete kernel, every conductor)**.** *Let $q \geq 3$, $\mathcal{G}_q = (\mathbb{Z}/q\mathbb{Z})^\times/\{\pm 1\}$, and $(c_a)_{a\in\mathcal{G}_q} \in \mathbb{Q}^{\mathcal{G}}_q$. Then*

$$\sum_{a\in\mathcal{G}_q} c_a\beta_q(a) = 0 \quad\Longleftrightarrow\quad c_a = c_{a^{-1}} \quad \text{for every } a \in \mathcal{G}_q. \tag{11.9}$$

In particular,

$$\beta_q(a) = 0 \quad\Longleftrightarrow\quad a^2 \equiv \pm 1 \pmod q, \tag{11.10}$$

$$\dim_{\mathbb{Q}} \operatorname{span}\{\beta_q(a) : a \in \mathcal{G}_q\} = \frac{|\mathcal{G}_q| - |\mathcal{G}_q[2]|}{2}. \tag{11.11}$$

Choosing one a from each inverse pair $\{a, a^{-1}\}$ with $a \neq a^{-1}$ gives a basis of the symbol space.

The proof is short after the correct positive matrix is identified. Its use of *all* nontrivial roots, including those of smaller order, is essential at composite conductor.

11.6.1 Cyclotomic logarithms and a positive Gram matrix

For $1 \leq k < q$, define the real vector

$$v_k = \big(\log|1-\zeta_q^{ak}|\big)_{a\in\mathcal{G}_q} \in \mathbb{R}^{\mathcal{G}}_q, \qquad A_q = \sum_{k=1}^{q-1} v_k v_k^{\mathsf{T}}. \tag{11.12}$$

The entries are independent of the representative of a modulo sign. Let P_b be the permutation matrix $(P_b v)_a = v_{ab}$. Then

$$P_b v_k = v_{bk}, \qquad P_b A_q P_b^{\mathsf{T}} = A_q, \qquad P_b A_q = A_q P_b. \tag{11.13}$$

Lemma 11.10 (Positive definiteness)**.** *The vectors v_k span $\mathbb{R}^{\mathcal{G}}_q$. Hence A_q is positive definite.*

Proof. Complexify the regular representation of the finite abelian group $\mathcal{G}_q$. Each character occurs once. The span of the v_k is $\mathcal{G}_q$-invariant by (11.13), so it suffices to show a nonzero Fourier component for every character.

Let χ be nontrivial and let $f \mid q$ be its primitive Dirichlet conductor. It is an even character, so $f > 2$. Write χ^* for the primitive character modulo f. Reduction gives a surjection $\mathcal{G}_q \longrightarrow G_f$. For the vector with $k = q/f$, its Fourier coefficient is a nonzero positive integer multiple of

$$\sum_{a\in G_f} \chi^*(a)\log|1-\zeta_f^a| = -\frac{1}{2}\tau(\chi^*)L(1,\overline{\chi^*}) \neq 0. \tag{11.14}$$

The equality follows by Abel-regularizing $\log|1-re^{i\theta}| = -\sum_{n\geq 1} r^n\cos(n\theta)/n$, summing over a, using the primitive Gauss-sum identity, and letting $r \uparrow 1$. The Gauss sum is nonzero and Dirichlet's nonvanishing theorem gives $L(1,\overline{\chi^*}) \neq 0$. This is a standard cyclotomic-logarithm identity; cf. [47, 48].

For the trivial character, choose a prime $\ell \mid q$. If ℓ is odd, use $k = q/\ell$ and

$$\sum_{a\in G_\ell} \log|1-\zeta_\ell^a| = \tfrac{1}{2}\log\ell \neq 0.$$

If $\ell = 2$, the vector $v_{q/2}$ is the nonzero constant vector $(\log 2)_{a\in\mathcal{G}_q}$. Thus every character component occurs. This proves the spanning assertion, and $w^{\mathsf{T}}A_q w = \sum_k (w^{\mathsf{T}}v_k)^2 > 0$ for $w \neq 0$. □

11.6.2 From exterior symbols to skew matrices

There is a $\mathbb{Q}$-linear logarithm map

$$\ell : \mathcal{A}(\mathbb{Q}(\zeta_q)) \longrightarrow \mathbb{R}_q^{\mathcal{G}}, \qquad \ell(\langle u \rangle) = (\log |\sigma_a(u)|)_{a \in \mathcal{G}_q}.$$

Composing its exterior square with the passage to the real exterior power and then identifying $v \wedge w$ with $vw^{\mathsf{T}} - wv^{\mathsf{T}}$ gives a $\mathbb{Q}$-linear map to real skew matrices. We do *not* assume this map is injective on the whole exterior power. Its value on (11.7) is explicitly

$$\ell_\wedge(\beta_q(b)) = P_b A_q - A_q P_b^{\mathsf{T}} = A_q(P_b - P_{b^{-1}}). \tag{11.15}$$

Proof of Theorem 11.9. If the left side of (11.9) vanishes, apply (11.15) and multiply by the invertible matrix A_q^{-1}. We get

$$0 = \sum_{b \in \mathcal{G}_q} c_b (P_b - P_{b^{-1}}) = \sum_{b \in \mathcal{G}_q} (c_b - c_{b^{-1}}) P_b.$$

The regular permutation matrices P_b are linearly independent, even over $\mathbb{R}$: their values on a single coordinate vector are the distinct coordinate vectors. Therefore $c_b = c_{b^{-1}}$.

Conversely, if these equalities hold, group the sum into inverse pairs and use Lemma 11.8; self-inverse classes have $2\beta_q(b) = 0$ and hence contribute zero. This proves (11.9). Applying it to a coefficient vector supported at one class proves (11.10), and counting nontrivial inverse pairs proves (11.11) and the basis assertion. □

Remark 11.11 (Why the composite case is not a numerical rank claim). All matrix entries involving logarithms are used only in the proof of positive definiteness. The conclusion follows from exact character nonvanishing, not from a numerical determinant. In particular, no lower bound for a numerically computed singular value is required. The resulting coordinate model $\beta_q(a) \mapsto P_a - P_{a^{-1}}$ is exact on the symbol span, even though its derivation used a real logarithm map.

11.6.3 Explicit rank formula and zero conductors

Corollary 11.12. *For $q \geq 3$,*

$$d(q) := \dim_{\mathbb{Q}} \operatorname{span}\{\beta_q(a)\} = \frac{\varphi(q) - r_+(q) - r_-(q)}{4}.$$

Write $q = 2^e \prod_{j=1}^{s} p_j^{e_j}$ with distinct odd primes p_j. Then

$$r_+(q) = 2^s \begin{cases} 1 & e = 0, 1, \\ 2 & e = 2, \\ 4 & e \geq 3, \end{cases}$$

and $r_-(q) = 0$ if $e \geq 2$ or some $p_j \equiv 3 \pmod 4$; otherwise $r_-(q) = 2^s$.

Proof. The self-inverse elements of $\mathcal{G}_q$ are the sign classes of roots of $u^2 = 1$ or $u^2 = -1$. These two sets are disjoint for $q > 2$, and every sign class has two representatives. Thus $|\mathcal{G}_q[2]| = (r_+ + r_-)/2$. The stated root counts follow from the Chinese remainder theorem and the elementary square-root counts at prime powers. □

Corollary 11.13 (Complete list of identically zero conductors)**.** *With the conventions at $q = 1, 2$, the function β_q is identically zero exactly for*

$$q \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}. \tag{11.16}$$

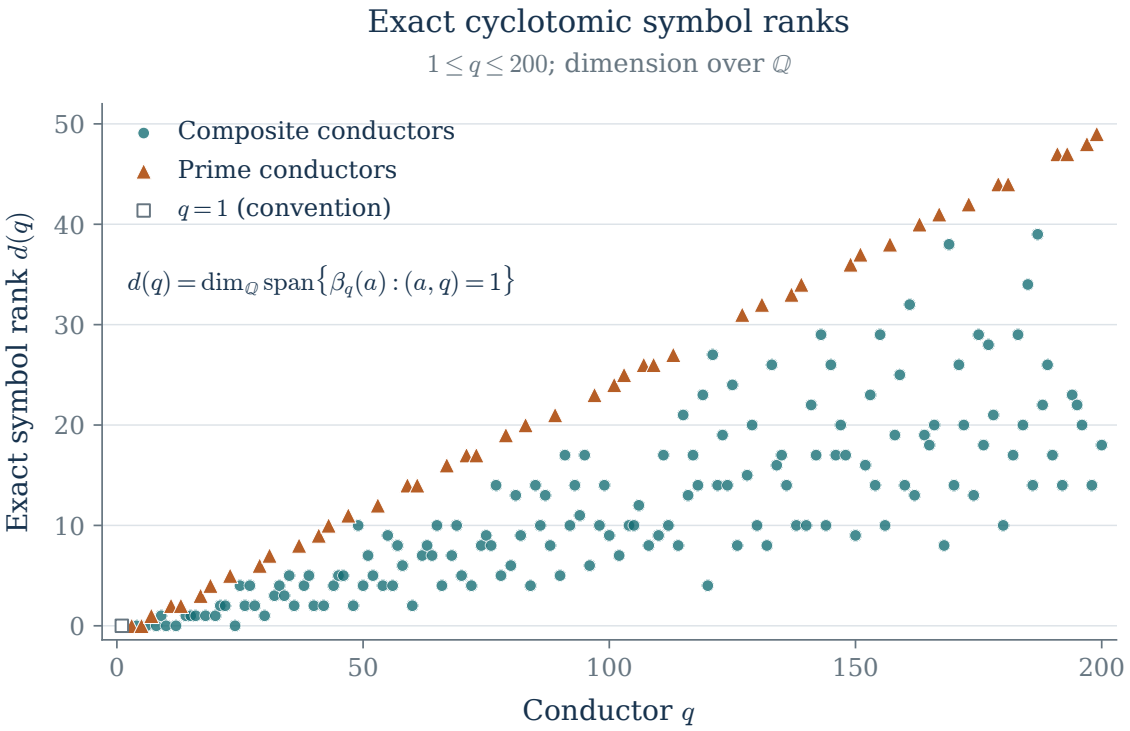


Figure 11.1: Exact ranks of the cyclotomic symbol spaces for conductors $1 \le q \le 200$. The plotted values come from integer residue counts and the proved rank formula. They are not numerical estimates of linear independence among evaluated periods.

Proof. The condition is that $\mathcal{G}_q$ have exponent at most 2. Projecting to each odd prime-power factor forces that factor to be 3 or 5 to the first power, since its unit group is cyclic and has order dividing 4. The 2-part is at most 8: the square of 3 modulo 16 is neither 1 nor -1. If 5 occurs with 3 or with 4, the Chinese remainder theorem produces a unit whose square is -1 at 5 and 1 at the other factor; its square is not a global sign. Hence the cases containing 5 are only $5, 10$. Without 5, all divisors of 24 work. This is exactly (11.16). □

q	7	9	13	15	16	24	25	60	105
$d(q)$	1	1	2	1	1	0	4	2	10

The rank concerns symbols, not the dimension of the numerical span of Herglotz values. Formula (11.11) makes the distinction computable without a height-bounded integer-relation search.

11.7 Exact reduction criteria and constructive rational cores

The two cyclotomic pieces in (11.8) cannot cancel across coprime conductors. This needs a norm on the *individual exterior factors*; averaging an exterior symbol under Galois action would not work, because the symbols themselves are invariant.

Lemma 11.14 (Separation by normalized field norms)**.** *Let* $(p, q) = 1$ *and regard both symbols in* $\Lambda^2\mathcal{A}(K)$*,* $K = \mathbb{Q}(\zeta_{pq})$*. Then*

$$\beta_q(p) - \beta_p(q) \in \Lambda^2\mathcal{A}(\mathbb{Q}) \quad \Longleftrightarrow \quad \beta_q(p) = \beta_p(q) = 0. \tag{11.17}$$

Proof. Let $E = \mathbb{Q}(\zeta_p)$ and define the linear normalized norm

$$N_p : \mathcal{A}(K) \longrightarrow \mathcal{A}(E), \qquad \langle u \rangle \longmapsto \frac{\langle \mathrm{Norm}_{K/E}(u) \rangle}{[K : E]}.$$

It fixes the vectors from E. The fields $\mathbb{Q}(\zeta_p)$ and $\mathbb{Q}(\zeta_q)$ are linearly disjoint over $\mathbb{Q}$. In each summand of $\beta_q(p)$, the two factors are Galois conjugate in $\mathbb{Q}(\zeta_q)$, so N_p sends them to the same rational vector. Hence

$$(\Lambda^2 N_p)\beta_q(p) = 0, \qquad (\Lambda^2 N_p)\beta_p(q) = \beta_p(q).$$

The analogous statements hold with p, q exchanged. If the difference equals a rational wedge W, applying $\Lambda^2 N_p$ gives $W = -\beta_p(q)$. Applying $\Lambda^2 N_q$ to this equality gives $W = 0$, and both symbols vanish. The converse is immediate. The conventions for conductor 1 or 2 make the same argument valid in the degenerate field cases. □

Theorem 11.15 (Complete formal reduction criterion)**.** *For coprime positive integers p, q, the following are equivalent:*

1. *$\xi_{p/q}$ has a formal rational-dilogarithm reduction.*
2. *$\beta_q(p) = \beta_p(q) = 0$.*
3. *$p^2 \equiv \pm 1 \pmod{q}$ and $q^2 \equiv \pm 1 \pmod{p}$.*

Moreover, $\xi_{p/q}$ has a formal logarithmic reduction if and only if $p = 1$ or $q = 1$.

Proof. If $\xi_{p/q}$ is represented by rational dilogarithms, its differential is a rational wedge. Equations (11.8) and (11.17) imply (2). Conversely, if (2) holds, the differential is $-\langle p \rangle \wedge \langle q \rangle$. By (11.6) choose $\eta \in \mathcal{P}(\mathbb{Q})$ with this differential. Then $\xi_{p/q} - \eta$ is a Galois-invariant Bloch-group element and hence zero. The equivalence with (3) is Theorem 11.9.

For the last assertion, a normalized norm to $\mathbb{Q}$ sends every β-summand to zero on the exterior square. Thus $\partial\xi_{p/q} = 0$ implies $\langle p \rangle \wedge \langle q \rangle = 0$. If $p, q > 1$ are coprime, their prime-exponent vectors have disjoint nonempty supports, and their wedge is nonzero. If $p = 1$ or $q = 1$, all terms in (11.8) vanish; invariant Bloch descent gives the reduction. □

Remark 11.16 (Interpretation of a failed test). The theorem gives a necessary and sufficient condition for a certificate in a specified algebraic calculus. A nonzero symbol is not a proof that the corresponding real number satisfies no other numerical identity. Such a statement would require additional information about the kernel of period evaluation. In particular, the theorem should not be restated as an unconditional transcendence theorem.

11.7.1 Checking the boundary formula directly

Because the reduction criterion depends on (11.8), it is useful to audit that formula algebraically. The finite evaluation uses the formal summand

$$f(\alpha, \beta) = \left[\frac{\beta}{\beta - 1}\right] - \left[\frac{\alpha - \beta}{1 - \beta}\right],$$

summed over $\alpha^p = 1$, first over nontrivial qth roots β, and then with the analogous sum over nontrivial pth roots subtracted. Terms with $\alpha = 1$ or $\alpha = \beta$ have zero differential and are set aside before taking multiplicative logarithms. For the remaining roots of unity α, β, put $A = \langle 1 - \alpha \rangle$,

$B = \langle 1-\beta\rangle$, $C = \langle \alpha-\beta\rangle$. Away from the terms $[0], [1]$, which contribute zero, direct expansion gives

$$\partial f(\alpha,\beta) = -C\wedge A + C\wedge B + B\wedge A.$$

The differential of the subtracted summand is antisymmetric under $\alpha \leftrightarrow \beta$, so that sum has zero differential. In the first sum use

$$\prod_{\substack{\beta^q=1\\ \beta\neq 1}} (\alpha-\beta) = \frac{\alpha^q-1}{\alpha-1}, \qquad \prod_{\substack{\alpha^p=1\\ \alpha\neq 1}} (\beta-\alpha) = \frac{\beta^p-1}{\beta-1}.$$

The first two wedge terms then give $-\beta_p(q)$ and $\beta_q(p)$. The last gives $\langle q\rangle\wedge\langle p\rangle$. This proves (11.8) independently of any finite-field cocycle claim.

11.7.2 A recurrence parametrization of the reducible pairs

The congruence criterion has an elementary arithmetic description that connects it to the golden-ratio ladder thread of the repository.

Theorem 11.17 (Quadratic recurrence families)**.** *Apart from* $(1,1)$*, all coprime pairs* $1 \le p < q$ *satisfying* (11.10) *and Theorem 11.15 are exactly the adjacent pairs in the following three families. Their union need not be disjoint.*

1. *For each integer* $h \ge 2$*,* $u_0 = 1$*,* $u_1 = h$*,* $u_{j+1} = hu_j - u_{j-1}$*.*
2. $u_0 = 1$*,* $u_1 = 2$*,* $u_{j+1} = 3u_j - u_{j-1}$*.*
3. *For each integer* $h \ge 1$*,* $u_0 = 1$*,* $u_1 = h$*,* $u_{j+1} = hu_j + u_{j-1}$*, retaining pairs in increasing order.*

The first family has both square congruences $+1$*, the second both* -1*, and the third permits opposite signs, with the usual coincidences of signs modulo* 1 *or* 2*.*

Proof. Suppose $1 < a < b$ and write $a^2 \equiv s \pmod b$, $b^2 \equiv t \pmod a$ with $s,t \in \{\pm 1\}$. Set

$$c = \frac{a^2 - s}{b}.$$

Then $1 \le c < a$, $(c,a) = 1$, and the new pair (c,a) satisfies the same conditions with signs (t,s). Indeed $bc \equiv -s \pmod a$ implies $c^2 \equiv b^{-2} \equiv t \pmod a$, while $a^2 \equiv s \pmod c$. This gives a strictly descending process.

If $s = t$, the positive integer $h = (a^2+b^2-s)/(ab)$ is preserved under descent and satisfies $b = ha - c$. For $s = 1$, the terminal pair is $(1,h)$ with $h \ge 2$. For $s = -1$, a terminal pair $(1,b)$ has $b \mid 2$, so it is $(1,1)$ or $(1,2)$ and the invariant is $h = 3$.

If $s = -t$, the positive integer $h = (b^2-a^2-t)/(ab)$ is preserved and satisfies $b = ha + c$. The terminal pairs are seeds $(1,h)$, with the terminal $(1,2)$ also lying in the $h = 1$ sequence $1,1,2,3,5,\ldots$. Reversing descent gives the three displayed recurrences. Conversely their preserved quadratic identities, or the same recurrence step in reverse, verify both congruences and coprimality at every stage. □

For example, the second family is $1,2,5,13,34,\ldots$, the odd-index Fibonacci numbers. The first family with $h = 3$ is $1,3,8,21,55,\ldots$, and the third with $h = 3$ is $1,3,10,33,109,\ldots$. This is an arithmetic parametrization of *which* rational arguments admit formal reduction, not an assertion that all their logarithmic correction terms are already tabulated.

11.7.3 An explicit rational-dilogarithm core

Theorem 11.18 (A short core with a boundary certificate)**.** *For every pair satisfying Theorem 11.15, one can produce a rational combination $C_{p/q}$ of symbols $[z]$, $0 < z < 1$, such that*

$$\partial C_{p/q} = -\langle p\rangle \wedge \langle q\rangle, \qquad \xi_{p/q} - C_{p/q} = 0 \quad \text{in } \mathcal{P}(K).$$

The following recursion suffices for $1 \le p < q$:

1. *If $p = 1$, use $C_{p/q} = 0$.*
2. *If $q = p + 1$, use $C_{p/q} = [p/q]$.*
3. *Otherwise choose $s \in \{\pm 1\}$ with $q \mid p^2 - s$, put $c = (p^2 - s)/q$, and use*

$$C_{p/q} = C_{c/p} + \begin{cases} -\frac{1}{2}[1 - p^{-2}], & s = 1, \\ \frac{1}{2}[p^2/(p^2+1)], & s = -1. \end{cases} \tag{11.18}$$

Put $C_{1/1} = 0$. For $p > q$, negate the core for q/p. After the consecutive-pair shortcut, the number of terms is $O(\log \max(p, q))$.

Proof. The first two cases have the required differential. In the last case $cq = p^2 - s$. Direct calculation gives

$$\begin{aligned} -\tfrac{1}{2}\partial[1 - p^{-2}] &= -\langle p\rangle \wedge \langle q\rangle - \langle p\rangle \wedge \langle c\rangle && (s = 1), \\ \tfrac{1}{2}\partial[p^2/(p^2+1)] &= -\langle p\rangle \wedge \langle q\rangle - \langle p\rangle \wedge \langle c\rangle && (s = -1). \end{aligned}$$

Adding $\partial C_{c/p} = -\langle c\rangle \wedge \langle p\rangle$ leaves the required $-\langle p\rangle \wedge \langle q\rangle$. Descent terminates by the previous theorem. Both $C_{p/q}$ and $\xi_{p/q}$ are invariant; their difference lies in the Bloch group and vanishes by (11.6).

For the length bound, the $h = 2$ minus recurrence consists of consecutive integers and is handled in one step. Every remaining minus recurrence has $h \ge 3$ and grows by a factor greater than 2 after its seed. Each plus recurrence grows at least as fast as the Fibonacci recurrence. Thus the number of descent steps is logarithmic in the larger input. □

Argument	A rational-dilogarithm core for $F(p/q) - F(1)$
2/5	$\frac{1}{2}\operatorname{Li}_2(4/5)$
3/8	$-\frac{1}{2}\operatorname{Li}_2(8/9)$
4/17	$\frac{1}{2}\operatorname{Li}_2(16/17)$
5/13	$\frac{1}{2}\operatorname{Li}_2(4/5) + \frac{1}{2}\operatorname{Li}_2(25/26)$
10/33	$\frac{1}{2}\operatorname{Li}_2(9/10) - \frac{1}{2}\operatorname{Li}_2(99/100)$

In this table the difference between $F(p/q) - F(1)$ and the displayed core is a combination of logarithm products and π^2. The core algorithm certifies the dilogarithm part; it does not yet print the full logarithmic correction. Constructing that correction efficiently and with a complete branch certificate is a further research task.

Corollary 11.19 (A lower bound for a family of formal values)**.** *Fix $q \ge 3$ and choose one positive numerator representing each nontrivial inverse pair in $\mathcal{G}_q$. The corresponding $d(q)$ elements $\xi_{a/q}$ are linearly independent modulo rational dilogarithms.*

Proof. Take a common cyclotomic field for all numerators and q, and apply the exterior square of its normalized norm to $\mathbb{Q}(\zeta_q)$. The numerator-conductor symbols disappear and the denominator symbols $\beta_q(a)$ remain. If their combination were a rational wedge, the normalized norm to $\mathbb{Q}$ would kill it, so it would be zero. The chosen $\beta_q(a)$ are independent by Theorem 11.9. □

11.8 Complete classification and evaluations of $J(2/q)$

Define

$$S(n) = \sum_{k=1}^{n-1} \log^2\left(2\sin\frac{\pi k}{n}\right), \qquad S(1) = 0, \qquad \phi = \frac{1+\sqrt{5}}{2}.$$

The reciprocal-integer identity, obtainable by pairing conjugate roots in the finite dilogarithm formula, is

$$F(1/n) - F(1) = -\frac{(n-1)(3n-2)}{24n}\pi^2 - \frac{1}{2}S(n), \qquad n \geq 1. \tag{11.19}$$

All logarithms here are real logarithms of positive algebraic numbers.

Theorem 11.20 (Complete formal classification)**.** *For integers $q \geq 2$:*

1. *$J(2/q)$ has a formal logarithmic reduction exactly when $q \in \{2,3,5\}$ or $4 \mid q$.*
2. *It has a formal rational-dilogarithm reduction exactly when $q \in \{2,3,5,6,10\}$ or $4 \mid q$.*

In every positive case the evaluations below give the corresponding analytic identity.

Proof. First suppose q is odd. Applying (11.8) to (11.1) at $x = 2/q$, the numerator symbols at conductors $1, 2, 4$ vanish and the rational wedges cancel. Thus the boundary is

$$\beta_q(4) - 2\beta_q(2).$$

Put $g = 2 \in \mathcal{G}_q$. By Theorem 11.9, this vanishes exactly when g has order at most 2. To check the assertion directly, an element of order 3 gives $\beta(g^2) - 2\beta(g) = -3\beta(g)$; order 4 gives $-2\beta(g)$; for order at least 5, the two relevant inverse pairs are distinct. Hence vanishing is equivalent to $4 \equiv \pm 1 \pmod{q}$, or $q = 3, 5$. A rational wedge cannot remove a nonzero combination of these symbols, since the normalized norm to $\mathbb{Q}$ kills the combination and fixes rational wedges. This proves both classifications for odd q.

If $4 \mid q$, all three arguments $4/q, 2/q, 1/q$ in (11.1) are reciprocal integers. Equation (11.19) gives a logarithmic reduction.

The remaining case is $q = 2m$ with m odd. Its boundary is

$$\beta_m(2) - \langle 2\rangle \wedge \langle m\rangle. \tag{11.20}$$

For $m = 1$ it is zero. For $m > 1$, applying the normalized norm to $\mathbb{Q}$ gives the nonzero rational wedge $-\langle 2\rangle \wedge \langle m\rangle$; there is no formal logarithmic reduction. Rational dilogarithms are allowed exactly when $\beta_m(2) = 0$, equivalently $m = 3, 5$. Invariant Bloch descent supplies the stated formal reductions. □

11.8.1 All positive evaluations

The isolated cases are

$$J(1) = \tfrac{1}{2}\log^2 2, \tag{11.21}$$

$$J(2/3) = -\frac{\pi^2}{144} + \frac{3}{4}\log^2 2, \tag{11.22}$$

$$J(2/5) = \frac{11\pi^2}{240} + \frac{3}{4}\log^2 2 - 2\log^2\phi, \tag{11.23}$$

$$J(1/3) = \frac{\pi^2}{9} + \frac{1}{2}\log^2 2 - \frac{1}{2}\log^2 3 - \mathrm{Li}_2(1/3), \tag{11.24}$$

$$J(1/5) = \frac{7\pi^2}{60} + \frac{1}{2}\log^2 2 - \frac{1}{4}\log^2 5 - \log^2\phi - \frac{1}{2}\mathrm{Li}_2(1/5). \tag{11.25}$$

For all even $m \geq 2$, the infinite family is

$$\boxed{J(1/m) = \frac{\pi^2(m^2-2)}{48m} + S(m) - \frac{1}{2}S(m/2) - \frac{1}{2}S(2m).} \tag{11.26}$$

Thus (11.26) evaluates $J(2/q)$ for every multiple q of 4, without a dilogarithm term.

Derivation of the evaluations. Equation (11.21) follows by integrating $\log(1+t)/(1+t)$. For (11.26), substitute (11.19) at $n = m/2, m, 2m$ in (11.1). The constant $F(1)$ cancels; the polynomial coefficient simplifies to $m/48 - 1/(24m)$, yielding the formula.

The small odd cases are obtained from the finite root-of-unity dilogarithm formula and the reflection, duplication, and Landen relations. For clarity, the two fractional F evaluations needed in the calculation can be taken as

$$\begin{aligned} F(2/3) - F(1) &= \frac{\pi^2}{36} + \log^2 2 - \frac{3}{4}\log^2 3 - \mathrm{Li}_2(1/3), \\ F(2/5) - F(1) &= \tfrac{1}{2}\,\mathrm{Li}_2(4/5) - \frac{\pi^2}{5} + \log^2\!\left(2\sin\frac{\pi}{5}\right) + \log^2\!\left(2\sin\frac{2\pi}{5}\right) + \log 2\log(2/5). \end{aligned}$$

The second is the previously known evaluation [35]; the first follows by using $\cot^2(\pi/3) = 1/3$ in the $F(2/q)$ sum and $\mathrm{Li}_2(-1/3) = 2\,\mathrm{Li}_2(1/3) - \pi^2/6 + \frac{1}{2}\log^2 3$. Substitution in (11.1), together with (11.19) and the two standard functional equations

$$\begin{aligned} F(x) + F(1/x) &= 2F(1) + \tfrac{1}{2}\log^2 x - \frac{\pi^2(x-1)^2}{6x}, \\ F(x) - F(x+1) - F(x/(x+1)) &= -F(1) + \mathrm{Li}_2(1/(1+x)), \end{aligned}$$

gives (11.22)–(11.25). Reflection of $\mathrm{Li}_2(4/5)$ supplies the $\mathrm{Li}_2(1/5)$ in (11.25). All manipulations use arguments in $(0,1)$ or the real negative axis, so no unstated complex logarithm branches are involved. □

The first member of (11.26) is particularly useful for auditing the earlier claim:

$$J(1/2) = \frac{\pi^2}{48} + \frac{1}{4}\log^2 2. \tag{11.27}$$

The complete classification proves a precise replacement for the unsupported numerical "if and only if" in the draft. It also shows that the distinction between logarithms and rational dilogarithms is real at the level of the boundary: the cases $q = 6, 10$ have a nonzero rational wedge, even though their cyclotomic obstruction vanishes.

11.9 Bridges to the golden ladders and the Clausen lattice

The new constants are not isolated; they touch two corpora we have already mapped.

The golden bridge ($q = 5$). The arguments $-\cot^2\frac{\pi}{5} = -\frac{5+2\sqrt{5}}{5}$ and $-\cot^2\frac{2\pi}{5} = -\frac{5-2\sqrt{5}}{5}$ lie in $\mathbb{Q}(\sqrt{5})$, the field of the golden polylogarithm ladders. Comparing Theorem 11.3 ($q = 5$) with eq. 24 gives the functional equation

$$\mathrm{Li}_2\!\left(\tfrac{4}{5}\right) + \mathrm{Li}_2\!\left(-\cot^2\tfrac{\pi}{5}\right) + \mathrm{Li}_2\!\left(-\cot^2\tfrac{2\pi}{5}\right) = 2\log^2 2 - 2\log 2\log\tfrac{2}{5} - 6\big(\log^2(2\sin\tfrac{\pi}{5}) + \log^2(2\sin\tfrac{2\pi}{5})\big),$$

hence the golden-sum closed form (with $\varphi = \frac{1+\sqrt{5}}{2}$)

$$\mathrm{Li}_2\Big(-\cot^2\tfrac{\pi}{5}\Big) + \mathrm{Li}_2\Big(-\cot^2\tfrac{2\pi}{5}\Big) = \mathrm{Li}_2\Big(\tfrac{1}{5}\Big) - \frac{\pi^2}{6} - 3\log^2\varphi + \tfrac{1}{4}\log^2 5.$$

The symmetric sum has the displayed reduction. The antisymmetric combination $X = \frac{1}{2}[\mathrm{Li}_2(-\cot^2(\pi/5)) - \mathrm{Li}_2(-\cot^2(2\pi/5))]$ did not reduce in the source's tested basket. Its independence is open. A formal symbol with nonzero Bloch boundary is not subject to a vanishing-rank conclusion for the Bloch group.

The Clausen bridge. The coefficients of $W(q)$ are real algebraic numbers contained in $\mathbb{Q}(\zeta_{\mathrm{lcm}(4,q)})^+$; at odd conductors the factor i implicit in the cotangent can require an extension of $\mathbb{Q}(\zeta_q)^+$. This is an algebraic coefficient issue, not a proof of numerical period independence. Applying the distribution identities of Chapter 2 gives

$$W(8) = 2\,\mathrm{Cl}_2(\pi/4) - 2\,\mathrm{Cl}_2(3\pi/4) = G, \qquad W(12) = \frac{11\sqrt{3}}{9}\,\mathrm{Cl}_2(\pi/3).$$

For the second formula, pair k with $12-k$:

$$W(12) = 2\sqrt{3}\,[\mathrm{Cl}_2(\pi/6) - \mathrm{Cl}_2(5\pi/6)] + \frac{2}{\sqrt{3}}\,[\mathrm{Cl}_2(\pi/3) - \mathrm{Cl}_2(2\pi/3)]\,.$$

Use $\mathrm{Cl}_2(\pi/6) - \mathrm{Cl}_2(5\pi/6) = V/2$ and $\mathrm{Cl}_2(2\pi/3) = 2V/3$. This proves the original report's coefficient $22\sqrt{3}/9$ multiplying the first difference. It is also checked against direct derivative quadrature in the fresh verification battery.

11.10 Positive measures and sharp approximation

Binet's kernel, already encountered in the gamma-integral chapter, gives the Herglotz function a positive-measure representation. The same representation certifies truncation errors and explains the optimal cutoff's arithmetic fluctuations. These analytic results complement the exact symbol calculus without relying on period conjectures.

11.11 A positive measure and certified remainder bounds

The analytic part begins with a representation already obtained by Radchenko and Zagier [35, §7.3]. We derive its consequences explicitly, keeping signs and normalizations visible. Put

$$G(x) = -F(x) - \frac{\zeta(2)}{2x}, \qquad \mathcal{L}(u) = \sum_{m=1}^{\infty} \sigma_{-1}(m) e^{-2\pi m u}, \qquad \sigma_{-1}(m) = \sum_{d\mid m} \frac{1}{d}. \tag{11.28}$$

Expanding a geometric series gives

$$\mathcal{L}(u) = \sum_{n=1}^{\infty} \frac{1}{n(e^{2\pi n u} - 1)}.$$

Binet's digamma integral, followed by a positive change of variables and Tonelli's theorem, yields

$$G(x) = 2\int_0^{\infty} \frac{u\mathcal{L}(u)}{x^2+u^2}\,du = \int_0^{\infty} \frac{d\mu(s)}{x^2+s}, \qquad d\mu(s) = \mathcal{L}(\sqrt{s})\,ds. \tag{11.29}$$

Thus $y \mapsto G(\sqrt{y})$ is a Stieltjes transform of a positive measure with a positive density on $(0,\infty)$.

Lemma 11.21 (Explicit moments)**.** *Every nonnegative moment exists, and*

$$\mu_j := \int_0^\infty s^j \, d\mu(s) = \frac{2\Gamma(2j+2)\zeta(2j+2)\zeta(2j+3)}{(2\pi)^{2j+2}} = \frac{|B_{2j+2}|}{2j+2}\zeta(2j+3). \tag{11.30}$$

Proof. Substitute $s = u^2$, integrate each positive exponential term, and use $\sum_m \sigma_{-1}(m)m^{-w} = \zeta(w)\zeta(w+1)$ for $w > 1$. Euler's formula for the even zeta values gives the second equality. □

Define the approximation and its first omitted term by

$$F_N(x) = -\frac{\zeta(2)}{2x} - \sum_{j=0}^{N-1} \frac{(-1)^j \mu_j}{x^{2j+2}}, \qquad T_N(x) = \frac{\mu_N}{x^{2N+2}}, \qquad N \geq 0. \tag{11.31}$$

The empty sum for $N = 0$ is zero. The series is asymptotic and divergent; we will work with its exact finite remainder.

Theorem 11.22 (Signed remainder and three-moment bounds)**.** *For $x > 0$ and $N \geq 0$,*

$$F(x) - F_N(x) = (-1)^{N+1} R_N(x), \tag{11.32}$$

$$R_N(x) = x^{-2N} \int_0^\infty \frac{s^N}{x^2+s} \, d\mu(s) > 0, \tag{11.33}$$

$$R_N(x) + R_{N+1}(x) = T_N(x). \tag{11.34}$$

Set

$$A = \frac{\mu_{N+1}}{\mu_N x^2}, \qquad B = \frac{\mu_{N+2}}{\mu_N x^4}.$$

Then the strict bounds

$$\frac{1}{1+A} < \frac{R_N(x)}{T_N(x)} < 1 - \frac{A^2}{A+B} < 1 \tag{11.35}$$

hold. In particular, two consecutive truncations bracket $F(x)$.

Proof. The finite geometric identity

$$\frac{1}{x^2+s} = \sum_{j=0}^{N-1} \frac{(-s)^j}{x^{2j+2}} + \frac{(-1)^N s^N}{x^{2N}(x^2+s)}$$

gives (11.32)–(11.33), and multiplying the remainder integrand by $1 + s/x^2$ gives (11.34).

Under the probability measure $s^N \, d\mu(s)/\mu_N$ let $U = s/x^2$. Then $R_N/T_N = \mathbb{E}(1+U)^{-1}$, $\mathbb{E}U = A$, and $\mathbb{E}U^2 = B$. Strict Jensen convexity gives the lower bound. Cauchy–Schwarz gives

$$\mathbb{E}\frac{U}{1+U} \; \mathbb{E}\big(U(1+U)\big) \geq (\mathbb{E}U)^2,$$

so $\mathbb{E}(1+U)^{-1} \leq 1 - A^2/(A+B)$. Both inequalities are strict because the measure has a nonconstant positive-density support. □

For interval computation, an equivalent and useful form is

$$L_N(x) = \frac{x^{-2N}\mu_N^2}{x^2\mu_N + \mu_{N+1}},$$

$$L_N(x) < R_N(x) < T_N(x) - L_{N+1}(x). \tag{11.36}$$

It uses only three adjacent moments, and hence only rational Bernoulli numbers and odd zeta values. No oscillatory integral is needed.

11.11.1 An exact rational certification scheme

For integer $s \geq 2$, integer $M \geq 2$, and $J \geq 1$, Euler–Maclaurin gives

$$Z_{M,J}(s) = \sum_{n=1}^{M-1} n^{-s} + \frac{M^{1-s}}{s-1} + \frac{1}{2M^s} + \sum_{j=1}^{J-1} \frac{B_{2j}}{(2j)!}(s)_{2j-1} M^{-s-2j+1}, \tag{11.37}$$

$$|\zeta(s) - Z_{M,J}(s)| \leq \frac{|B_{2J}|}{(2J)!}(s)_{2J-1} M^{-s-2J+1}. \tag{11.38}$$

Here $(s)_k$ is a rising factorial. A direct positive-kernel proof is useful because it fixes the endpoint convention in this bound. The partial fraction expansion of the hyperbolic cotangent gives

$$k(t) := \frac{1}{1-e^{-t}} - \frac{1}{t} - \frac{1}{2} = 2t \sum_{n \geq 1} \frac{1}{t^2 + (2\pi n)^2}, \qquad t > 0.$$

Expand each denominator through $J-1$ terms by the finite geometric identity. After summing, the result is

$$k(t) = \sum_{j=1}^{J-1} \frac{B_{2j}}{(2j)!} t^{2j-1} + \varepsilon_J(t), \qquad |\varepsilon_J(t)| \leq \frac{|B_{2J}|}{(2J)!} t^{2J-1}.$$

Indeed the remainder before summing is $(-1)^{J-1} 2t^{2J-1}/((2\pi n)^{2J-2}((2\pi n)^2 + t^2))$, whose magnitude is bounded by replacing the last denominator by $(2\pi n)^2$. Integrate against $t^{s-1}e^{-Mt}/\Gamma(s)$, using

$$\sum_{n=M}^{\infty} n^{-s} = \frac{1}{\Gamma(s)} \int_0^{\infty} \frac{t^{s-1}e^{-Mt}}{1-e^{-t}}\, dt.$$

This proves (11.37)–(11.38), including the first-omitted-term bound, rather than assuming a truncation-sign rule. Related sharper Hurwitz-zeta remainder bounds are discussed in [50].

All quantities in (11.37)–(11.38) are rational. Combining their enclosing intervals with (11.30), (11.31), and (11.36) yields exact rational endpoints for $F(x)$ when x is rational. The code uses interval operations with integer numerators and denominators throughout; decimal endpoints are rounded outward only for display.

x	N	Certified width of the interval for $F(x)$, at most
2	6	$4.18751510340 \times 10^{-7}$
4	12	$4.88272699850 \times 10^{-13}$
8	25	$2.15471915709 \times 10^{-24}$
16	50	$1.10232929493 \times 10^{-46}$

These are reproducible certificates, not estimates based on the number of floating-point digits. The file `data/rational_intervals.json` contains the complete exact endpoints. The parameters here are $M = J = 48$; no optimality of that computational choice is asserted.

11.11.2 Classical Stieltjes consequences

The representation also settles the qualitative approximation question without any new special-function identity.

Proposition 11.23 (Determinacy and positive rational approximation)**.** *The measure μ is uniquely determined by its moments. Gaussian quadrature for μ produces lower bounds that are rational functions of y for $G(\sqrt{y})$, $y > 0$; Gauss–Radau quadrature with a prescribed node at 0 produces rational upper bounds. These bounds converge pointwise to $G(\sqrt{y})$ as their orders tend to infinity.*

Proof. Stirling's formula in (11.30) gives $\mu_j^{-1/(2j)} \sim \pi e/j$. Thus the Stieltjes Carleman sum diverges, which implies determinacy. The quadrature error signs follow from the alternating derivatives of $(y+s)^{-1}$: Gaussian quadrature has a positive even-order remainder, while the Radau rule with left endpoint 0 has a negative odd-order remainder. For convergence, the discrete positive quadrature measures have the same total mass and, eventually, every fixed finite set of moments as μ. A first-moment bound gives tightness; higher moment bounds give uniform integrability of each fixed power. Every subsequential weak limit therefore has all the moments of μ and must equal μ by determinacy. Integration of the bounded continuous function $(y+s)^{-1}$ proves the claim. These are standard Stieltjes and Padé consequences; see [51]. □

The value of the proposition here is practical: it explains why positive rational approximations are available even in parameter regimes where an asymptotic series is not an efficient evaluator. An effective convergence rate for this particular measure remains a separate problem.

11.12 Uniform sharp asymptotics at the optimal scale

The scale $N \sim \pi x$ can be located from the ratio of consecutive moments. To obtain the error constant and its corrections, it is more efficient to convert the exact remainder into gamma expectations. Write

$$a = 2\pi x, \quad K = 2N + 2, \quad r = K - a = 2\delta + 2, \quad \delta = N - \pi x. \tag{11.39}$$

Let T have gamma density $t^{K-1}e^{-t}/\Gamma(K)$ on $(0,\infty)$, and define

$$E_K(c) = \mathbb{E}\frac{1}{1+(T/c)^2}, \qquad w_m = \sigma_{-1}(m)m^{-K}, \qquad Z_K = \sum_{m\geq 1} w_m = \zeta(K)\zeta(K+1).$$

Lemma 11.24 (Exact gamma mixture)**.** *For all $x > 0$ and $N \geq 0$,*

$$R_N(x) = \frac{2\Gamma(K)}{a^K}\sum_{m=1}^{\infty} w_m E_K(ma), \qquad T_N(x) = \frac{2\Gamma(K)}{a^K}Z_K. \tag{11.40}$$

For $K \geq 4$,

$$0 < Z_K - 1 \leq 4\,2^{-K}. \tag{11.41}$$

Proof. In each positive exponential summand of (11.33), written in the variable $u = \sqrt{s}$, substitute $t = 2\pi m u$. Collecting powers gives (11.40). For the tail, use $\sigma_{-1}(m) \leq m$ and

$$\sum_{m\geq 2} m^{1-K} \leq 2^{1-K} + \frac{2^{2-K}}{K-2} \leq 4\,2^{-K}.$$

□

The arithmetic terms $m \geq 2$ therefore do not affect any algebraic coefficient in the optimal-scale expansion. They do, however, move the exact transition point by an exponentially small amount; this effect is treated separately in Section 11.14.

11.12.1 Gamma moments and a uniform remainder mechanism

Let $f(v) = (1+v^2)^{-1}$ and $H = (T-a)/a$. The local Taylor expansion is

$$f(1+H) = \frac{1}{2} - \frac{H}{2} + \frac{H^2}{4} - \frac{H^4}{8} + \frac{H^5}{8} - \frac{H^6}{16} + \cdots,$$

and the moments are exactly

$$\mathbb{E}H^\ell = a^{-\ell} \sum_{j=0}^{\ell} \binom{\ell}{j} (-a)^{\ell-j} (a+r)_j. \tag{11.42}$$

This is an algebraic coefficient algorithm, not a fit to numerical data.

For rigor at any fixed order J, Taylor-expand f through degree $2J+1$. Every derivative of f is bounded on $[0,\infty)$, so the absolute remainder is bounded by a constant times $|H|^{2J+2}$. The centered gamma cumulants are $K(j-1)!$ in orders $j \geq 2$. Consequently each centered even moment of order $2J+2$ is a polynomial in K of degree at most $J+1$. Translating the center from K to $a = K - r$, with r bounded, gives

$$\mathbb{E}|H|^{2J+2} = O_J(a^{-J-1})$$

uniformly on each compact interval of r. This proves the required uniform remainder after taking expectations. In particular,

$$\begin{aligned} 2E_K(a) = 1 &+ \frac{1/2 - r}{a} + \frac{(r^2+r)/2 - 3/4}{a^2} + \frac{-3r^2/2 + r/4 + 13/8}{a^3} \\ &- \frac{4r^4 - 16r^3 - 66r^2 + 74r + 59}{16a^4} + O_D(a^{-5}). \end{aligned} \tag{11.43}$$

The subscript D signifies that the constant may depend on a fixed bound for $|r|$; the same convention will be used for δ.

The classical shifted Stirling expansion [49, §5.11.8] is

$$\frac{\Gamma(a+r)}{a^{a+r}} \sim \sqrt{2\pi}\, a^{-1/2} e^{-a} \exp\left(\sum_{j=1}^{\infty} \frac{(-1)^{j+1} B_{j+1}(r)}{j(j+1)a^j} \right). \tag{11.44}$$

It is uniform for r in compact sets. Combining its rigorous finite forms with the gamma-moment argument proves the following theorem.

Theorem 11.25 (Complete optimal-scale expansion)**.** *For each fixed $D > 0$ and each integer $J \geq 0$, there are explicitly computable rational polynomials $C_j(\delta)$ such that, uniformly for integers $N \geq 0$ with $|N - \pi x| \leq D$,*

$$R_N(x) = \frac{e^{-2\pi x}}{\sqrt{x}} \left(\sum_{j=0}^{J} \frac{C_j(\delta)}{(2\pi x)^j} + O_{D,J}(x^{-J-1}) \right). \tag{11.45}$$

The first three are

$$C_0(\delta) = 1,$$

$$C_1(\delta) = 2\delta^2 + \delta - \frac{5}{12}, \tag{11.46}$$

$$C_2(\delta) = 2\delta^4 + \frac{2}{3}\delta^3 - \frac{10}{3}\delta^2 - \frac{7}{12}\delta + \frac{205}{288}. \tag{11.47}$$

Moreover, $R_N(x)/T_N(x) \longrightarrow 1/2$ *uniformly in that regime.*

Proof. Use (11.42) to expand $2E_K(a)$, multiply by (11.44), and put $r = 2\delta + 2$. The frequency tail (11.41) is smaller than every power of a^{-1}. The leading factor is $\sqrt{2\pi}\, a^{-1/2}e^{-a} = x^{-1/2}e^{-2\pi x}$. For an explicit check of the first two coefficients, put $g_1 = B_2(r)/2$ and $g_2 = -B_3(r)/6$. Before substituting for r they are

$$g_1 + \tfrac{1}{2} - r, \qquad \tfrac{1}{2}g_1^2 + g_2 + g_1(\tfrac{1}{2} - r) + \tfrac{1}{2}(r + r^2) - \tfrac{3}{4}.$$

These simplify to (11.46)–(11.47). The ratio assertion follows directly from (11.40), since $E_K(a) \to 1/2$ and $Z_K \to 1$. □

The leading constant is therefore exactly 1. Replacing the true remainder by the first omitted term loses a factor of 2 at the optimal scale. The three-moment bounds also approach this factor: there $A \to 1$ and $B \to 1$, so both sides of (11.35) tend to $1/2$.

The program `code/truncation_coefficients.py` generates C_j symbolically using rational polynomial operations and Bernoulli polynomials. The theorem's bounded-δ condition is essential to the stated uniformity. A growing window, for example $N - \pi x$ of order $\sqrt{x}$, requires a different scaling.

11.13 The exact best cutoff and its transitions

An asymptotic nearest-integer rule is useful, but it cannot determine the best cutoff arbitrarily close to a transition. The positive measure supplies an exact answer.

Theorem 11.26 (Unique transitions and global minimizers)**.** *For every* $N \geq 0$*, there is a unique* $x_N > 0$ *such that* $R_N(x_N) = R_{N+1}(x_N)$*. The sequence* x_N *is strictly increasing and tends to infinity. With* $x_{-1} = 0$*, the index* N *is a global minimizer of* $R_j(x)$*,* $j \geq 0$*, exactly when*

$$x_{N-1} \leq x \leq x_N. \tag{11.48}$$

There are exactly two adjacent minimizing indices at an endpoint and exactly one elsewhere on $x > 0$*. An exact transition equation is*

$$x_N^2 \int_0^\infty \frac{s^N}{x_N^2 + s}\, d\mu(s) = \frac{\mu_N}{2}. \tag{11.49}$$

Proof. Put $y = x^2$ and $M_N(y) = \int s^N (y+s)^{-1}\, d\mu(s)$. Strict Cauchy–Schwarz gives

$$M_{N+1}(y)^2 < M_N(y) M_{N+2}(y).$$

Hence $R_N = y^{-N} M_N(y)$ is strictly log-convex as a sequence in N, and its consecutive ratios are strictly increasing. The recurrence $M_{N+1} = \mu_N - yM_N$ gives

$$\frac{R_{N+1}(x)}{R_N(x)} = \frac{\mu_N}{yM_N(y)} - 1. \tag{11.50}$$

By dominated convergence, $yM_N(y) = \int s^N y/(y+s)\,d\mu(s)$ increases continuously and strictly from 0 to μ_N as y increases from 0 to infinity. The ratio in (11.50) thus decreases continuously from infinity to 0, proving the unique transition and (11.49).

Strict increase of the ratios in N forces $x_{N+1} > x_N$. For fixed y, the moment ratios M_{N+1}/M_N tend to infinity. Indeed, if their increasing sequence were bounded by B, the moments would be at most $M_0 B^N$, contradicting positive mass on an interval strictly to the right of B. This also proves $x_N \to \infty$ and existence of a global minimum. The strict ratio monotonicity now gives (11.48) and the assertions about ties. □

To expand the transition, let α_K be the unique positive root of

$$E_K(\alpha_K) = \frac{1}{2}, \tag{11.51}$$

and write $\beta_K = 2\pi x_N$ for $K = 2N + 2$. The full transition is the unique root of

$$\Phi_K(a) := \sum_{m \geq 1} w_m \left(E_K(ma) - \frac{1}{2} \right) = 0. \tag{11.52}$$

The symbols α_K, β_K in this analytic section are real numbers; the subscript and argument distinguish β_K from the cyclotomic symbol $\beta_q(a)$.

Theorem 11.27 (Algebraic transition expansion)**.** *Put $b = K - \frac{1}{2} = 2N + \frac{3}{2}$. As $N \to \infty$,*

$$\alpha_K = b + \frac{3}{8b} - \frac{1}{b^2} + \frac{407}{128b^3} + O(b^{-4}). \tag{11.53}$$

The same expansion holds for $\beta_K = 2\pi x_N$, and the two quantities have identical expansions to every algebraic order.

Proof. The gamma law gives $T/K \to 1$ in probability. Evaluating at $a = cK$ shows that both roots have $a/K \to 1$. More precisely, substitution of $a = K + C$ for fixed constants on either side of $C = -1/2$ in (11.43) brackets both roots for large K; the frequency tail does not change these opposite signs. Therefore $r = K - a$ is bounded at the roots, and the uniform expansion applies.

Solve $2E_K(a) = 1$ from (11.43) to obtain

$$r = \frac{1}{2} - \frac{3}{8a} + \frac{1}{a^2} - \frac{425}{128a^3} + O(a^{-4}).$$

For example, putting $r = 1/2 + q_1/a + q_2/a^2 + q_3/a^3$ gives successively $q_1 = -3/8$, $q_2 = 1$, $q_3 = -425/128$. Since $\partial_a[2E_K(a)]$, with K fixed, is $1/K + O(K^{-2})$ near the root (see (11.56)), a residual of order a^{-j-1} locates the root to order a^{-j}. Thus the formal inversion has the asserted error control. Reexpressing it in $b = K - 1/2$ yields (11.53). Finally, the perturbing tail in (11.52) is $O(2^{-K})$, while its derivative is of order $1/K$; it changes the root by $O(K2^{-K})$. This proves the last assertion. □

To leading order, choose the integer nearest to $\pi x - 1/4$. The exact prescription remains (11.48); no rounding formula with an unspecified remainder resolves every near-tie case.

11.14 An arithmetic displacement beyond all algebraic orders

The comparison in this section is between two exactly defined transition points. Subtracting a finite approximation such as $b + 3/(8b) - 1/b^2$ would leave an algebraic error much larger than the effect we want to measure.

Theorem 11.28 (First arithmetic shift)**.** *For every $K = 2N + 2 \geq 2$, the full transition satisfies $0 < \beta_K < \alpha_K$. As $K \to \infty$,*

$$\boxed{\beta_K - \alpha_K = -\frac{9}{10} K 2^{-K} \left(1 + O(K^{-1}) + O((2/3)^K)\right).} \tag{11.54}$$

Equivalently, with $x_N^{(1)} = \alpha_K/(2\pi)$, $x_N - x_N^{(1)}$ has leading term $-9K2^{-K}/(20\pi)$.

Proof. The function $E_K(c)$ is strictly increasing in c. At $a = \alpha_K$, the $m = 1$ term of (11.52) vanishes, while every $m \geq 2$ term is positive. Hence $\Phi_K(\alpha_K) > 0$. Since Φ_K is strictly increasing, its root lies below α_K.

For the coefficient, $\alpha_K = K + O(1)$ and gamma concentration with its second-moment bound gives

$$E_K(2\alpha_K) = \frac{4}{5} + O(K^{-1}).$$

Also $w_2 = (3/2)2^{-K}$ and $\sum_{m \geq 3} w_m = O(3^{-K})$. Therefore

$$\Phi_K(\alpha_K) = \frac{9}{20} 2^{-K} \left(1 + O(K^{-1}) + O((2/3)^K)\right). \tag{11.55}$$

Differentiation under the expectation gives

$$E_K'(a) = \frac{1}{a} \mathbb{E}h(T/a), \qquad h(v) = \frac{2v^2}{(1+v^2)^2}.$$

On a bounded neighborhood of K, $h(1) = 1/2$ and a second-moment expansion yield $E_K'(a) = (2K)^{-1}(1 + O(K^{-1}))$. For any m, the derivative of $E_K(ma)$ with respect to a is $a^{-1}\mathbb{E}h(T/(ma))$, and $0 < h \leq 1/2$. Thus

$$\Phi_K'(a) = \frac{1}{2K}\big(1 + O(K^{-1}) + O(2^{-K})\big) \tag{11.56}$$

uniformly in that neighborhood. In particular it has a positive lower bound of order K^{-1}. The mean value theorem first gives $\beta_K - \alpha_K = O(K2^{-K})$ and then, using (11.55)–(11.56), gives (11.54). □

The constant has a transparent decomposition:

$$\frac{9}{10} = \underbrace{\frac{3}{2}}_{\substack{\text{divisor weight} \\ \text{at } m=2}} \underbrace{\left(\frac{4}{5} - \frac{1}{2}\right)}_{\text{kernel gap}} \Big/ \underbrace{\frac{1}{2}}_{\text{scaled derivative}}.$$

It records the second Lambert frequency, rather than a correction to the Stirling expansion.

Remark 11.29 (A transferable positive-mixture calculation). The same argument applies to a positive exponential mixture with frequencies $1, \rho, \ldots$, $\rho > 1$, and coefficients $c_1, c_\rho > 0$. If its higher-frequency Dirichlet tail is $o(\rho^{-K})$, the corresponding normalized transition displacement is

$$-\frac{c_\rho}{c_1}\frac{\rho^2 - 1}{\rho^2 + 1} K \rho^{-K}(1 + o(1)).$$

This follows by replacing the kernel limit $4/5$ by $\rho^2/(1 + \rho^2)$ in the proof. It is a useful extension of the calculation, rather than a claim to a new general method of exponentially improved asymptotics; see [49, §2.11] for the broader established theory.

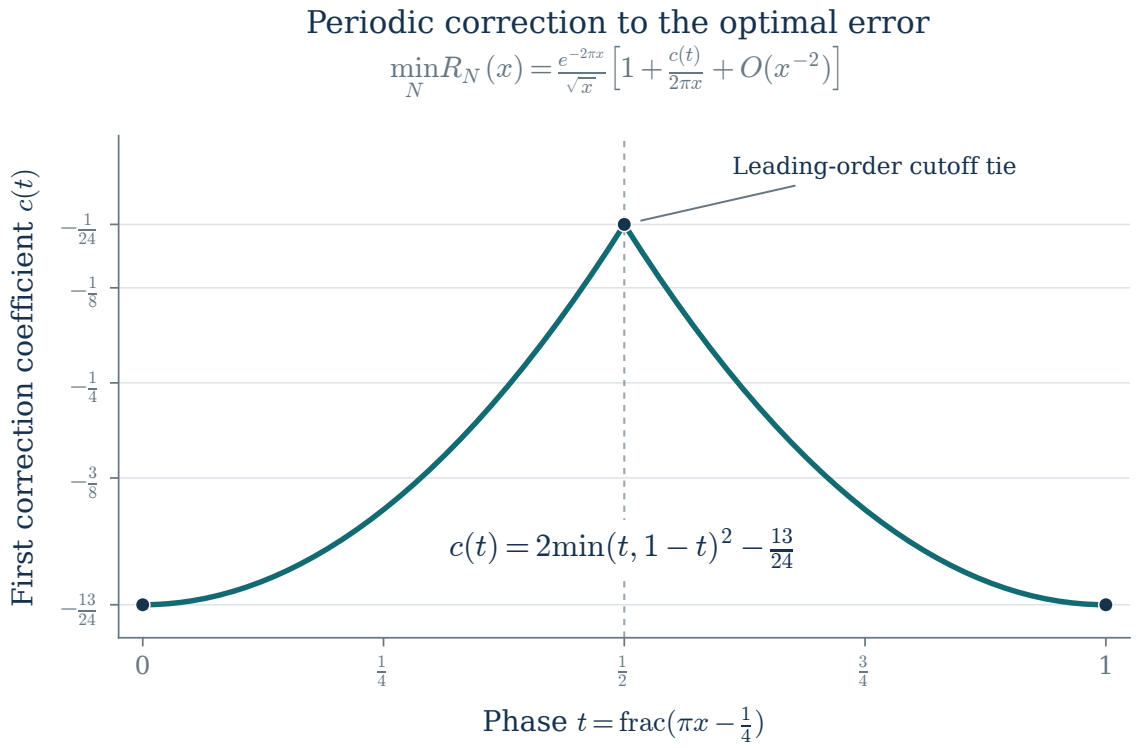


Figure 11.2: The explicit first correction in (11.57), as a function of the fractional part of $\pi x - 1/4$. This plots a proved coefficient function, not fitted numerical data. The two endpoint phases represent the same point of the unit circle.

11.14.1 The periodic correction to the best attainable error

Corollary 11.30 (Integer-cutoff oscillation)**.** *Uniformly for real $x \to +\infty$,*

$$\min_{N\geq 0} R_N(x) = \frac{e^{-2\pi x}}{\sqrt{x}}\left[1 + \frac{2\,\mathrm{dist}(\pi x - \frac{1}{4}, \mathbb{Z})^2 - \frac{13}{24}}{2\pi x} + O(x^{-2})\right]. \tag{11.57}$$

The numerator of the first correction ranges from $-13/24$ to $-1/24$.

Proof. Complete the square in (11.46): $C_1(\delta) = 2(\delta + 1/4)^2 - 13/24$. The exact transition expansion shows that the optimizing index differs from a nearest integer to $\pi x - 1/4$ only within $O(x^{-1})$ of a tie phase. There the squared distances for the two competing integers differ by $O(x^{-1})$; after multiplication by $(2\pi x)^{-1}$ this contributes only $O(x^{-2})$. The bounded-δ uniformity of Theorem 11.25 completes the proof, including arbitrarily close approaches to transition points. □

11.15 Rational Herglotz derivatives and an exponentially small oscillation

Let

$$F(x) = \sum_{n\geq 1} \frac{\psi(nx) - \log(nx)}{n}, \qquad D_r(x) = \frac{(-1)^{r+1}x^{r+1}}{r!}F^{(r)}(x) \quad (r \geq 1, x > 0).$$

The all-orders assertion that rational jets reduce to depth-one cyclotomic polylogarithms is already stated by Radchenko–Zagier [35], immediately after Proposition 4 of their preprint, p. 9. It must not be presented as a new closure theorem. The explicit Bernoulli coefficient formula below gives a constructive version; the large-order estimate and its effective remainder bound are derived independently here. The limit considered below is derivative order tending to infinity at a fixed positive argument.

11.15.1 An explicit finite rational-jet formula

Write $B_j(X)$ for the Bernoulli polynomials with $B_1(X) = X - 1/2$. For positive integers p, q (coprimality is unnecessary for this formula), put

$$c_{a,b} = \frac{a}{q} + \frac{b}{p}, \qquad d_{r,j}(a,b) = \binom{r}{j} B_{r-j}(1 - b/p) - \mathbf{1}_{j=0} B_r(a/q),$$

where $1 \le a \le q$, $0 \le b < p$, and $0 \le j < r$.

Theorem 11.31 (Finite Bernoulli–Hurwitz evaluator)**.** *For $r \ge 1$,*

$$F^{(r)}(p/q) = \frac{(-1)^{r+1}(r-1)!\,q^{r-1}}{p^{r+1}} \left\{ \sum_{a=1}^{q} \sum_{b=0}^{p-1} \left[-\psi(c_{a,b}) + \sum_{j=0}^{r-1} d_{r,j}(a,b)\,\zeta(r+1-j, c_{a,b}) \right] + pq \left(\frac{1}{r} - \gamma - \log q \right) \right\}.$$

In particular every Hurwitz zeta value in the formula has integer order between 2 and $r+1$; no spectral derivatives occur.

Proof. Use the classical integral representation (Radchenko–Zagier, eq. (6))

$$F^{(r)}(x) = (-1)^{r+1} \int_0^\infty t^r \operatorname{Li}_{1-r}(e^{-xt}) \left(\frac{1}{1-e^{-t}} - \frac{1}{t} \right) dt.$$

After $t = qu$ introduce, initially for $\Re s > 0$,

$$I(s) = \int_0^\infty u^{r+s} \operatorname{Li}_{1-r}(e^{-pu}) \left(\frac{1}{1-e^{-qu}} - \frac{1}{qu} \right) du.$$

The combined integral is holomorphic on $\Re s > -1$ since its integrand is $O(u^{\Re s})$ at zero. In $\Re s > 0$ the two terms can be integrated separately, giving

$$I(s) = \Gamma(r+1+s) \sum_{n \ge 1, m \ge 0} \frac{n^{r-1}}{(pn+qm)^{r+1+s}} - \frac{\Gamma(r+s)}{qp^{r+s}} \zeta(1+s).$$

Set $n = qk + a$, $m = p\ell + b$, and $h = k + \ell$. Faulhaber's identity gives

$$\sum_{k=0}^{h} (qk+a)^{r-1} = \frac{q^{r-1}}{r} \left[B_r(h+1+a/q) - B_r(a/q) \right].$$

Writing $v = h + c_{a,b}$ expands the bracket as $\sum_{j=0}^{r} d_{r,j}(a,b) v^j$, with $d_{r,r} = 1$. Consequently

$$\sum_{n \ge 1, m \ge 0} \frac{n^{r-1}}{(pn+qm)^{r+1+s}} = \frac{q^{r-1}}{r(pq)^{r+1+s}} \sum_{a,b} \sum_{j=0}^{r} d_{r,j}(a,b)\,\zeta(r+1+s-j, c_{a,b}).$$

The $j = r$ terms have Laurent expansion $\zeta(1+s, c) = s^{-1} - \psi(c) + O(s)$. Their combined residue cancels that of the second term of $I(s)$. The finite part of the two gamma factors contributes $pq[1/r - \gamma - \log q]$ after factoring out $(r-1)!/(p^{r+1}q^2)$. Finally $F^{(r)}(p/q) = (-1)^{r+1}q^{r+1}I(0)$. All rearrangements took place in an absolutely convergent half-plane; holomorphy of the combined integral justifies taking the finite part. □

Cyclotomic interpretation. For coprime p, q, the rational function

$$R_{r,p,q}(z) = \frac{\mathrm{Li}_{1-r}(z^p)}{1 - z^q}$$

has poles only at pth and qth roots of unity. Its coefficient sequence has a unique finite decomposition

$$[z^N]R_{r,p,q}(z) = \sum_{\omega^p=1 \text{ or } \omega^q=1} \sum_{j=0}^{r} a_{\omega,j} N^j \omega^N \quad (N \geq 1).$$

The coefficients lie in $\mathbb{Q}(\mu_{pq})$, $a_{1,r} = 1/(rp^r q)$, and $a_{\omega,r} = 0$ for $\omega \neq 1$. Partial fractions therefore give the equivalent formula

$$F^{(r)}(p/q) = (-1)^{r+1}(r-1)!(q/p)^r \left(\frac{1}{r} + \log p\right) + (-1)^{r+1} r! q^{r+1} \sum_{\omega} \sum_{j=0}^{r-1} a_{\omega,j} \,\mathrm{Li}_{r+1-j}(\omega).$$

This proves the already known closure statement and gives all its coefficients. The special $a_{1,r}\zeta(1+s)$ term is the only pole: its finite-part contribution is exactly the displayed elementary term.

For example,

$$F''(1) = -\frac{1}{2} - \zeta(2), \qquad F'''(1) = \frac{2}{3} + 3\zeta(2) + \zeta(3),$$
$$F''(1/2) = -2 - 8\zeta(2) - \frac{7}{2}\zeta(3), \qquad F'''(1/2) = \frac{16}{3} + 48\zeta(2) + 32\zeta(3).$$

At $p = q = 1$ the complete formula reduces to

$$F^{(r)}(1) = (-1)^{r+1}(r-1)! \left[\frac{1}{r} + \sum_{j=1}^{r-1} \binom{r}{j} B_{r-j}(1)\zeta(r+1-j)\right].$$

For high r this exact expression can suffer substantial floating-point cancellation; exact coefficients do not make low-precision evaluation reliable.

11.15.2 The complete algebraic large-order expansion

Define

$$E_r(x) = D_r(x) - \zeta(2) - \frac{x}{r}(\log x - H_{r-1}).$$

The following result concerns $r \to \infty$ with x fixed, not the large-x Bernoulli expansion of F.

Theorem 11.32 (Large derivative order with effective remainder)**.** *For $x > 0$, integers $r \geq 2$, real $1 < A < r$, and $\pi/4 < \theta < \pi/2$,*

$$|E_r(x)| \leq \frac{2x\zeta(A)\zeta(A+1)}{\pi(2\theta - \pi/2)\cos\theta} \left(\frac{x}{2\pi\cos^2\theta}\right)^A \frac{\Gamma(r-A)\Gamma(A)\Gamma(A+1)}{\Gamma(r+1)}.$$

Consequently, uniformly for x in each compact subinterval of $(0,\infty)$,

$$E_r(x) = O\left(r^{-1/2}\exp[-2\sqrt{\pi r/x}]\right).$$

In particular, the exact harmonic-number expression $\zeta(2) + (x/r)(\log x - H_{r-1})$ contains every algebraic order in $1/r$.

Proof. Binet's digamma integral and Mellin inversion give, for $1 < c < 2$,

$$F(x) + \frac{\zeta(2)}{2x} = -\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{\pi\Gamma(s)\zeta(s)\zeta(s+1)}{(2\pi x)^s \sin(\pi s/2)}\,ds.$$

One derivation is to Mellin-transform $\psi(y) - \log y + 1/(2y)$ in $1 < \Re s < 2$ using $\int_0^\infty v^{s-1}/(1+v^2)\,dv = \pi/[2\sin(\pi s/2)]$; the sum over n supplies $\zeta(s+1)$. Absolute convergence of Binet's integral and exponential decay on vertical lines justify inversion and fixed-order differentiation. After r differentiations,

$$D_r(x) = \frac{\zeta(2)}{2} + \frac{1}{2\pi i}\int_{(c)} \frac{\pi\Gamma(r+s)\zeta(s)\zeta(s+1)x^{1-s}}{\Gamma(r+1)(2\pi)^s \sin(\pi s/2)}\,ds.$$

Move the contour to $\Re s = -A$. The only crossed poles are $s = 1$ and $s = 0$. At the negative even integers the zero of $\zeta(s)$ cancels the sine denominator; gamma poles lie to the left since $A < r$. The residue at 1 is $\zeta(2)/2$. Using $\zeta'(0) = -\frac{1}{2}\log(2\pi)$ and $\psi(r) = H_{r-1} - \gamma$, the residue at 0 is $x(\log x - H_{r-1})/r$. Thus the integral on the new line equals $E_r(x)$.

Apply the zeta functional equation twice to its kernel:

$$\frac{\pi\zeta(s)\zeta(s+1)}{(2\pi)^s\sin(\pi s/2)} = 2(2\pi)^s\cos(\pi s/2)\Gamma(1-s)\Gamma(-s)\zeta(1-s)\zeta(-s).$$

On $s = -A + it$, use $|\zeta(1-s)\zeta(-s)| \le \zeta(A+1)\zeta(A)$ and $|\Gamma(r-A+it)| \le \Gamma(r-A)$. Rotating the Euler gamma integral through angle $\theta\,\mathrm{sgn}(t)$ gives the elementary inequality

$$|\Gamma(a+it)| \le \Gamma(a)(\cos\theta)^{-a}e^{-\theta|t|} \qquad (a > 0, 0 < \theta < \pi/2).$$

Use it for the other two gamma factors, with parameters A and $A+1$, and use $|\cos(\pi s/2)| \le e^{\pi|t|/2}$. The remaining integral is $\int_{\mathbb{R}} e^{-(2\theta-\pi/2)|t|}dt = 2/(2\theta - \pi/2)$, yielding the stated effective bound. The same estimates justify the horizontal-contour limits.

For the stretched-exponential estimate take an integer $m = \sqrt{\pi r/x} + O(1)$ and $\theta = \pi/4 + 1/(2m)$. For r sufficiently large, $1 < m < r$ and the bound applies. Uniformly on compact positive x-intervals,

$$\frac{\Gamma(r-m)}{\Gamma(r+1)} = r^{-m-1}\exp\left[\frac{m(m+1)}{2r} + O(r^{-1/2})\right], \qquad \Gamma(m)\Gamma(m+1) = 2\pi m^{2m}e^{-2m}(1+O(m^{-1})).$$

Also $(2\pi\cos^2\theta)^{-m} = \pi^{-m}e^{1+O(m^{-1})}$, and the prefactor is $O(xm)$. The power and exponential factors reduce to $O(r^{-1/2}e^{-2\sqrt{\pi r/x}})$, as required. □

An elementary factorial form. At $\theta = \pi/3$ and integer $2 \le m < r$ the estimate becomes

$$|E_r(x)| \le \frac{24x}{\pi^2}\zeta(m)\zeta(m+1)\left(\frac{2x}{\pi}\right)^m \frac{(r-m-1)!(m-1)!m!}{r!}.$$

It is useful as a directly computable error certificate even without asymptotic optimization.

Positivity and a convergent Taylor certificate. The derivative integral has a positive kernel with $1/2 < 1/(1-e^{-t}) - 1/t < 1$, so

$$\frac{\zeta(2)}{2} < D_r(x) < \zeta(2).$$

Therefore the Taylor expansion of $F(x+h)$ about $x > 0$ has, for $\rho = |h|/x < 1$, the explicit remainder bound

$$\left| F(x+h) - \sum_{j=0}^{N} \frac{F^{(j)}(x)}{j!} h^j \right| \le \frac{\zeta(2)}{x} \frac{\rho^{N+1}}{1-\rho}.$$

Its radius is exactly x, since $D_r(x) \to \zeta(2)$.

11.15.3 Exact arithmetic sectors and the leading oscillation

The divisor-weighted eta/Binet representation in Radchenko–Zagier [35, Section 7.3] is classical context for this calculation. The following identity is a differentiated and reciprocity-transformed form useful for large derivative order. The quantitative sum estimates are needed in addition to a one-sector expansion.

Write $\sigma_{-1}(n) = \sum_{d|n} d^{-1}$. Let U denote Tricomi's confluent hypergeometric function with its principal branch. Its integral representation is DLMF 13.4.4 [82]. The large-first-parameter asymptotics used below are classical (DLMF 13.8.11–16; Temme [83]); a direct saddle calculation is included to fix the branch and the first correction in this application. The arithmetic sum over all sectors and its aggregate error bound must be justified in addition to that classical one-sector asymptotic.

Theorem 11.33 (Exact divisor-weighted expansion)**.** *For every integer $r \ge 2$ and $x > 0$,*

$$\begin{aligned} E_r(x) &= 2x\Gamma(r) \sum_{n\ge 1} \sigma_{-1}(n) \operatorname{Re} U\left(r, 0, \frac{2\pi i n}{x}\right) \\ &= 2x \sum_{n\ge 1} \sigma_{-1}(n) \int_0^\infty \frac{u^{r-1}}{(1+u)^{r+1}} \cos\left(\frac{2\pi n u}{x}\right) du. \end{aligned}$$

The series of the integrals is absolutely convergent. The assertion does not require absolute convergence after moving the infinite sum inside the oscillatory u-integral.

More precisely, for $1 < A < r$ and $\pi/4 < \theta < \pi/2$, the contribution of an arbitrary subset $\mathcal{N} \subset \mathbb{N}$ is bounded by the previous effective estimate with $\zeta(A)\zeta(A+1)$ replaced by $\sum_{n\in\mathcal{N}} \sigma_{-1}(n) n^{-A}$.

Proof. Set $w = -s$ in the post-functional-equation contour representation:

$$E_r(x) = \frac{2x}{2\pi i \Gamma(r+1)} \int_{(A)} \Gamma(r-w)\Gamma(w)\Gamma(w+1)\zeta(w)\zeta(w+1)\cos(\pi w/2) \left(\frac{x}{2\pi}\right)^w dw.$$

The Dirichlet series $\zeta(w)\zeta(w+1) = \sum_{n\ge 1} \sigma_{-1}(n) n^{-w}$ converges absolutely on this line. The effective gamma bound proves absolute convergence of the resulting series of contour integrals.

For $\Re z > 0$ put

$$M_r(z) = \int_0^\infty e^{-zu} \frac{u^{r-1}}{(1+u)^{r+1}} du = \Gamma(r) U(r, 0, z).$$

For $0 < \Re w < r$, taking its Mellin transform in z gives

$$\int_0^\infty z^{w-1} M_r(z)\, dz = \frac{\Gamma(w)\Gamma(r-w)\Gamma(w+1)}{\Gamma(r+1)}.$$

Mellin inversion is valid in that strip. Both the defining integral and the Barnes integral extend continuously to $z = \pm i\tau$, $\tau > 0$: the first by dominated convergence, since its rational weight is integrable, and the second by its exponential vertical decay. Averaging the two boundary values replaces z^{-w} by $\tau^{-w}\cos(\pi w/2)$. Substitution proves the formula and the termwise bounds. □

A finite-sum tail bound. For an integer cutoff $N \geq 1$ and $A > 1$,

$$\sum_{n>N} \sigma_{-1}(n) n^{-A} \leq N^{1-A}\left[\frac{1+\log N}{A-1} + \frac{1}{(A-1)^2}\right],$$

because $\sigma_{-1}(n) \leq H_n \leq 1 + \log n$ and $(1+\log t)t^{-A}$ is decreasing for $t \geq 1$. Thus the exact series comes with a computable truncation bound.

An alternative bound avoids the auxiliary contour parameters. Rotate the defining u-integral for $M_r(i\tau)$ to the negative imaginary ray. There are no poles in that quadrant; the small and large arcs vanish. Then

$$|M_r(i\tau)| \leq \int_0^\infty e^{-\tau y} \frac{y^{r-1}}{|1-iy|^{r+1}}\, \mathrm{d}y \leq \Gamma(r)\tau^{-r}.$$

It follows that the arithmetic tail after N terms is at most

$$2x\Gamma(r)\left(\frac{x}{2\pi}\right)^r N^{1-r}\left(\frac{1+\log N}{r-1} + \frac{1}{(r-1)^2}\right), \qquad r \geq 2. \tag{11.58}$$

This is often the simplest finite-sum certificate; the earlier gamma-factor bound is more useful for the sharp large-r exponent.

Theorem 11.34 (Two-term oscillatory asymptotic)**.** *For fixed $x > 0$ define*

$$\begin{aligned}
\mathcal{A}_r(x) &= 2^{5/4}\pi^{3/4}x^{3/4}r^{-3/4}e^{-2\sqrt{\pi r/x}},\\
\Phi_r(x) &= 2\sqrt{\pi r/x} - \frac{\pi}{x} - \frac{\pi}{8},\\
\kappa(x) &= \frac{3}{16}\sqrt{\frac{x}{2\pi}} + \frac{1}{12}\left(\frac{2\pi}{x}\right)^{3/2}.
\end{aligned}$$

Then, uniformly for x in compact subintervals of $(0,\infty)$,

$$E_r(x) = \mathcal{A}_r(x)\left[\cos\Phi_r(x) + \frac{\kappa(x)}{\sqrt{r}}\cos\left(\Phi_r(x) + \frac{\pi}{4}\right) + O(r^{-1})\right].$$

This is an additive oscillatory expansion; a relative-error claim after division by $\cos\Phi_r(x)$ would be invalid near its zeros.

Proof. We first prove, uniformly for $z = i\tau$ with τ in any compact positive interval,

$$M_r(z) = \sqrt{\pi}\, z^{1/4} r^{-3/4} e^{z/2 - 2\sqrt{rz}}\left[1 + \frac{1}{\sqrt{r}}\left(\frac{3}{16\sqrt{z}} - \frac{z^{3/2}}{12}\right) + O(r^{-1})\right]. \tag{11.59}$$

All powers use their principal values.

Rotate the u-contour in the integral defining $M_r(i\tau)$ to argument $-\pi/4$. There are no poles in that quadrant. On the large arc the rational factor is $O(R^{-2})$, while $|e^{-i\tau u}| \le 1$; a small arc of radius ϵ contributes $O(\epsilon^r)$. Hence the rotation is justified directly. Put

$$u = \sqrt{r/z}\, v, \qquad \lambda = \sqrt{rz}, \qquad \epsilon = \sqrt{z/r}.$$

This gives the exact representation

$$M_r(z) = \sqrt{z/r} \int_0^\infty v^{-2} \exp\left[-\lambda v - (r+1)\log(1+\epsilon/v)\right] dv.$$

On any fixed compact positive v-interval,

$$\begin{aligned} -\lambda v - (r+1)\log(1+\epsilon/v) = & -\lambda(v+v^{-1}) + \frac{z}{2v^2} \\ & + r^{-1/2}\left(-\frac{\sqrt{z}}{v} - \frac{z^{3/2}}{3v^3}\right) + O(r^{-1}), \end{aligned}$$

with the same estimates after any fixed number of v-derivatives.

For completeness, localization at $v = 1$ is uniform. Write $a = \sqrt{\tau/(2r)}$, so $\epsilon = a(1+i)$ and $\Re\lambda = ra$. On a compact v-interval the real part of the exponent is $-ra(v+v^{-1}) + O(1)$, and $v + v^{-1}$ has its unique minimum 2 at 1. For the small tail $0 < v < \delta < 1/2$, use $|1+\epsilon/v| \ge 1 + a/v$ and substitute $y = a/v$:

$$\int_0^\delta v^{-2}|1+\epsilon/v|^{-r-1} dv \le \frac{1}{ar}(1+a/\delta)^{-r} = O\left(r^{-1/2} e^{-ra/\delta + O(1)}\right).$$

This is exponentially smaller than e^{-2ra}. For $v > M > 2$, the factor $e^{-\lambda v}$ gives the analogous bound $O(e^{-Mra})$. The remaining compact interval outside any fixed neighborhood of 1 also has a strict exponential gap.

Apply complex Laplace expansion at $v = 1$ to $\phi(v) = v + v^{-1}$ and $g(v) = v^{-2} e^{z/(2v^2)}$. It is legitimate since $\arg\lambda = \pi/4$ lies strictly inside the right half-plane. The Gaussian integral along the real v-axis gives $e^{-2\lambda}\sqrt{\pi/\lambda}$. The coefficient of $1/\lambda$, divided by $g(1)$, is

$$\frac{g''(1)}{4g(1)} + \frac{3g'(1)}{4g(1)} + \frac{3}{16} = z + \frac{z^2}{4} + \frac{3}{16}.$$

Adding the explicit $r^{-1/2}$ correction in the exponent at $v = 1$ gives

$$-\sqrt{z} - \frac{z^{3/2}}{3} + \frac{1}{\sqrt{z}}\left(z + \frac{z^2}{4} + \frac{3}{16}\right) = \frac{3}{16\sqrt{z}} - \frac{z^{3/2}}{12}.$$

This proves (11.59); further terms follow by the same finite Taylor and Gaussian-moment calculation, with standard local remainder estimates. As an independent classical check, in DLMF 13.8.11 with $b = 0$ one has $p_0(z) = 1$ and $q_0(z) = -z/12$; the standard large-argument expansions of K_1 and K_0 give exactly the same first correction.

At $z = 2\pi i/x$, taking $2x$ times the real part of (11.59) gives the stated main term and correction. Indeed $z^{1/4}$ has argument $\pi/8$, and the correction equals $\kappa(x)e^{-\pi i/4}$.

It remains to control all the other arithmetic sectors uniformly. For $A \ge 2$,

$$\zeta(A)\zeta(A+1) - 1 = \sum_{n\ge 2} \sigma_{-1}(n) n^{-A} \le 4\big(\zeta(2)\zeta(3) - 1\big)2^{-A}.$$

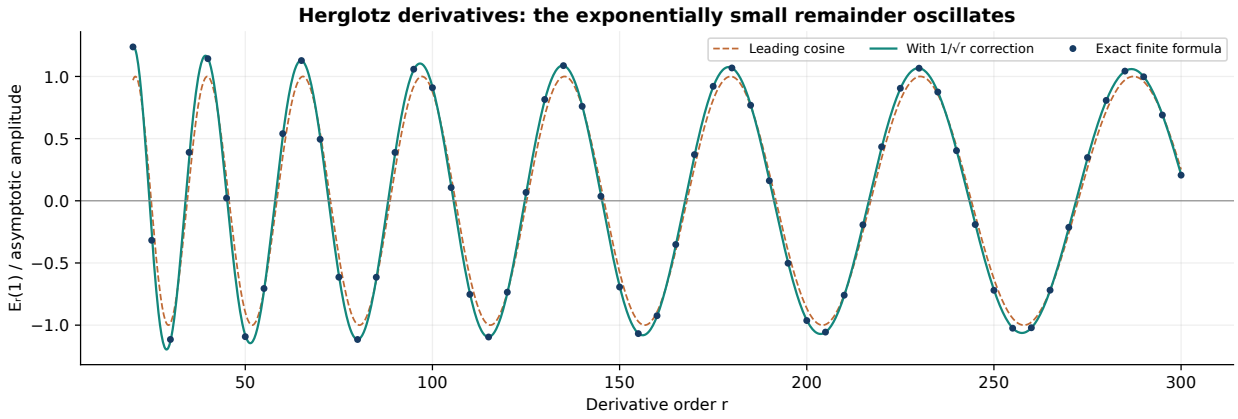


Figure 11.3: The Herglotz remainder divided by its asymptotic amplitude at $x = 1$. The leading cosine and its first correction track the exact finite-formula values. The asymptotic is additive, so no division by a small cosine is used.

Use the effective sector bound with $A = \sqrt{2\pi r/x} + O(1)$ an integer and $\theta = \pi/4 + 1/(2A)$. Exactly the previous optimization yields

$$\sum_{n\geq 2} 2x\sigma_{-1}(n)\,\mathrm{Re}\,M_r(2\pi i n/x) = O\left(r^{-1/2}e^{-2\sqrt{2\pi r/x}}\right).$$

This is absorbed into $\mathcal{A}_r(x)O(r^{-1})$, uniformly on compact positive x-intervals. □

Corollary 11.35 (Sharp exponential rate and infinitely many signs)**.** *For each fixed $x > 0$ the real sequence $E_r(x)$ takes positive and negative values infinitely often. More precisely,*

$$\limsup_{r\to\infty} \frac{E_r(x)}{\mathcal{A}_r(x)} = 1, \qquad \liminf_{r\to\infty} \frac{E_r(x)}{\mathcal{A}_r(x)} = -1,$$

and

$$\limsup_{r\to\infty} \frac{\log|E_r(x)|}{\sqrt{r}} = -2\sqrt{\pi/x}.$$

Here $\log 0 = -\infty$ if an exact zero occurs.

Proof. The phase $\Phi_r(x)$ increases to infinity, with successive differences tending to zero. Consequently it has subsequences approaching any fixed phase modulo 2π. Apply the additive expansion and choose phases 0 and π. The same subsequences establish the lower bound for the limsup of the logarithmic rate; the asymptotic envelope gives the upper bound. □

Further research specific to this section. Compute and certify several further half-integer corrections in (11.59); build an interval implementation of the exact sector sum; locate the sign-transition indices beyond the leading $\sqrt{r}$ phase law; and obtain a uniform transition theory when x also varies with r. The fixed-x theorem above does not justify such a double-scaling limit automatically.

11.16 Quadratic units and Kronecker-limit coordinates

Define the normalized functions

$$\mathcal{F}(x) = F(x) - F(1) + \frac{\pi^2}{12}(x - 2 + x^{-1}) - \frac{1}{4}\log^2 x, \quad \mathcal{J}(x) = J(x) - \frac{1}{2}\log^2 2 + \frac{\pi^2}{24}(x - x^{-1}).$$

For a real quadratic order of discriminant D, a narrow ideal class $\mathcal{B}$ has a partial zeta function. Its Kronecker constant is

$$\varrho(\mathcal{B}) = \lim_{s\to 1}\left[D^{s/2}\zeta(\mathcal{B}, s) - \frac{\log \varepsilon_D}{s-1}\right],$$

where $\varepsilon_D > 1$ is the least norm-one unit. In the reduced-form cycle representation, $w_j > 1 > w_j' > 0$, Radchenko–Zagier give

$$\varrho(\mathcal{B}) = 2\gamma \log \varepsilon_D - \ell(\mathcal{B})\frac{\pi^2}{6} + \sum_{j \bmod \ell}\left[\mathcal{F}(w_j) - \mathcal{F}(w_j') + L(w_j'/w_j)\right], \tag{11.60}$$

where L is the unnormalized Rogers dilogarithm. Class characters turn the ϱ coordinates into Hecke L-values. Their Theorem 5 expresses $\mathcal{J}$ at specified even-trace units in these coordinates. The source's independent cycle engine reported checks at $n = 2, 4, 6, 8$ with residuals about 5×10^{-38}; the engine was not imported into this checkout.

11.17 Regulator candidates and field degrees

The published Stark examples and the source's follow-up fits are conjectural or numerical evaluations. A cubic polynomial over $\mathbb{Q}$ defines a degree-three field E. It cannot itself be a Hilbert or ring class field containing a quadratic field K. In the examples at issue, its degree-six splitting field can be a cyclic cubic extension of K; E is a cubic subfield. The distinction corrects a repeated field-degree error in the report.

For $u = 16 + \sqrt{257}$ the source reports

$$\mathcal{J}(u) \overset{\text{num}}{=} 4\zeta(2) + \log 2 \log u + 2\,\mathrm{reg}(E), \quad E = \mathbb{Q}[x]/(x^3 - 2x^2 - 3x + 1). \tag{11.61}$$

The determinant of the two selected unit-logarithm columns is signed; with the source's ordering it is $-\mathrm{reg}(E)$. Vertical bars around a matrix denote its determinant, not an absolute value. Replacing this negative determinant by its absolute value without changing its external coefficient reverses the regulator term.

The other published example concerns $u = 10 + 3\sqrt{11}$ and the order of discriminant 396. Its unit-logarithm expression is a Stark minor rather than a full field regulator. We retain that structural distinction; the unimported algebraic-unit definitions do not provide a locally replayable formula for the minor.

The source reports that $n = 6, 10, 14, 16, 26$ are the narrow-class-number-three cases in its scan of $2 \le n \le 5000$. This is a bounded census, not a proof that the family is finite. Negative fits at class numbers five and seven likewise do not establish a universal "single regulator if and only if cyclic cubic" theorem.

There is a precise conditional route to a leading-L interpretation. Suppose a non-Galois totally real cubic field E has quadratic resolvent K, and its normal closure supplies a nontrivial order-three class character χ of K. Artin induction gives $\zeta_E(s) = \zeta(s)L_K(s, \chi)$. The analytic class-number formula then gives

$$L_K(s, \chi) = h_E\,\mathrm{reg}(E)s^2 + O(s^3),$$

n	D_0	cubic field polynomial	c_1	c_2	c_3
6	37	$x^3 - x^2 - 3x + 1$	3/2	2/3	2/3
10	101	$x^3 - x^2 - 5x - 1$	5/2	2/3	2/3
14	197	$x^3 - x^2 - 7x - 3$	7/2	2/3	2/3
16	257	$x^3 - x^2 - 4x + 3$	4	1	2
26	677	$x^3 - x^2 - 11x - 7$	13/2	2/3	2/3

Table 11.1: Historical numerical candidates $\mathcal{J}(n + \sqrt{n^2+1}) = c_1\zeta(2) + c_2 \log 2 \log u + c_3 \operatorname{reg}(E)$. The source reports residuals below 1.2×10^{-109}. These are not unconditional theorems.

because $\zeta_E(s) = -h_E \operatorname{reg}(E) s^2/2 + O(s^3)$ and $\zeta(0) = -1/2$. Thus the coefficient equals $\operatorname{reg}(E)$ only after $h_E = 1$ and the field-character identification have been established. This argument does not convert the numerical Herglotz fits into proofs of Stark's conjecture.

11.18 Exact class-number-one certificates

Proposition 11.36 (Class-number certificates for the five cubic fields)**.** *Each of the five totally real cubic fields in Table 11.2 has class number one. In the table, α denotes a root of the polynomial in its row, d_H is the field discriminant, and $B = \lfloor 2\sqrt{d_H}/9 \rfloor$ is the integer Minkowski bound. The final column lists a generator of a principal ideal of norm p for each indicated rational prime p; the field norm of the generator may be $-p$.*

Polynomial for α	d_H	B	p : generator
$X^3 - X^2 - 3X + 1$	148	2	$2 : \alpha - 1$
$X^3 - X^2 - 5X - 1$	404	4	$2 : \alpha + 1$; $3 : \alpha^2 - 2\alpha - 2$
$X^3 - X^2 - 7X - 3$	788	6	$2 : \alpha + 1$; $3 : \alpha$; $5 : \alpha^2 - 2\alpha - 4$
$X^3 - X^2 - 4X + 3$	257	3	$3 : \alpha$
$X^3 - X^2 - 11X - 7$	2708	11	$2 : \alpha + 1$; $3 : \alpha + 2$; $7 : \alpha$
			$5 : -\alpha^2 + 2\alpha + 10$; $11 : 2\alpha^2 - 4\alpha - 17$

Table 11.2: Exact generators used in the class-number-one proof.

Proof. The rational-root test proves irreducibility of each polynomial f. Their polynomial discriminants are, in the displayed order, $148, 404, 788, 257, 2708$, all positive. Thus all five fields are totally real. At discriminant 257, squarefreeness immediately gives $\mathcal{O}_{H_0} = \mathbb{Z}[\alpha]$. In each other row only 2 could divide the index. In all four cases $\overline{f} = (X+1)^3$ modulo 2, while

$$\left.\frac{f(X) - (X+1)^3}{2}\right|_{X=1} \in \{-5, -7, -9, -13\}.$$

These values are odd. Dedekind's index criterion therefore excludes 2 from the index. Equivalently, each shifted polynomial $f(X+1)$ is Eisenstein at 2, so its root generates the local ring of integers [55, Proposition 7.55 and Remark 7.56]. This proves maximality and the asserted field discriminants. The use of Dedekind factorization below is consequently valid also at 2.

Minkowski's theorem gives an integral ideal in every ideal class with norm at most $2\sqrt{d_H}/9$ for a totally real cubic field [55]. It suffices to prove principalness for every prime ideal whose norm is at most B. In the four even-discriminant rows, 2 is totally ramified; the displayed norm-2 element

generates its unique prime ideal. At discriminant 257, the polynomial is irreducible modulo 2, so the prime above 2 has norm $8 > B$.

For each relevant odd rational prime $p \leq B$, direct factorization of f modulo p has type $1 + 2$. There is therefore exactly one prime ideal of norm p. The corresponding element in the table has norm $\pm p$, so this degree-one prime is principal. Its degree-two partner is principal as well, since their product is (p). This last observation covers in particular the prime of norm 9 in the field of discriminant 2708. The table accounts for all rational primes $p \leq B$, hence for all prime ideals within the Minkowski bound. Every ideal class is therefore trivial.

All finite calculations in this proof—discriminants, index tests, modular splitting types, and generator norms—are reproduced with integer arithmetic by `../reports/corpus-corrections/code/12-relation-spaces-verify_cubic_class_numbers.py`. □

For the other polynomial used in the 257 example, $x^3 - 2x^2 - 3x + 1$, the same bound is three: it is irreducible modulo two, and its unique norm-three prime is generated by $\alpha - 1$. Its squarefree discriminant is 257, so its order is maximal. Thus that cubic also has class number one. The leading primitive Artin coefficient consequently specializes to the regulator in these cases. Inflating a primitive character to a ray modulus can remove Euler factors; at the 257 modulus-two example the factor at zero is three. The primitive and ray-modulus Dirichlet series must be identified before comparing their leading coefficients. This still does not turn the fitted Herglotz evaluations into proved Stark identities.

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[108] ProveIt Contributors, *A Convolution Algebra of Stieltjes Functions*, research continuation, 10 October 2026, prepared against `0efdbaa694`. Preserved under `reports/stieltjes-correlation-calculus/continuations/convolution-algebra/`.

[109] ProveIt Contributors, *Shifted Hurwitz-Zeta Correlations: Exact Stieltjes Closure, Polygamma Contact Terms, and Log-Gamma Antiderivatives*, research continuation, 10 October 2026, observed source `0e5ab2f60f`. Preserved under `reports/stieltjes-correlation-calculus/continuations/shifted-hurwitz-jets/`.

[110] ProveIt Contributors, *Bilinear Closure of Generalized Stieltjes Correlations*, research continuation, 10 October 2026. Individual source observations and scope are retained in the original provenance note. Preserved under `reports/stieltjes-correlation-calculus/continuations/bilinear-correlation-closure/`.

[111] ProveIt Contributors, *A Convolution Calculus for Stieltjes Functions*, research continuation, 10 October 2026, prepared against `570b0567f3`. Preserved under `reports/stieltjes-correlation-calculus/continuations/convolution-calculus/`.

[112] NIST Digital Library of Mathematical Functions, Eq. 15.4.20, Gauss summation of the hypergeometric function at one. https://dlmf.nist.gov/15.4.E20.

[113] J. Choi, Summation formulas involving binomial coefficients, harmonic numbers, and generalized harmonic numbers, *Abstract and Applied Analysis* **2014** (2014), Article 501906, printed p. 6, Conjectures 4 and 6 and Eqs. (40)–(46). https://doi.org/10.1155/2014/501906.

[114] ProveIt Contributors, *Exact Stieltjes and Harmonic Identities*, research continuation, 10 October 2026, prepared against `fc4d3bf80534ad7c901d3b8c9e71baf2df0064ed`. Preserved under `reports/stieltjes-harmonic-resonance/continuations/stieltjes-harmonic-identities/`.

[115] NIST Digital Library of Mathematical Functions, §25.14, Lerch's transcendent. https://dlmf.nist.gov/25.14.

[116] NIST Digital Library of Mathematical Functions, §25.15, Dirichlet L-functions. https://dlmf.nist.gov/25.15.

[117] S. Hu and M.-S. Kim, On the Stieltjes constants and gamma functions with respect to alternating Hurwitz zeta functions, *Journal of Mathematical Analysis and Applications* **509**(1) (2022), 125930. https://arxiv.org/abs/2106.14674.

[118] H. Zhu, S. Hu and M.-S. Kim, On the properties of alternating invariant functions, arXiv:2505.10985v3 (2025), Theorem 1.13. https://arxiv.org/abs/2505.10985.

[119] ProveIt Contributors, *Twisted Stieltjes Convolutions and Harmonic Laurent Identities*, research continuation, 10 October 2026, prepared against `e0d9463bdee9685dfb1dddb819059cc738540c57`. Preserved under `reports/stieltjes-correlation-calculus/continuations/twisted-harmonic-laurent/`.

[120] ProveIt external research continuation, *Exact Identities: Collision Calculus, Shifted Harmonic Jets and Weighted Arctangent Families*, 10 October 2026. Original package retained at `reports/stieltjes-correlation-calculus/continuations/collision-and-weighted-identities/`. The weighted real evaluations are proved in Chapter 3; the remaining claims retain separate intake and audit status. Global priority is not established.

[121] ProveIt research continuations, *Periodic Stieltjes Collisions: Exact Contact Terms, Finite-Part Fubini Laws, and Gamma–Polylogarithm Primitives* and *Periodic Stieltjes Contact Calculus: All-Index Collision Corrections, Harmonic Derivative Anomalies, and Exact Zeta-Jet Integrals*, 10 October 2026. Independent routes retained in the correlation spine as `continuations/periodic-collision-contacts/` and `continuations/periodic-stieltjes-contact/`; their immediate coincident source is preserved as `continuations/coincident-stieltjes/`. The fixed-coordinate periodic comparison is reconciled in Chapter 9; the derivative anomaly was already established there.

[122] NIST Digital Library of Mathematical Functions, §5.7, *Series Expansions of the Gamma Function*, especially formula 5.7.3. https://dlmf.nist.gov/5.7.E3. Inspected 10 October 2026.

[123] ProveIt research continuation, *Mixed Spectral Identities*, 10 October 2026, prepared against `dcd95baeb7c0e914b138bddef6829c5b1d52b726`. Preserved at `reports/stieltjes-harmonic-resonance/continuations/mixed-spectral-identities/`. The mixed Gamma identity is derived from the classical Rogers–Dougall sum in Chapter 8. Its further resonance, harmonic and collision claims retain separate audit scopes. No global priority is asserted.

[124] NIST Digital Library of Mathematical Functions, §16.4, *Rogers–Dougall Very Well-Poised Sum*, formula 16.4.9. https://dlmf.nist.gov/16.4.E9. Inspected 10 October 2026.

[125] ProveIt research contribution, *A certified obstruction in the weight-seven reduction search*, in the extremal cyclic-descent continuation, `article/s6_obstruction.tex`. Pinned repository commit `5a790187c8e186e41e2b990b4941cb7a1a3c7b6b`. Source article.

[126] ProveIt research contribution, *A Cayley normal form that preserves series depth*, in the polylogarithm–Stieltjes critical-transition continuation, `article/05_cayley_depth.tex`. Pinned repository commit `5a790187c8e186e41e2b990b4941cb7a1a3c7b6b`. Source article.

[127] D. E. Radford, A natural ring basis for the shuffle algebra and an application to group schemes, *Journal of Algebra* **58** (1979), no. 2, 432–454. doi:10.1016/0021-8693(79)90171-6.

[128] F. Patras, La décomposition en poids des algèbres de Hopf, *Annales de l'Institut Fourier* **43** (1993), no. 4, 1067–1087. doi:10.5802/aif.1365.

[129] ProveIt research continuation, *Exact Resonant Identities Beyond the Diagonal*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/exact-resonant-identities/`. The complete-Cayley formal obstruction is audited in Chapter 12; the numerical S_6 conjecture remains open.

[130] ProveIt research continuation, *Harmonic Parity and Resolvent Identities*, 11 October 2026. Original package preserved under `reports/stieltjes-harmonic-resonance/continuations/harmonic-parity-resolvents/`. The centered trigamma-square moments and their polylogarithmic generator are audited in Chapter 8; the broader classification and resolvent claims retain separate scopes.

[131] ProveIt research continuations, *Higher Digamma Powers and Ordered Gamma Sums*, *Canonical Digamma Polynomials and the Fifth Harmonic Bridge*, and *Mixed Resolvent Products, Higher Digamma Powers, and Stieltjes Collisions*, 11 October 2026. Original packages are preserved as `reports/stieltjes-harmonic-resonance/continuations/higher-digamma-powers/`, `reports/stieltjes-harmonic-resonance/continuations/canonical-digamma-fifth/`, and `reports/stieltjes-correlation-calculus/continuations/mixed-resolvents-harmonic-powers/`. Their all-power and fifth-moment cores are reconciled in Chapter 8; other Gamma, resolvent, collision and ordered branches retain separate audit scopes.

[132] NIST Digital Library of Mathematical Functions, Sections 5.5 (digamma recurrence and reflection), 5.7 (Taylor and reciprocal-Gamma series), and 5.11 (sectorial asymptotics). In particular formulas 5.5.2, 5.5.4, 5.7.4 and 5.11.2. https://dlmf.nist.gov/5. Inspected 11 October 2026 UTC.

[133] K. Ihara, M. Kaneko and D. Zagier, Derivation and double shuffle relations for multiple zeta values, *Compositio Mathematica* **142** (2006), 307–338. https://doi.org/10.1112/S0010437X0500182X.

[134] P. T. Young, Series of height one multiple zeta functions, *Integers* **24** (2024), A43, published 20 May 2024. https://math.colgate.edu/~integers/y43/y43.pdf. https://doi.org/10.5281/zenodo.11221606. The coefficient-description check in Appendix A concerns the inspected publisher PDF, printed page 12; its ordinary Stieltjes formula is retained.

[135] M. Can, L. Kargın, M. Cenkci and A. Dil, *On certain harmonic zeta functions*, arXiv:2403.07123v1, 11 March 2024, printed page 1. https://arxiv.org/pdf/2403.07123v1. The initial-index correction in Appendix A is scoped to that version's introductory Stieltjes-limit display. Inspected 11 October 2026 UTC.

[136] ProveIt research continuations, *Bernoulli Determinants, Cubic Tornheim Jets, and Canonical Digamma Integrals*, *Differential Polygamma and Lerch Identities*, and *Exact Polylogarithm Parity, Harmonic-Tail Reconstruction, and Higher Polygamma Moments*, 11 October 2026. Original packages preserved under `reports/stieltjes-harmonic-resonance/continuations/bernoulli-tornheim-digamma/`, `reports/stieltjes-harmonic-resonance/continuations/differential-polygamma-lerch-jets/`, and `reports/stieltjes-harmonic-resonance/continuations/polygamma-tail-identities/`. Their common differential-polynomial analytic core is reconciled in Chapter 8; their remaining coordinate, rank and Lerch claims retain separate audit scopes.

[137] ProveIt research continuation, *Beta Mixtures Across Integer Twists: Finite spectral corrections, harmonic logarithmic moments, and Stieltjes–polylogarithm primitives*, 10 October 2026. Original article, PDF and verification evidence preserved under `reports/stieltjes-correlation-calculus/continuations/beta-crossing/`. The affine crossing, integer-total mixture and balanced-jet cores are reconciled in Chapter 8; the later logarithmic Mellin and primitive families retain their separate integration scope.

[138] ProveIt research continuations, *Alternating Tails and Spectral Defects*, *Quartic Spectral Reconstruction*, *Bernoulli Tails and Spectral Faces*, *Bilinear Tails and Harmonic Determinants*, and *Tail Kernels and Boundary Products*, 11 October 2026. Original sources and exact evidence preserved on the harmonic-resonance spine and under `reports/tail-boundaries/`. The common centered coefficient model, retained determinant and effective eventual proof are reconciled in Chapter 8. Complete orders ten and twelve retain report attribution; the collective order-fourteen extension has its full exact certificate under `verification/prefix-order-fourteen/`.

[139] ProveIt research continuations, *Universal Gamma Normal Forms: Rational Kernels, Harmonic Sums, and Stieltjes Primitives*, and *A Universal Beta Formula for Gamma Normal Forms*, 11 October 2026. Original articles, PDFs and exact evidence preserved under `reports/stieltjes-harmonic-resonance/continuations/universal-gamma-coefficients/` and `reports/stieltjes-harmonic-resonance/continuations/universal-beta-normal-forms/`. Their common all-index coefficient formula and ordinary finite projections are reconciled in Chapter 8. The collective analytic-jet extension gives an actual locally convergent infinite normal generator.

[140] ProveIt research continuations, *Higher Identities*, *Cubic Gamma Normal Form*, and *Universal Gamma Coefficients*, 11–12 October 2026. Delivered articles and exact evidence preserved on the harmonic-resonance continuation spine. The compensator construction uses the classical Euler homotopy operator: D. Poole and W. Hereman, *The homotopy operator method for symbolic integration by parts and inversion of divergences with applications*, https://arxiv.org/abs/0908.0399; W. Hereman, B. Deconinck and L. D. Poole, *Continuous and Discrete Homotopy Operators: A Theoretical Approach made Concrete*, https://arxiv.org/abs/nlin/0510064. The hierarchy specialization, analytic primitive domain and independent normal-generator proof are reconciled in Chapter 8. The collective continuation proves every primitive coefficient and full support/sign, with convergence on the full input disk; fixed-gap analytic completion remains a separate pending proof scope.

[141] ProveIt research continuation, *A Finite Polylogarithm Alphabet for Rational Hankel Sums*, 11 October 2026. Original article, compiler and recorded evidence preserved under `reports/stieltjes-harmonic-resonance/continuations/rational-hankel-polylogarithms/`. The confluent weight budget, anchored two-letter compiler and quadratic denominator evaluation are reconciled in Chapter 8. The classical tools retain their attribution: M. E. Hoffman, *Quasi-shuffle products*, J. Algebraic Combin. **11** (2000), 49–68, https://arxiv.org/abs/math/9907173; E. Panzer, *Algorithms for the symbolic integration of hyperlogarithms with applications to Feynman integrals*, Comput. Phys. Commun. **188** (2015), 148–166, https://arxiv.org/abs/1403.3385.

[142] ProveIt research continuations, *Resonant Polylogarithmic Jets and Convergent Inverse Transseries* and *Polylogarithm Order Jets and Resonant Inversion*, 11–12 October 2026. Original sources and evidence are preserved under `reports/uniform-transition-continuation/continuations/`, in `normalized-resonant-jets/` and `order-jets-resonant-inversion/`. The

fixed-sector theorem extends R1 of the preserved `resonance-correlations-nonseparable-jets/` report. Reciprocal-Gamma normalization is credited to both overlapping continuations. Section 1.6 develops the exact order-lowering and arbitrary repeated-primitive identities from that normalization. Inverse geometry and Borel reconstruction retain separate review scopes.

[143] ProveIt research continuations, *Resonant Polylogarithm Transseries* and *Boundary Completion for Quadratic Polylogarithm Feedback*, 11–12 October 2026. Original articles, PDFs and evidence are preserved under `Analysis/Transseries/docs/series-and-transseries/`, in `Resonant_Polylogarithm_Transseries/` and `Quadratic_Polylogarithm_Feedback_Boundary_Completion/`. The first report establishes the interior fold and poses the boundary question; the second proves its completion. Section 1.7 reconciles that classification and develops finite Bessel-polynomial coefficient and order-spectrum identities. Nonmarginal critical charts and crossover refinements retain separate integration scopes.

[144] NIST Digital Library of Mathematical Functions, §18.34, *Bessel Polynomials*, especially (18.34.1)–(18.34.5), https://dlmf.nist.gov/18.34. The convention here is $y_n(x) = y_n(x; 2)$.

[145] P. Flajolet and A. Odlyzko, *Singularity analysis of generating functions*, SIAM J. Discrete Math. **3** (1990), 216–240.

[146] NIST Digital Library of Mathematical Functions, §4.13, *Lambert W-function*, https://dlmf.nist.gov/4.13.

[147] A. G. Pakes, *Affine relation between an infinitely divisible distribution function and its Lévy measure*, J. Math. Anal. Appl. **499** (2021), article 125008, https://doi.org/10.1016/j.jmaa.2021.125008. Related lower-branch Laplace-transform literature; the normalization and identities used in Section 1.8 are proved there.

[148] ProveIt research continuation, *Mixed Hurwitz-Tail Moments, Zeta-Anchor Reductions, and Stieltjes Transport*, 11 October 2026, preserved article and verification evidence in `reports/stieltjes-harmonic-resonance/continuations/mixed-tail-anchor-reductions/`. Its third research question asks for rational polynomial combinations for $(2, 2, 3)$ and $(2, 3, 4)$. Section 8.48 develops the first case within a precisely specified polynomial certificate class.